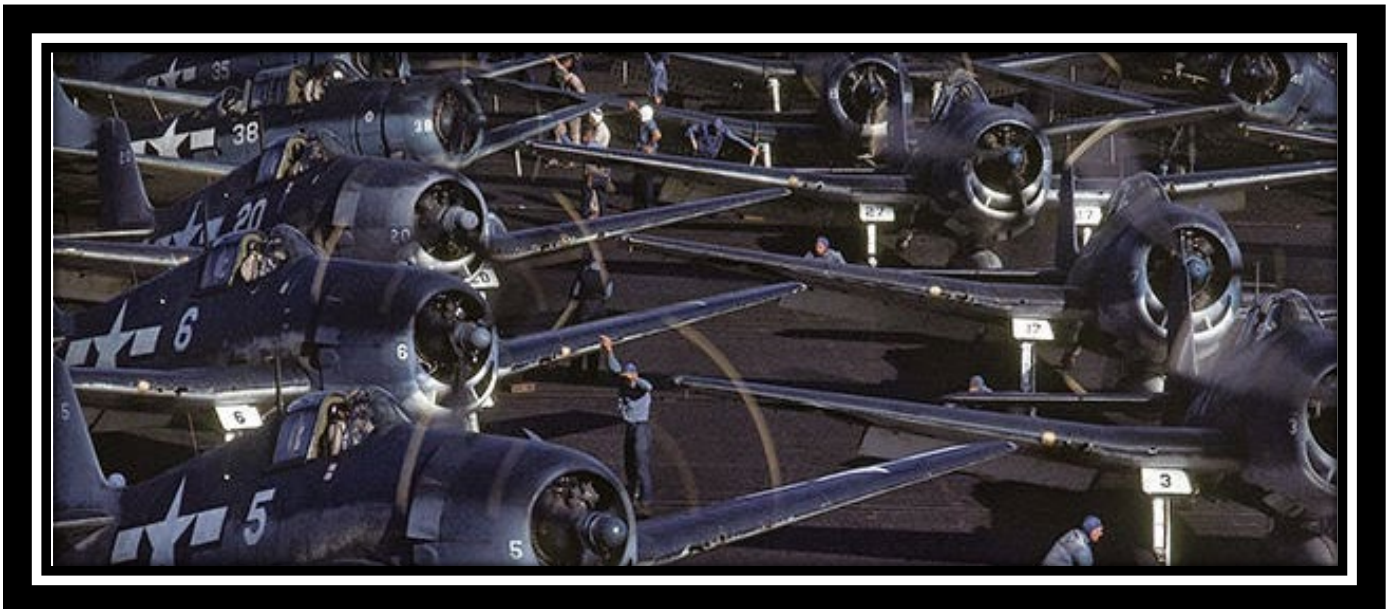


THE UNITED STATES NAVAL WAR COLLEGE

Joint Military Operations Department



War at Sea Instruction Book



Instructions and Rules to Simulate Naval Battles from World War II into the Near Future

“The future U.S. admirals of the Pacific War had reached their late 20s and early 30s...they were sent off to the Naval War College in Newport, where they would study, debate, and plan for prospective future naval wars. Scenarios were played out on tabletop game boards, with model ships to represent the contending fleets and throws of the dice to decide the fortunes of battle.”

-Ian Toll, *Pacific Crucible*.

“The war with Japan had been reenacted in the game room here [Naval War College] by so many people in so many different ways that nothing that happened during the war was a surprise—absolutely nothing except the Kamikaze tactics toward the end of the war; we had not visualized those.”

-ADM Chester Nimitz

Tacit Knowledge: The knowledge, skills, and abilities an individual gains through experience that is often difficult to put into words or otherwise communicate.

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–Introduction–



If this is your first time playing this game, read Appendix C first.

Using This Book

If this is your first time playing, it is advisable to read Appendix C and/or watch the how-to-play videos.

This book covers the instructions and rules for playing three different wargames, which build upon each other as they increase in their complexity. They are differentiated throughout by color association:

When learning...	Font	Rules	Table Outlines
...Leyte	Read only those main body words in black font. Ignore the blue and purple .	<i>Read only the rules that are in this format.</i> Ignore the blue and purple .	Pay attention to only tables outlined in this thick black border.
...Falklands / Malvinas (F/M)	Add the words in bold blue font to your knowledge from Leyte. Ignore the purple .	Add those rules that are in this format. Ignore the purple .	Now also pay attention to tables outlined in this thick blue border.
...Borneo	Add the words in purple Times New Roman font to your knowledge from the first two games.	Add those rules that are in this format.	Now also pay attention to only tables outlined in this thick purple border.

There are five Appendices to help make this book more digestible.

- Appendix A is a breakdown of what each die does, for each game, for each Team.
- Appendix B is the Fire Damage Tables for each game, which tell you how lethal each die is.
- Appendix C is the Quick Start Guide. Appendix C will get the novice up to speed quicker and make the rest of this book more understandable.
- Appendix D is the Quick Start Guide for the F/M game. It quickly covers the major changes you need to know as a player.
- Appendix E is the Quick Start Guide for the Borneo game.

War at Sea Background

War at Sea was designed by the Joint Military Operations (JMO) and Wargaming Departments at the United States Naval War College to support JMO curricula. The game builds upon JMO course material and is designed to cement student understanding of that material, specifically *operational art* and *naval warfare theory*. In general terms, the game allows Teams of players to make tactical and operational decisions and gain experience using the concepts they learned in class to achieve operational outcomes in a maritime warfare context.

While not an exclusive list, the following are some of the major topical areas that *War at Sea* is designed to address:

- The relationship between the strategic, operational, and tactical levels of war and their objectives.
- The operational factors of space, time, and force, and the interrelations between them.
- The relationship between theater structure, theater geometry, and military objectives.
- The importance of operational and joint functions to support operations to achieve military objectives.

- The key elements of operational design, including center of gravity, critical vulnerabilities, lines of operation, etc.
- The importance of the commander's estimate of the situation and the operational idea to developing and conducting military operations.

Purpose of Wargaming

War at Sea forces its players to deal with uncertainty, make decisions in response to that uncertainty, and then deal with the consequences of those actions. Wargames force students to not only discuss critical thinking and problem solving but to actually practice those skills.

The purpose of this wargame is not to "win," although the desire to win can (and should) be highly motivating and contributes to the learning experience. The true purpose is to put players in a stressful situation where they are required to make decisions to achieve objectives. A key component of *War at Sea* is that students play against thinking opponents who are also striving to achieve their own objectives. A key aspect of wargaming is the post-game Debrief, where students discuss their decisions and the processes by which they arrived at those decisions. Education happens in game preparation, during the game, and during the Debrief.

In designing *War at Sea* to be used as an educational tool in support of Professional Military Education objectives, the designers and maintainers balance three often-competing factors:

- Historical and Technical Accuracy (Credibility). Based on the historical examples provided by the US Philippines Campaign in World War II and the 1982 Falklands/Malvinas (F/M) War between the United Kingdom and Argentina, the designers strove to accurately model the numbers, capabilities, and relative combat power of various naval, air, and ground forces. The Borneo scenario is more conjectural, based in the near-future, but also uses a standard model to simulate the relative combat power of belligerents.
- Ease of Play (Utility). Designed as a method to support pre-existing educational requirements within a constrained schedule, a key goal of *War at Sea* was that it not require a great deal of time to understand game play or learn its mechanics. A simple game, learned and played quickly, is better than a complex game that requires a great deal of effort to play.
- Student Buy-In (Immersion). *War at Sea* is not meant to *simulate* warfare or naval combat but merely to *model* them. The difference between simulating and modeling is important; the game must model real world conflict *accurately enough* to ensure students remain immersed in the scenario while also maintaining focus at the operational level of war.

These factors were balanced as an over-emphasis on one could severely detract from the others, (*i.e., ensuring ease of play limits the accuracy*) there will always be tension between these factors, and there is no perfect balance. The current balance has been deemed appropriate to maximize the game's educational objectives over the course of hundreds of iterations.

It bears repeating that *War at Sea* is meant to model the *operational* level of war but certain aspects of the tactical and strategic levels unavoidably find their way into game play and decisions. This in no way lessens the educational benefit of the game but in fact, enhances it by exposing students to the ties the operational level has both up and down the levels of war. However, just as the operational commander should limit their interest in the tactical details of their subordinates, *War at Sea* strays away from the tactical level as much as possible.

–General Overview–

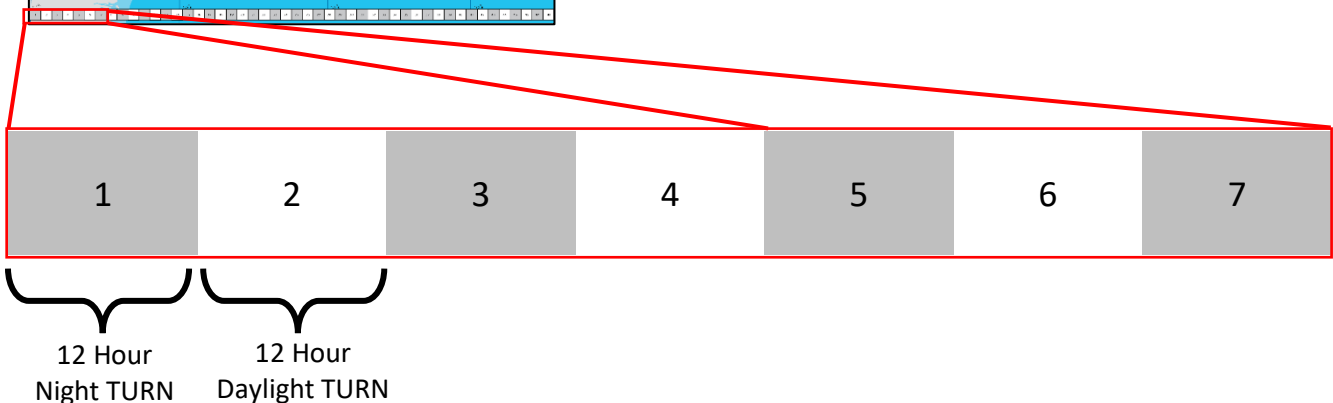
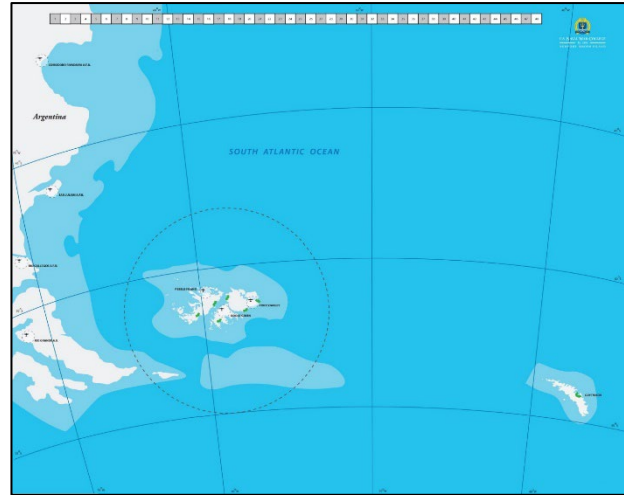
Simplified Overview

The game is conducted via 12-hour TURNS divided as Day and Night. Depending on the scenario nighttime may affect some capabilities (*i.e., American WWII scout planes cannot do Nighttime reconnaissance*).

Leyte Gulf Game Map



F/M Game Map



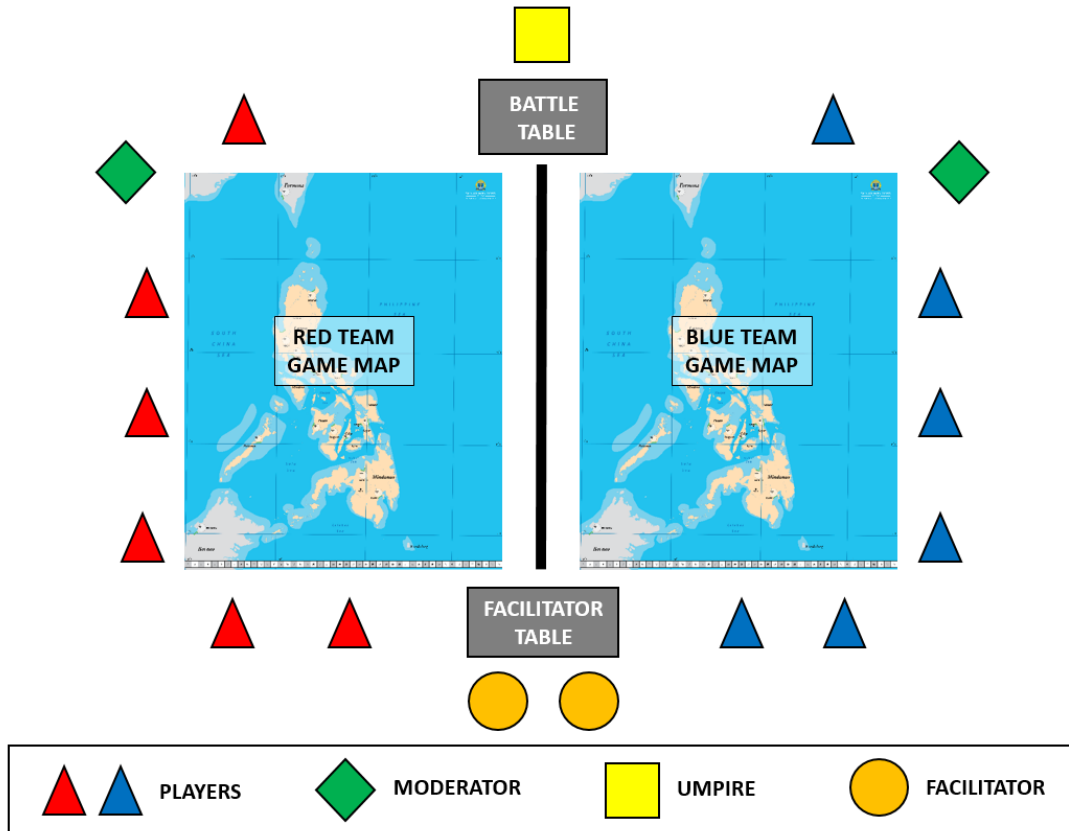
There are two phase during each TURN, Planning and Execution.

- In Planning, Teams will answer three basic questions: 1) Where do I want to move my forces, 2) Where do I want to look for the enemy, and 3) Where/what do I want to attack? Teams will attempt to find solutions to those questions using mainly two things: the Commander's Operational Idea and what they learned from the previous TURNS.
- In Execution, (what this book refers to as the "Operations Phase") Teams will get to see the fruits or pitfalls of their planning. Much like combat at any level of war, good planning, bad planning, and chance will go hand in hand throughout each TURN.

The *War at Sea* instructions outlined in this book are intended to reinforce the JMO curriculum and provide an opportunity to apply the course concepts.

Typical Room Setup

The game takes place in a room divided by a partition denoted by the black vertical line below. Two tables, each with an identical Game Map, are located on either side of the partition. During the game, players remain on their side and do not see the opposing Team's Game Map. They will stand at their respective Game Maps, plan and oversee their forces' actions, receive reports from the Facilitator, and communicate with their Teammates as necessary to plan and conduct operations.



Battle Box. Set at one end of the partition with a pair of open boxes placed on it, is the Battle Box. When combat occurs between opposing forces, each Team's respective Tactical Commander, and only the Tactical Commander controlling the units engaged in that particular combat event, will report to the Battle Box and Umpire. Combat in *War at Sea* is modeled by rolling specific dice capabilities for Damage Points (DP) against the other. Battle happens in two Battle Rounds (BR) – each being a throw of the dice against your opponent.

Facilitator Table. Set at the opposite end of the partition and never seen by players. Here, the Facilitators (NWC Staff) roll their own set of dice that are not tied to a capability, only chance. They consult tables, and compare results as necessary to run the game. The Facilitators will spend time between the Facilitator Table and the two Game Maps to ensure the free flow of information between all parties.

Roles

Players are divided into two Teams, labelled generically as the Red Team and Blue Team. Each Team has a “Team Captain,” referred to as the **Operational Commander**. Other players act as **Tactical Commanders** and fulfill other roles. These, and the roles of **Moderator**, **Umpire**, and **Facilitator**, are described below.

Operational Commander. Responsible for developing the Operational Idea (Op Idea) for the Team, the Operational Commander organizes their players, maintains their Team’s “big picture” during the game, and ensures that their players are operating their forces in accordance with their Op Idea.

Tactical Commander. The single, specific player on a Team who is responsible to make decisions for a given military force or forces in the game. The Tactical Commander moves forces around the Game Map, decides when they engage with enemy forces, and, during combat, fights their forces at the Battle Box.

Moderator. The Seminar Moderator oversees Team actions and decision making, and interacts with players to ensure that learning outcomes are achieved. The Moderator is mainly an observer but will interact with students as necessary to ensure an effective educational experience. The Moderator is concerned with questions such as:

- Is the Team sticking with its Op Idea?
- Did the Team’s actions properly take into account the factors of time, space, and force?
- Did the Team appropriately employ the joint functions to accomplish its objective?
- Did the Team properly sequence its operations?
- Did the Team properly analyze its and the enemy’s Centers of Gravity and act accordingly?

Moderators consult with Umpires/Facilitators as necessary to ensure the game runs smoothly. Following the game, Moderators guide the Debrief.

Umpire. The Umpire oversees the Battle Box (discussed below) and ensures that the game’s combat mechanics are properly conducted. The Umpire is an expert in game combat and will consult with Moderators and Facilitators to ensure success of the game.

Facilitator. The Facilitator “runs” the game and is primarily concerned with game mechanics. They are an impartial arbiter of the game “reality” and represent its “laws of nature.” It is through the Facilitator that Teams interact with the game world and with their opponent. Players will inform the Facilitator of their intentions, and the Facilitator will report the outcomes of those actions back to the players.

Game Components

War at Sea uses various components to support play, including:

- Dice of different colors and sizes that represent different operational capabilities. Dice may be held in Cups or placed directly on the Game Map.
- Cups and Tiles associated with each Team.
- Planning Cards. Pieces of clear plexiglass upon which Teams draw their Movement, ISR, and Fires plans.
- Planning Maps. A smaller copy of the Game Map on which to plan Team actions.
- Thin markers for use on Planning Cards.
- Thick markers for use on the large Game Map (when covered by a clear sheet).
- Stickies to annotate intelligence or other information on the Game Map.
- Rulers to measure distances on the Game Map.
- Brass Rings to assign Search Areas on the Game Map.



—Game Scenarios—

War at Sea requires the use of *scenarios*, maritime warfare situations in which the two Teams challenge each other to achieve opposing **operational objectives**. Two scenarios are based upon historical case studies, the US Philippines Campaign in World War II, and the **1982 Falklands/Malvinas War**. It also uses the fictional situation of the **Borneo Conflict**. In it, the situation itself is fictional and set in the near future, but all forces are based on real-world military capabilities. All scenarios involve a case study in some form.

Case Studies

It is common educational practice to read about the decisions of the military commanders in historical case studies, this is no different. The purpose of the case study is to provide an overview of the strategic background and “set the stage” for the scenario at all three levels of war. This ensures that the larger educational outcomes are achieved while also (typically) influencing player decisions in the game.

Objectives and Forces (OAF) Document (Team Specific Information)

The **OAF Document** exists because operational security is always an important part of any contest and there are some things you want to keep from the enemy. Moderators decide who will be on, and in charge of, Team Red and Team Blue. They will pass on their respective **OAF Doc** to those leaders to distribute to their Team. It consists primarily of five key pieces of information:

- **First, Objectives and Intel:** This lays out your operational-level OBs and any available intel. This part is written as if it is an actual report and is differentiated by its cut lines and typewriter font.
- **Second, Composition:** This is their Order of Battle (OOB). It will tell the Team what forces/units they have at their disposal as well as what capabilities they possess.
- **Third, Disposition:** This describes the forces’ locations at the start of the game. A Map is provided in the **OAF Doc** that indicates where each can go with a given shaded area or line.

- **Fourth, Logistics:** This describes how much fuel and supplies your Team will from the beginning and throughout. This does not become a factor until the F/M game and the cards you track logistics on come with the F/M OAF so you can start planning upon receipt.
- **Lastly,** any major information or asset that was known only to one side at the start of the scenario.

One notable category of information that does not fall in the **OAF Doc**, but could be considered an operational security risk (and which you find in **Appendix A** of this book), is the capabilities and limitations of each of the dice, for each game. While it is true that there are always specific aspects of an enemy's force capabilities an opponent may not know, it is an aspect of war's nature and one that leans more towards a tactical-level problem than an operational one. Since this game is meant to lean more towards the operational level, it is acceptable. This Sun Tzuian problem is made fair by **both sides** knowing much of both their own and their enemy's capabilities.

–Student Preparation–

Readings

To play, students should be familiar with this book and their specific Team **OAF Doc** described above. Knowledge of the case study will also be helpful for players to understand mistakes already made - and hopefully to avoid repeating them. Finally, there are videos that cover the more salient points of game play, which many students will find helpful. A mixture of reading and watching will generate the best understanding.

Team Formation

Moderators will assign students to each Team and select each Operational Commander. The Operational Commander will determine the roles of their Teammates. Typical player roles include:

- **Tactical Commanders.** Players are typically assigned like forces to command. For instance, one Tactical Commander may own all Third Fleet forces while another owns all Seventh Fleet forces. One may own all land-based air forces, while another owns all ground forces, and a third owns all submarine forces. This will be driven by the Team Order of Battle (OOB) and student leadership.
- **Chief of Staff / Deputy Commander.** A player to assist the Operational Commander as they see fit.
- **Intelligence Officer (J2).** The J2 could be responsible for tracking all intelligence data and coordinating all intelligence, surveillance, and reconnaissance (ISR) collection efforts as well as providing possible enemy courses of action per TURN. The J2 will also typically draw the ISR card.
- **Logistician (J4).** One player may be assigned the role of J4, responsible for tracking fuel and special munitions expenditures.
- **Cyber / SOF.** One or two players could be assigned a role in which they oversee the Cyber Warfare and SOF Cards. These become a more critical factor in the Borneo game and F/M provides their introduction.

Planning

Once the Team is formed and all players have read their Team Scenario Guide, the Team, under the leadership of its Operational Commander, must plan its operations.

This planning process, is as detailed as the Team wants to make it and should be based upon current student understanding of the elements of Operational Art as they relate to the given scenario. Remember, start with the Objective.

Planning Products

The final product of the Team's planning is an Operational Idea, derived from the Commander's Estimate of the Situation (CES). It should include a sequence of major events and actions of the principal subordinate forces, and it should be detailed enough to allow subordinate commanders to decide the operational scheme of their respective forces. The products to support implementing the team's Operational Idea in the War at Sea game may include:

- **Narrative**. Stating the *Commander's Intent* and describing the *Commander's Operational Idea* (including a summary of both friendly and enemy Center of Gravity Analysis & Critical Factor Analysis as they relate to the Team's objective(s).)
- **Diagram**. A Map showing the major movements and relative positioning of all forces.
- **Task and Purpose Listing**. Tasks assigned to specific forces and the purpose for each. This serves as overall written guidance to the Tactical Commanders during the operation.

Day-Prior Preparations.

The day before game play, the seminar will arrange the classroom (or game room). Each Team will be given the Cups, tiles, and dice that make up its military forces to arrange on the Game Map. The Team's OOB found in its **OAF Doc** describes the starting location of a given force. Once the Game Map is prepared, Team members from opposing sides are not allowed to see each other's Map.

Each Team will also conduct preliminary planning for the first TURN of the game. The seminar is now ready to commence the game at the start of the next class day. Different seminars will have different ways of doing this, so follow your Moderators' guidance over this section.

–General Game Play–

Overview

As previously stated, the length of a TURN (in game time) is 12 hours. In a TURN, each Team will do the following:

- **Plan / Prepare.** The Team will plan and prepare its movement and intelligence gathering activities for that TURN and report those plans to a Facilitator. Teams get 12 minutes per TURN for this, [and 15 minutes in Borneo](#).
- **Roll for Initiative.** This determines the order in which Teams conduct these actions in a given TURN.
- **Move.** The Team, following guidance from the Facilitator, will move its forces around the Game Map.
- **Look.** Receive information concerning the environment and enemy forces from ISR efforts.
- **Shoot.** The Team will engage in combat as appropriate.
- **Close.** The Team will finish the TURN and prepare for the next.

Detailed Sequence of Play.

Plan / Prepare	PLANNING PHASE - 17
	Set Background Traffic - 17
	Movement and Maneuver Planning - 17
	Logistics Planning - 22
	ISR Planning - 24
	Fires Planning - 27
	Facilities Planning - 29
	Cyber Planning - 30
	SOF Planning - 34
	Set Deception / Denial Cups - 36
Move	Designate Air Cargo Transfer - 36
	Aggregate/Disaggregate Forces - 37
	Set EMCON Status - 37
	ROLL FOR Initiative - 38
	OPERATIONS PHASE - 39
	Cyber Effects Step - 39
	SOF Mission Step - 39
	Movement and Maneuver Step - 40
	Move Forces (Conduct Organic ISR)
	Conduct Trip-Over Fires
Look	Conduct VBSS
	Conduct At-Sea Resupply
	Account for Logistics
	ISR Report Step - 41
	Space Based ISR
	Airborne ISR
	SOF ISR
	Fires Step - 41
	(Finalize Plan and) Conduct Deliberate Fires
	Account for Logistics (again)
Shoot	Activate Minefields
	TURN CLOSE-OUT PHASE - 58
	Recover ISR Assets
	Facility Repair Phase
	Place Newly Arriving Forces
	Advance TURN Marker
Close	



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You may have noticed that the instructions to the left have you doing the Move Step before the ISR Step, and this is clearly against the ideal situation of looking before you move. Fear not. The ISR step is intended to prepare you for your Move in the next TURN and the Fires Step of the current TURN.

Chance and Dice

As in the real world, chance is an aspect of the nature of war, whether as the “fog of war” or some other circumstance. Chance in the game is modeled through the use of six-sided dice. Dice are used in two ways: 1) to emulate chance via probability-adjusted outcomes for most game events, and 2) to represent operational capabilities by quantities of specifically-colored dice.



Different color and sized dice model the capabilities deemed most relevant to each scenario as the games increase in complexity. Leyte uses 10, **F/M uses 13**, while **Borneo has 21 different dice modeling capabilities**. Each color and size (dice are either **large** or **small**) represents a different operational capability and each is described, by game, thoroughly in Appendix A.

Dice capabilities can be separated into the two categories of **reusable** and **expendable**. Reusable dice, like the capability of naval guns to fire repeatedly, do not run out. Expendable dice (like those representing Kamikazes) once thrown, do not come back for reuse. **This lesson becomes more acute in F/M when dice representing torpedoes change from being reusable to expendable.**

Due to the character of warfare in World War II, aircraft capabilities in the Leyte scenario are lumped into three basic dice colors despite the fact that there were numerous classes of fighter/attack-, bomber-, and reconnaissance aircraft-. At the operational level of war in the 1940s, these three basic categories of air capabilities are what matters. **Due to the growing sophistication and specialization of aircraft during the Cold War and beyond, the F/M and Borneo scenarios capture aircraft capabilities with greater specificity, but remain essentially categorical.**

Cups and Tiles

A given number of dice represents a specific amount of military **capability**. A given naval platform (ship, aircraft, submarine) or ground unit holds those specific capabilities (colors of dice) of various types. A collection of naval platforms grouped together is called a “force” or “cup.” A force is typically represented by a plastic Cup that contains some number of dice. Dice do not necessarily represent individual platforms but the aggregated total of military capability contributed *by all platforms* that make up that force.

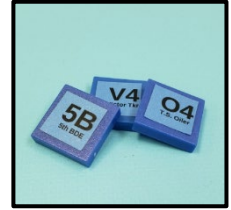
Leyte Example: A group of ships (a maritime “force”), made up of Battleships, Cruisers, and Destroyers, is represented by a plastic Cup holding some number of , , , , and . Their strength in combat is determined by the number and colors of dice in the Cup. A large number of  indicates a strong ability to shoot at other ships and the land. The number of  indicates the ability to shoot down aircraft attacking that force; a small number of  indicates a lesser ability to attack submarines. The  represents the ability to send out a portion of the force on its own, and the  model the ability to conduct ISR from the air.

During combat, as the force is damaged, it loses dice, representing a loss in capability. When all dice are eliminated, that force is no longer combat-capable. It can, however, move and detect enemies as just an empty Cup. When the Cup is eliminated, the force is completely destroyed.



In addition to Cups, Tiles represent forces in two different cases:

- A ground force can be represented by a Tile since Tiles use up less table space than an empty Cup and are easier to manipulate during game play. The dice associated with that Tile are maintained in a Cup off the Game Map and enter play: 1) at the Battle Box, or 2) on the Game Map if that force splits into its component dice.
- **An air or maritime force that has no inherent combat capability (i.e., no dice) or cannot carry dice (transport shipping) is represented on the Game Map by its Tile.**



Game Mechanics

The remainder of this document defines the game mechanics across the three scenarios that build upon one another. To reiterate, the basic instructions (Leyte) serve as the base for the other two (F/M and Borneo) through an increasingly sophisticated approach to wargaming.

- The **Basic instructions** support the Leyte Gulf scenario and are foundational to the later games. These mechanics emulate the following joint functions: Command and Control, Intelligence, Fires, Movement and Maneuver, and Protection.
- **The Intermediate instructions support the F/M scenario. In addition to the joint functions listed above, these mechanics model Sustainment and Information joint functions, the use of missiles, and the importance of emissions control (EMCON) in naval warfare. All things that are specific to F/M and beyond are in blue font, throughout.**
- The **Advanced instructions** support the Borneo scenario set in the near future. These additions more thoroughly model the Information function as well as modern military capabilities the U.S. and its allies are likely to possess and encounter.

–Phases of Play–

This chapter describes the phases that get a Team ready for the upcoming TURN, how to fight through it, and what to do at the conclusion of each one. **While this may seem like a lot to accomplish, the expectation is that the planning phase for each TURN lasts no longer than 12 minutes (15 minutes).**

PLANNING PHASE

Planning is the means by which a Team determines and then informs the Facilitator what actions it intends its forces to undertake during the current TURN. What is most important is that the Facilitator understands the **intent** of what the Team has attempted to convey to them. **A completed plan takes the form of Move and ISR Cards (and SOF & Cyber Cards) submitted to the Facilitator in a timely manner.** The Team also plans Deliberate Fires and creates a Fires Card that *may be* submitted to the Facilitator (if necessary) at this time, but does not have to be until the next phase, after Move and ISR are complete. The Move, ISR, and Fires Cards are pieces of plexiglass collectively known as *Planning Cards*.

PLANNING PHASE

Set Background Traffic - 17

Movement and Maneuver Planning

Of Maritime Forces - 17

Move Card - 17

Submarine Planning - 18

Mine Warfare Planning - 18

Of Air Forces - 18

Move Card - 19

Air-to-Air Refuel Planning - 19

Aircraft Dice Pairing - 20 (Also found in Appendix A)

Set Combat Air Patrols (CAP) - 20

Of Ground Forces - 20

Fortifications - 21

Of Logistics - 22

Fuel Costs - 22

Ground Resupply - 22

Rearming - 23

Log Tables - 24

ISR Planning

ISR Card - 24

Airborne ISR - 25

Armed Reconnaissance - 25

Space-Based ISR – MWAS - 26

AWACS - 27

Fires Planning

Fires Card - 27

Missiles - 28

Facilities Planning

Airfield Max on Ground (MOG) - 29

Repair of Facilities - 30

Cyber Planning

Offensive Cyber Planning - 31

Defensive Cyber Planning - 31

Cyber Cards - 33

SOF Planning

Special Ops Mission Menu - 34

Special Ops Cards - 35

Set Deception / Denial Cups - 36

Designate Air Cargo Transfer - 36

Escort Aircraft - 36

Aggregate/Disaggregate Forces

Aggregating Forces - 37

Cup Stacking Guidelines - 37

Disaggregating Forces - 37

Set EMCON Status - 38



If you are using this book in its PDF form, everywhere you see a Page Number, Rule Number, or Appendix Letter, you will find that the Number or Letter, itself, is hyperlinked to the Page it references. You can use Alt+← as your back button.

Purpose of Planning cards: The Facilitators simultaneously see both Game Play Maps through the use of Move and ISR plexiglass “cards” when overlaid (as seen on the right →). These must be drawn as accurately as possible since they will decide (not the large Game Maps) what forces see or miss one another.

In the hands of the students, the Planning Card glasses are laid on a small version of the Game Map and both are clipped together on a clipboard. Each Team marks its plans on the plexiglass using thin markers of a Team specific color (Blue for Team Blue, Black for Team Red). Teams draw the same movement patterns on the acetate covering their large Game Map as they did on their cards but with the thicker markers. The Planning Phase is a timed event that lasts **12 minutes** (15 in Borneo) in real time. It ends once both the Move and ISR Cards are submitted to the Facilitators from each Team.



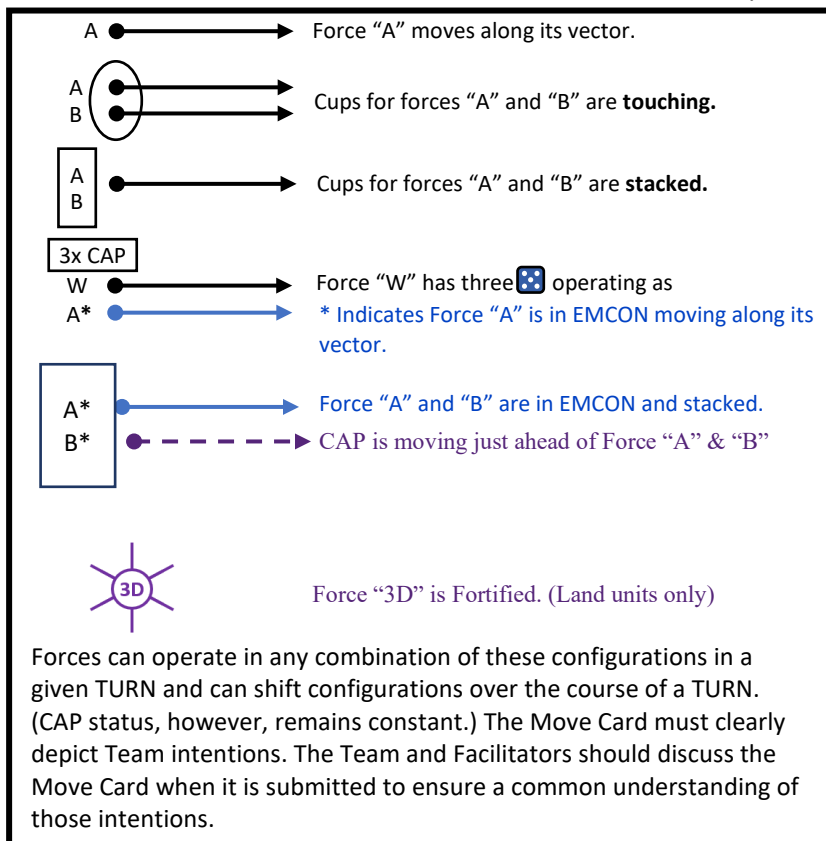
Set Background Traffic on Game Map (Facilitator Action).

This is the first action of the Planning Phase and the Facilitator will position background (“White”) traffic contacts in and around the Red Team Map. This is done prior to the Red Team turning in its Move Card.

Movement and Maneuver Planning of Maritime Forces

Move Card.

Each Team plots its scheme of Movement and Maneuver. Teams must use consistent notation on their Move Card to ensure that Facilitators understand their intentions. Teams will need to know how far and fast each of its forces can travel. The speed of each force (Cup) is listed on each Team’s **OAF Doc** but they generally follow a specific pattern. The speeds at which the different



Maritime forces are assigned a speed of FAST, SLOW, or CREEP:

*Littorals = Light Blue Water on Map

Surface Forces:

FAST = 25 kts: up to 15” per TURN*

*Red Surface ships in littorals = 12”

*Blue Surface ships in littorals = 9”

*Red & Blue Surface in littorals = 9”

SLOW = 13 kts: up to 8” per TURN

CREEP = 5 kts: up to 3” per TURN

(acoustic penalties for traveling >8” per turn)

Conventional Submarines:

Surfaced = SLOW 8”

Submerged = CREEP 3”

*Leyte – Surfaced in Littorals

*FI/MV – Periscope Depth in Littorals, CREEP

Nuclear Submarines:

Surfaced = SLOW 8”

Submerged = FAST 15”

(acoustic penalties for traveling >8” per turn)

*Periscope Depth in Littorals, SLOW

types of forces move is quite uniform with few

exceptions, the vast majority of surface forces can move FAST; the exceptions being transports, support, and amphibians that move SLOW. Once you know a force's maximum speed, you can know how far it can move per TURN. The boxes on the previous page provides those distances and gives examples of the standard notations for drawing the movement of forces on cards.

Rule 1: During Planning, forces are NOT yet moved on the Game Map, only their planned path is drawn on both the cards and the clear sheet covering the Game Map.

Rule 2: The EXACT path of each force is annotated on the Move Card, taking into account appropriate distances in accordance with their designated speeds.

Rule 3: Forces travel at a constant speed for the duration of the TURN. Touching shallow water brings that surface force down to its shallow water speed for the entire turn.

Rule 4: Forces are labelled on the Move Card by their Card Code found on the Cup lid.

Rule 5: The positions of stationary Maritime and Ground forces must also be annotated on the Move Card.

Rule 6: A Newly Arriving Force's movement onto the Game Map commences from the point where it touches the actual playing surface.

Rule 7: Maritime forces (surface forces and submarines) moving >8" per turn (high speed) incur acoustic penalties. Their own acoustic sensors are degraded, and they are more easily detected by enemy forces.

Submarine Planning.

Submarines are great as both hunters and killers in this game. Diesel subs move faster on the surface, but they are also more detectable and vulnerable. Diesel Subs are assumed to be submerged (and managing their own battery recharging) unless they are moving greater than 3" in a turn which required them to surface.

Nuclear powered subs are faster when submerged which provides greater flexibility for submarine missions without the increased risk of surfacing.

Rule 8: Submarines do not pay Fuel Points (FP) for movement.

Mine Warfare Planning.

Mine-capable forces are annotated in the Team OAF Doc. They are capable of detecting, clearing, and/or laying minefields. More detail on this is, by necessity, found within individual Team OAF Docs.

Rule 9: Maritime surface forces capable of laying mines can lay a 1" circle minefield in one TURN but must be in the position where they want the mines placed at the start of that TURN. The minefield will be emplaced and active at the conclusion of

the Fires Phase. The minefield will be marked with a 1" circle marker on that Team's Game Map.

Rule 10: The side that emplaced the minefield can move its forces through it unharmed since it would know the clear transit path (Q route) through the mines.

Rule 11: Mines can only be placed in littoral waters (light blue on game map).

Sea mines can be delivered by more than just surface forces, they can now be emplaced by submarines and some aircraft as well. Instructions within your OAF Doc will inform you on the specifics for your Team.



Helpful Hint: Following a zig-zag route will make your force's trip take longer but it can also add some deception to your plan. If your force travels through an enemy ISR search area, the opposing Team will have a more difficult time guessing where your route ended, and thus, where they can attack it.

Movement Planning of Aircraft

Move Card.

Moving aircraft around the Game Map and Move Card is similar to what is done with maritime forces, with some key caveats.

- There are no speeds involved, only distances.
- Those distances are separated into three different categories:
 - Range
 - Radius (half the distance of Total Distance)
 - Reconnaissance Distance.

If you know the total distance of any airframe dice capability, you *can figure out* the other two with simple math. Appendix A, your **OAF Doc**, and posters make this easier by providing each Team member all they need to know about aircraft movements at a quick glance. This also saves them from having to do “simple” math in front of peers and Moderators.

- **Range** is useful to know when the plan is to move an aircraft dice between carriers and/or airfields, or land at a different airfield after a mission. These distances strike a balance between historical accuracy, ease of game facilitation, and simplicity of game play. It represents how far an aircraft can go before running out of gas. **With the introduction of air-to-air fueling capability. An aircraft's range is reset from the last point of refueling (typically the max range of the tanker from its own base).**
- **Radius** is half the range to account for going somewhere and coming back on the same tank of gas.
- **Reconnaissance Distance** is dependent on the size of the area to search modeled by the size of the Search Ring used. The **6" diameter ring costs 10" of Range** while the **9" diameter ring costs 30" of Range**. This book goes into greater depth on how to do this in the section, ISR Planning, starting on Page 24.

Planning for Air-to-Air Refuel of Aircraft.

The use of refueling aircraft became widespread during the course of the Cold War. Red or Blue Tiles model this capability.

Rule 12: Long-range, land-based tanker aircraft have a range of 1400nm (70"), which equates to an operational radius of 700nm (35").

Rule 13: Aircraft supported by Tankers accompany those tankers up the Tankers' total operational radius (700nm or 35") and can fly up to the additional operational range of that aircraft beyond the final position of the tanker.

Rule 14: Aircraft that do not accompany tankers, but instead rendezvous with them, can fly up to their operational radius en route to the rendezvous and then their range is reset from that linkup point.

Rule 15: Regardless of the refueling tanker's Tile color, they will be treated as ☒ on the Damage Table when adjudicating what constitutes a "hit" against them.

Rule 16: If the tanker is shot down and the supported aircraft lacks the range to return to a friendly airfield, the supported aircraft will be lost as well.

*Rule 17: **A single Air Refueling Tile can support one air-refuellable dice.*

*Rule 18: **A single Tile can support up to 3x air-refuellable dice.*

Rule 19: When "Buddy Tanking," a single ☒ with this capability can refuel 2x ☒ but has no offensive combat capability.

Aircraft Dice Pairing.

Aircraft that carry a particular capability with them are said to have a “pairing.” The chart below breaks down possibilities for Falkland’s. Pairings for the Borneo scenario are too complex for this table and listed individually by dice color and Team (as are the F/M ones) in Appendix A.

Aircraft	Munition	Details
Lg Super Étendards	1x Exocet Missile	This pairing is only for Argentina and the rules for it are spelled out in chart form in more detail than room will allow here on Page 52.
Lg	1x ASW	If it detects a submarine, it can attack with a roll of the at the Battle Box for 2x BRs. If the Lg is intercepted by enemy aircraft, only 1x Damage Point is required to kill the pairing.
	1x or or or	Helicopters can move only 1x ground die or 1x SOF Chip at a time. A Damage Point over water will kill the pairing. If brought down over land, the ground dice or SOF team survives at the point of the kill.

BR = Battle Rounds

Rule 20: When a “paired airframe-and-capability” is targeted in combat, the airframe’s dice color determines the effects of enemy attacks against it.

Rule 21: The carried-capability die is automatically terminated with the loss of the airframe. This rule does not apply to helicopters carrying ground forces.

Set Combat Air Patrols (CAP)

Rules for CAP vary widely across the three scenarios. For that reason, each is listed separately below by its Game color font.

Set Combat Air Patrols

CAP is airborne fighter aircraft that can protect a surface force or base. To model this on the Map, place the desired number of dice on top of Cup (left), or on top of a yellow Tile on airfields (right). CAP can only be flown during day TURNS in the Leyte game. Keep in mind during planning that any established as CAP cannot then be used in a Strike Package later in that TURN. (See Rule 83:)



Rule 22: A CAP station covers a 3” diameter area around the center point of the force or airfield it is protecting.

Rule 23: **CAP can be established over a Force or Airfield only. (Superseded in the next game by Rule 28:)

Rule 24: CAP can be established over Force or Airfield other than its point of origin, but origin and CAP location must be within 10” of each other. (Half of its 20” combat radius)

Rule 25: **CAP can fly only during Day TURNS regardless if the are carrier- or land-based. (Superseded in the next game by Rule 30:)

Rule 26: If an aircraft is on CAP and the center point of its CAP station is within 10” of the current battle, it can enter combat during BR#1. (See Page 45 for more on CAP during combat)

Rule 27: in a CAP station can conduct 2x Battle Rounds worth of fighting per TURN.

Set Combat Air Patrols

Due to advances in air combat, CAP stations are no longer restricted to protecting only a Force or Airfield, they can now fly over any position Night or Day.

Rule 28: CAP can be established over any geographic position over land or sea, but still within 10" of its origin.

Rule 29: CAP stations operate Emitting, however, the force that launched the CAP station can remain in EMCON.

Rule 30: CAP can now fly Day or Night.

Rule 31: *CAP cannot engage  that transit its station. (Superseded in the next game by Rule 32:)

Set Combat Air Patrols

CAP stations can still be set at any location (land or sea) and must be annotated on the Team's Move Card. The station will have a 6" diameter of organic ISR (6" circle) due to increased detection capabilities. CAP can commit out to a total of 6" from their CAP point in order to intercept an incoming strike package. CAP stations can be set adjacent to a force they are protecting and move in parallel along with it. This is annotated on the Move Card with a dashed line.

Rule 32: Within Strike Packages that transit a CAP station, CAP can now also engage ,  Lg, and  missiles.

Rule 33: CAP attacking a strike package greater than 5" from its intended target are allowed 1xBR per TURN; CAP choosing to remain within 5" of the intended target are allowed 2x BR per TURN.

Rule 34: If required to refuel,  and  Lg must refuel within 10" of their intended CAP station. If the supporting tanker is lost and there is no other available place to land, then the CAP aircraft is lost as well.

AAR = Air-to-air refuel.

Movement Planning of Ground Forces

Rule 35: All Ground Forces travel over land at 2" per TURN. Ground forces within 1" of an opposing force and engaged in Combat cannot move. (This changes with the two rules below)

Rule 36: Infantry and Air Defense Ground Forces travel over land at 1" per TURN but Field Artillery still moves at 2" per TURN.

Rule 37: All Ground Forces travel over land at 2" per TURN when traveling along the Trans-Borneo Highway, and 1" per TURN when they deviate off of it (unless otherwise modified by the OAF).

Fortifications.

Historically, ground units that have fortified their positions have been much harder to destroy and/or uproot. Advanced conventional munitions make lethal attacks against ground units possible but well camouflaged and dug in units in rough terrain will still resist rapid destruction. Teams may denote that their ground units are "Fortified" during the Planning Phase.

Units denoted as "Fortified" on the movement card may not move during that TURN. Only on ground capabilities (, , ) can be "Fortified." It allows them to limit the hits attained against them through Rule 39:.. Owning players may attempt a saving dice roll for each capability dice destroyed. Once a fortified position is attacked for 2 battle rounds, it is assumed that the fortifications are destroyed at that position. Forces may not "Fortify" in the same position in subsequent turns, but surviving ground forces can move to other positions to reinforce fortified forces already established there. Surviving forces may also move to another location and "Fortify" there, but they cannot move and "Fortify" a new position on the same turn.

Rule 38: Fortified units marked as such on the Move Card cannot move for that TURN but they do retain their Organic ISR, the ability to defend themselves, and launch missiles.

Rule 39: For each Damage Point earned by an attacker against a fortified ground unit, the ground dice gets a "saving roll." Rolls of 5-6 result in a nullification of that Damage Point.

Rule 40: Saving Rolls are only available for 2 battle rounds per fortified position.

Logistics Planning

Logistics is one of the most important functions in warfare on top of being one of its hardest to get right. For that reason, it is omitted from the most basic of the of the triad of wargames (Leyte) to allow players to learn the game without the additional significant undertaking that is logistics. **Logistics considerations are included in F/M and Borneo. In addition to colored dice to represent operational capability, Teams will track fuel (via Fuel Points, FP (🎲)) and consumable supplies (via Logistics Points, LP (🎲)).**

The success of military operations can often hinge on properly tracking and allocating the right supplies, in the right amounts at the right times. Logistics is incorporated into *War at Sea* in three ways:

- **Fuel Points (🎲)** account for fuel use by modern warships to conduct operations. (*War at Sea* does not track fuel for most small vessels, aircraft, or subs.
- **Logistics Points (🎲)** account for general consumable supplies, such as food, repair parts, etc. They can be attacked and reduced by your opponent.
- The use of certain expendable munitions, (i.e., missiles, torpedoes, etc. represented by dice of various colors, are removed from game play once employed).

As forces maneuver, they expend 🎲. This is tracked during the Movement and Maneuver Step of the Operations Phase. Additionally, as forces engage in combat, they expend 🎲. This is tracked during the Fires Step of the Operations Phase.

🎲 can be carried by cargo vessels, and ground forces consume 🎲 on a By-TURN basis; tracking this is found in your OAF Doc with the fuel tracking charts.

Logistics Movement Planning. The two tables within this section are simplified looks at the details and rules throughout the Logistics Portion. The OAF Docs provide a much more thorough look at how to run logistics.

Fuel Costs Associated with Moving and Combat.

	Moving > 8"	Moving 8" – 0"	Taking Damage	Engaging in Combat	Zero 🎲 Cannot...	Zero 🎲 Can still...
Force Cup*	2x 🎲	1x 🎲	1x 🎲	1x 🎲	Move, Offensive actions	Organic ISR, Defend self
Oiler	-	1x 🎲	Up to 3x 🎲 per DP	1x 🎲	Move	Organic ISR
Shuttle Tanker	-	1x 🎲	Up to 3x 🎲 per DP	1x 🎲	Move	Organic ISR
Cargo Ship	-	-	1x 🎲 per DP	-	-	-

DP= Damage Point

A Force Cup can only expend up to 3x 🎲 per TURN (Rule 54:). This does not apply to Oilers or Shuttle Tankers. Also, not all Force Cups are required to track fuel.

Maritime Refueling.

The game consists of maritime Oilers and Tankers, typically paired with combat forces at the start. There is no requirement for that pairing to be permanent. Tankers supply only Oilers and Oilers supply all other ships.

Rule 41: Forces with zero  cannot move during that TURN and cannot initiate combat. Forces with zero  can conduct Organic ISR and can defend themselves in combat.

Rule 42: In order to transfer fuel, forces must be stacked. During replenishment, stacked forces are allowed to move together at up to SLOW speed.

Rule 43: Cups touching those refueling are not taking on fuel themselves.

Rule 44: Forces with zero  but being resupplied may move with the resupply force at up to SLOW.

Rule 45: Forces that are given  during At-Sea Resupply at the end of Movement and Maneuver may conduct Organic ISR and may attack and defend in combat during that TURN.

Rule 46: An Oiler can transfer fuel to up to 3 combatant forces in a TURN.

Rule 47: An Oiler can transfer fuel to other Oilers.

Rule 48: A Shuttle Tanker can only transfer fuel to Oilers (but not combatants) and can transfer fuel to up to 3 Oilers in a TURN.

Rule 49: There is no limit to the number of  that can be transferred in a TURN.

Rule 50: Forces that do not move or that move up to 8" (SLOW) expend 1x  for the TURN.

Rule 51: A FAST force that moves greater than 8" expends 2x  for the TURN.

Rule 52: Only expended munitions can be rearmed, not munitions lost to damage.

Simplified Rules Governing Refueling and Rearming:

Vessels Receiving Fuel and/or Arms

		Receiving →					
Providing ↙		Friendly Oilers	Friendly Shuttle Tankers	UK Surface Forces	UK Subs	ARG Surface Forces	ARG Subs
Providing Fuel and/or Arms	Friendly Surface Forces	Refuel	✗	✗	✗	- **	✗
		Rearm	✗	✗	✗ †*	✗ †*	✗
	Friendly Oilers	Refuel	✓ _{3/TURN}	✗	✓ _{3/TURN}	- **	✓ _{3/TURN}
		Rearm	✗	✗	✗	✗	✗
	Friendly Shuttle Tankers	Refuel	✓ _{3/TURN}	✓	✗	- **	-
		Rearm	✗	✗	✗	-	-
	Argentina Mainland Ports	Refuel	✓	-	✗	- **	✓
		Rearm	✗	-	✗	✗ ***	✓ †

** Submarines do not track Fuel Points.

*** Can only rearm a force's  &  , and only up to its starting numbers annotated on the ARG OAF Doc.

† Can only rearm a submarine's  up to the amount it started with.

†* Cannot pass missiles or fuel between vessels, only ground units can transfer between vessels.

Fuel Point Damage.

Fuel Points (🔥) (carried for distribution by Tankers / Oilers) and Logistics Points (📦)(carried by Cargo or Merchant ships) can be destroyed by fires just as dice can be. For the purposes of the 1-1-1 Rule (Rule 137:), a single Damage Point removes 3x🔥 from an Oiler or Tanker, or 1x📦 from the targeted cargo vessel.

Rule 53: Engaging in combat costs 1x🔥 ; taking damage in combat costs 1x additional 🔥 .
Rule 54: Regardless of all factors that cost 📦 , a maritime force Cup will not lose more than 3x🔥 in one TURN. This does not apply to Oilers and Tankers, which may lose more than 3x🔥 if the situation dictates.
Rule 55: Every Damage Point allocated against an Oiler or Tanker Tile removes 3x🔥 from the targeted vessel and a Survivability Point. This is in addition to the expenditure of 2x🔥 for getting into combat and moving 8" or less.
Rule 56: A single point of damage against a cargo ship or shore installation removes 1x📦 from the target.
Rule 57: For purposes of damage allocation, 📦 count as dice and are therefore subject to the 1-1-1 Rule.

Log Tracking Tables.

Each Team’s OAF Doc has its Log Tables included. The OAF doc provides all the information, including an example for how to use them. This is one of the pieces of intel kept from opponents.

ISR Planning

ISR Card.

Just like the Move Card, the ISR Card is a piece of plexiglass that allows the Facilitators to overlay one side’s ISR card over the other side’s Move Card, giving the Facilitator a view of what each ISR asset sees of the other side’s disposition and movements. **This is the reason that even static forces must be on the Move Card.** The Facilitators roll to determine if an ISR asset actually saw what was in its assigned Search Area / path using the chart below. The facilitator rolls for every ISR die allocated to a Search Area. The more ISR assets allocated to a search area, the greater the likelihood of success. A No Report could be the result of poor weather, losing the ISR asset to operational factors (such as a crash), having the ISR aircraft shot down by CAP, or by simply failing to detect the enemy force.

ISR Detection Table.

ISR & Trip-Overs		Die Roll					
		1	2	3	4	5	6
Outcomes		No Report (Lost/Shot Down*)	No Report	No Report	Report	Report	Report

*Whether an asset is “lost” (due to factors such as equipment failures or weather) without further explanation to the Team, or shot down will be situation dependent and determined by the facilitator

Rule 58: Search Area gold rings with their associated 🎲, 🎲, and/or 🎲 are placed on the Game Map.

Organic ISR.

All forces have inherent ISR capability in a circular area surrounding them. This circle is characterized by its radius from its center and is the area in which a force can detect enemy forces. This is applicable to maritime and air forces.

Rule 59: Forces always conduct organic ISR.

*Rule 60: **Day ISR Range for all maritime surface forces is **3" Radius**.*

*Rule 61: **Night ISR Range for all forces is **1.5" Radius**.*

*Rule 62: Aircraft, submarines, and ground forces possess an ISR Range of **1.5" Radius**, Day or Night.*

****Superseded in next game by EMCON rules on Page 38.**

Airborne ISR.

A specified route allows aircraft to fly a track and conduct ISR while en route. The area searched by that aircraft is 3" wide, centered on the route (1.5" to either side). The route can be any length up to the total allowable range of that aircraft and may include 6" or 9" rings.

Daytime Airborne ISR can be conducted by 🎲, 🎲, or 🎲 dice. For each mission, the Team designates the number of dice. Multiple air dice can be applied to an ISR mission to increase the likelihood of success. Airborne ISR missions are conducted as a Specified Route, a Search Area, or a combination of the two.



Helpful Hint: Typically, an ISR sortie can cover more area if it takes a different path back to its point of origin than it took going out to its search area.

Search Area.

Each Team uses brass rings in either 6" or 9" diameters to convey the area within which they want intelligence gathered, as seen in the photos on the right.

Search Areas cost aircraft range based upon their diameters as

noted in the rules below. An aircraft can have more than one Search Area, but **each ring** is a cost in that total range.

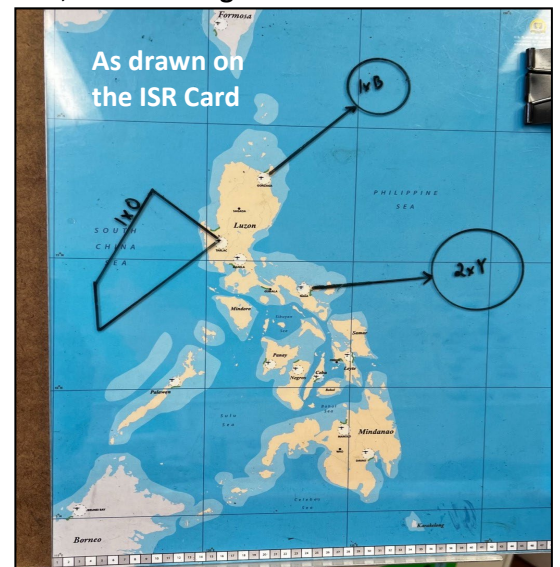
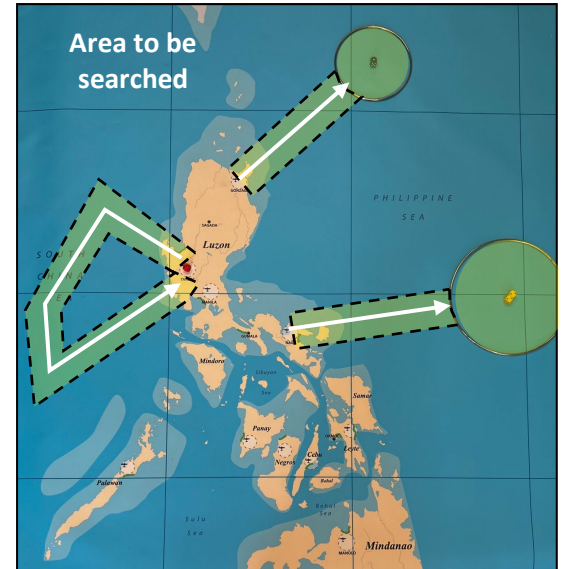
Rule 63: A 6" diameter Search Area costs 10" of total range for that aircraft.

Rule 64: A 9" diameter Search Area costs 30" of total range for that aircraft.

Rule 65: Aircraft conducting Airborne ISR must return to a landing location (airfield or aircraft carrier) but that landing location is NOT required to be the aircraft's point of origin.

Rule 66: Detections are conducted on each leg of the flight path and the Search Area.

Rule 67: Argentinian ISR suffers a -1 penalty for its Facilitator rolls during Night TURNS since its capabilities were greatly reduced by night, when compared to the UK.



Armed Reconnaissance.

This is a special case of Airborne ISR that can only be flown by 🎲 or 🎲 dice. Armed Reconnaissance follows the rules of Airborne ISR. However, if it detects enemy forces

those dice conducting the mission can conduct attacks as a Trip-Over (To be discussed on Page 42). A maritime force attacked by armed recon still completes its movement that TURN after the attack. **When a UK 🇬🇧 Lg is paired with a 🇺🇸, it is said to be sub hunting, which is a form of armed reconnaissance, with the exception that it cannot engage in surface fires or fight back in air-to-air combat. British 🇬🇧 Lg cannot conduct armed reconnaissance.**

Space-based ISR Assets.

Satellite assets may provide additional reporting information to the UK. This asset models the information the British received from American intel assets, which the Argentinians did not. In the Borneo scenario, these assets are available to both sides. They come in three forms:

- **1) Electro-Optical/Infrared (EO/IR) Satellites. (Day Only)** These are reconnaissance satellites used to take pictures of the Earth beneath them, a great asset but their utility was occasionally degraded due to poor weather conditions. These satellites can only detect forces on land but what they do find is accurate enough to be considered targetable.

Rule 68: EO/IR reports provide “targetable” intel; forces so detected can be engaged by Deliberate Fires.

*Rule 69: **During each Day TURN, the Facilitator will roll for each ground force, air force, airport, and seaport. On a roll of 1 or 2, the Facilitator will make no report and state the reason for no report as “Weather Interference.” On a roll of 3, 4, 5, or 6, the Facilitator will provide the Team with a Standard Report.*

*Rule 70: **During each TURN, the Facilitator will make a Standard Report for each enemy ground force and air force on land and each airport and seaport that is not in a Denial / Deception Cup.*

- **2) ELINT/SIGINT Satellites. (Day/Night)** Reconnaissance satellites “listen” to the electromagnetic emissions of military forces. The best defense against them is to eliminate emissions (be in EMCON). The report gives a general area where the detection was made. The info is not accurate enough to be targetable (unless the Team elects to attack in the blind) but may help focus your ISR planning and movements.

Rule 71: The Facilitator will provide the Team with a Standard SIGINT Report for each maritime force at sea that is not in EMCON each TURN, including Surfaced submarines.

Rule 72: ELINT/SIGINT reports do not provide “targetable” intel; forces so detected cannot be accurately engaged by Deliberate Fires without additional location information. (Superseded in next game below)

Rule 73: ELINT/SIGINT reports provide targetable intel. (TACSIT 1)

- **3) Maritime Wide Area Search Satellites.** Advanced imaging satellites that provide intelligence focused on the maritime domain. MWAS search a 9” diameter area and provide targetable data. MWAS can also follow a specific force found in its search area for subsequent TURNS, if tasked to do so. The team must inform the facilitator if they want to follow a specific force during subsequent TURNS, in which case the satellite Field of View will automatically center on that designated force each TURN until retasked. The designated force remains targetable even if it would otherwise have departed the 9” search area. Other forces that are within the 9” circle at the end of the TURN remain targetable. Other Forces that depart the field of view during the TURN are not targetable. Only the designated force remains targetable regardless of where it finishes its movement.

Rule 74: Teams place the 9” Orange ring in any location of their choosing on the Game Map, however, ONLY targets at sea are detectable by this asset.

Rule 75: The Facilitator will make a standard ISR roll against the Detection Table and provide the Team with a Standard Report for each detected force at sea.

Rule 76: The MWAS Satellite can track a high-value unit (once found) if designated to do so by the student Team. The ring will move with the designated enemy through the next TURNS while still detecting other forces based on the ring’s final position.

Airborne Warning and Control System (AWACS).

Many modern militaries possess some level of AWACS capability. These aircraft are modeled through  Lg.

Rule 77: AWACS aircraft operate with a Search Area around them at all times. Within the Search Area, it receives an Auto-Detect on all but  Lg aircraft.

Rule 78:  Lg are NOT subject to an Auto-Detect; the  Lg must make a standard ISR roll against the Detection Table.

Rule 79: Within the  Lg Search Area, the Facilitator will make a standard ISR roll against the Detection Table and provide the Team with a standard report for each surfaced maritime force at sea.

Rule 80:  Lg and their escorts always operate Emitting. If  Lg AWACS is found by enemy fighters while unescorted, they are automatically shot down.

Rule 81:  Lg are vulnerable to Space-based ISR and are always an Auto-Detect to the opposing Team's ELINT.

Fires Planning (Targeting)

Fires planning in the Game typically entails air strikes from carrier- or land-based air **as well as missile strikes** against enemy forces. This occurs during the Deliberate Fires Phase later, (all other combat occurs as a result of Trip-Overs). This section describes the target planning that can be done prior to the current TURN's Moves and ISR reports. While there is not much on which to go in the first couple of TURNS of a game, subsequent TURNS reveal a clearer picture of your opponent's disposition and intentions – making Fires Planning less ambiguous.

Fires Card.

Of the three pieces of plexiglass turned in by each Team to the Facilitators, the Fires Card comes in last and separate to the other two (Move & ISR). Though some of the Fires Card could be done beforehand, Teams typically do not finalize them until after Trip-Over combat is complete and they have learned everything they can from the Move and ISR phases, giving their planned strike(s) the best chance of success. Each Team annotates its fires plan on the Fires Card plexiglass, indicating which air forces **and/or missiles** are conducting fires on which targets. The card must include a dice count of air **and missile** capabilities to be employed and their planned route to the targeted area.

Rule 82: Fires can be combined from multiple forces within range of a target. All dice shall be rolled concurrently unless the attacking force specifically intends to phase its attacks.

Rule 83: Aircraft that were engaged in combat or flying CAP earlier in a TURN or moved from one airfield to another cannot then conduct Deliberate Fires in that same TURN.

Rule 84: A force cannot be allocated to attack multiple target areas, but it can still defend itself en route back following its attack on that single targeted area if necessary.

Rule 85: Forces that Tripped-Over one another earlier in the TURN but did not engage in combat can do so during the Deliberate Fires step.

You can see more on what will happen in the actual execution of your Deliberate Fire Plan on page 50.

An individual aircraft die can only do one task per TURN. Choose wisely.

ISR	CAP	Strike (Only one Strike per die per TURN)	Move
-----	-----	---	------



Fires example: These photos show both the Game Map (left) and the Fires Card (right) for the Blue Team conducting three Deliberate Fires missions at potentially three targets. Following Movement and Maneuver and ISR, the Team members decided to strike the airfields at Tarlac (1), Naga (2), and Leyte (3). They decided to conduct an Attack in the Blind using 3x launched from a carrier in force 'P' against Tarlac (1). Also from P, will be a strike (based on intel) using 3x against Naga (2). 2x from a carrier in force "N" will also strike Naga (2) and the Team informed the Facilitator that they want the total of 5x to arrive at Naga (2) at the same time. Also, from force "N" will be 3x more to attack the



airfield at Leyte (3). They elected to draw a circuitous route avoiding known enemy on the island of Samar. The eleven total dice being used in the Team's deliberate strikes cannot be the same dice that are currently on CAP above 'P' and 'N'. (4x above each.) Dice that are supporting a Combat Air Patrol (CAP) mission are unable to conduct strikes during that TURN.

Missiles.

The first missiles come into play in the second game represented by and . Much is written on these two munitions in both Appendix A of this book and in the OAF Docs for both teams. In Borneo, you will go from two different types of missiles up to seven. Along with them will come increased range and lethality. Though the Game Map is physically bigger, you may find that the missiles and their ranges make it seem much smaller. Below is a quick reference table to help you plan what can and cannot be done with each, but Appendix A provides much more detail:

Missile	Game(s)	Can be intercepted by...	Targets...	Fired in...
	F/M		Maritime Cups*	BR#0
	F/M	No defense against it	Aircraft, Maritime Cups*,	Def-BR#0, Off-BR#1-2
	Borneo	& CAP	Maritime Cups*	BR#0
	Borneo	No defense against it	Aircraft, Maritime Cups*, , Lg,	Def-BR#0, Off-BR#1-2
Lg	Borneo	& CAP	Maritime Cups*	BR#0
	Borneo	& CAP	Facilities, Ground Units	BR#0
Lg	Borneo		Maritime Cups*	BR#0
	Borneo	No defense against it	Aircraft, Maritime Cups*, , Lg, , Lg, Lg	Def-BR#0, Off-BR#1-2
Lg	Borneo		Facilities, Ground Units	BR#1-2
	Borneo	No defense against it	Aircraft, , Lg, , Lg, Lg	BR#1-2

*Refers to Maritime Cups that are on the surface, including surfaced subs.

- **Anti-Ship Ballistic Missiles - Lg (ASBM).**

ASBMs are a recent development designed to attack warships in the open ocean. Their long range, ballistic trajectory, high velocity, and powerful warhead make them a lethal threat to any high-value unit on the surface on the ocean. In the game they are modeled through Lg and fired during BR#0. The only thing that can intercept a Lg is a Ballistic Missile Defense (BMD) capability modeled through the . The does not have

to come from the targeted Cup – only one within range of intercepting the incoming Lg. The Lg is not maneuverable, following a ballistic path. Like all other missiles, both Lg and  are expendable.

- **Tomahawk Land Attack Missile - (TLAM) aka Land Attack Cruise Missile (LACM).**

As the “L” suggests in both TLAM and LACM, this is a missile that is used only against targets on land. Any maritime Cup possessing  can only fire them as a Deliberate Fire and only if that Cup did not see combat that TURN.¹ Teams fire LACMs during BR#0 and the missile is maneuverable - meaning you can create waypoints along their 50” of flight path. Firing a TLAM produces a detectable signature. The location of the force firing the TLAM(s) will be reported to the opposing side. However, this location is not targetable until the subsequent TURN.

Rule 86: LACMs are only fired during BR#0 of a strike during the Fires Phase.

Rule 87: A force that came in contact during a TURN prior to its planned LACM strike can no longer execute it that TURN.

Rule 88: A force that fires LACMs during a Fires Phase automatically comes out of EMCON when it does so.

Rule 89: Any enemy missile defense system an LACM flies within range of can engage it.

Rule 90: A force firing a TLAM(s) that was not otherwise detected cannot be engaged during the fires phase, but their location will be reported to the opposing force.

Facilities Planning

Modern precision weapons can significantly damage enemy bases, limiting Facilities’ ability to support operations. Bases can be attacked during the Fires Step by LACMs, SOF, or conventional forces. Bases are split, categorically, between being either airfields or ports. Each one has a designated number of Damage Points it can sustain before it can no longer support operations.

Ports

Both sides have port facilities where they can rearm, refuel, and embark/disembark troops. It is important to note that a ship in port doing any of these three activities is vulnerable to attack and unable to defend itself.

Rule 91: A maritime force in port is in EMCON and cannot defend itself.

Airfield Maximum on Ground (MOG) Capacity.

Like aircraft carriers, many ground Facilities have a maximum number of aircraft they can support. For that reason, a firm line has been put in place at most sites.

Rule 92: Airfields with a MOG value can hold/support only up to that number of aircraft dice/Tiles.

Rule 93: If an airfield is at MOG capacity, no other aircraft are able to land there. Aircraft attempting to land in this situation must divert to another airfield or carrier. If there is no other destination within range of the aircraft, it is lost.

Rule 94: The MOG value only applies to aircraft dice; it does not apply to ground forces, air defense dice, or other missiles.

Rule 95: “1-1 Rule” When allocating Damage Points against a facility, you must take a Survivability Point for every dice until one or the other is depleted.

When a base is attacked during the SOF or Fires Step, the attacking player will inflict Damage Points (DP) on the Facility as well as removing capability dice at the base location. For an airfield, its number of Survivability Points (SP) is on its Airfield Card. The attacking Team must use a 1-1 Rule (not to be confused with the 1-1-1 Rule) when allocating earned DP. For every dice removed, they must also take an SP until one or the other is depleted. After that, any remaining DP will go towards whatever remains, just like the 1-1-1 Rule.

¹ An LACM strike is a complicated, detailed, and rehearsed operation that could easily be disrupted if the striking force was recently engaged in combat. For that reason, a maritime force engaged in combat during a trip-over will have to wait until the next TURN to execute this kind of pre-planned strike.

Hits achieved on a Facility are marked on the owning players Facility display sheet. Players check off the damage boxes by writing in the TURN the Facility was damaged, much like checking off boxes on the  tracking sheets with the TURN number.

The following table defines operational capacity by damage percentage to an airfield or port:

Airfield			Port		
100% - 51%	50% - 1%	0%	100% - 51%	50% - 1%	0%
Fully operational	Can only support a number of aircraft equal to ½ of its MOG.	Cannot support any air operations until repaired.	Fully operational	Can no longer rearm forces at port.	Can no longer rearm or refuel forces. Can no longer load or unload ground forces.

Repair of Facilities.

Facilities may be repaired and restored back to fully operational. Each has a repair capacity that can be used to get back SP lost during the Fires Step. Players repair the Facility by erasing the marked-out damage boxes. When the damage is within the percentage levels noted in the table above, the Facility operates as indicated. Facility SPs are restored during the Facility Repair Phase of a TURN. Those DPs incurred during a TURN cannot be restored until the end of that same TURN, (i.e., *Facility damage incurred during the fires step of TURN 2 would not be repaired until the Facility Repair Phase of TURN 3*).

Rule 96: At the end of each TURN the Facility can remove the number of Damage Points equal to its repair capacity found on its card. Damage Points taken in one TURN cannot be repaired until the following one.

Cyber Warfare Planning

Cyber Warfare has quickly become an important consideration in modern warfare. While the specifics of the cyber capabilities of most nations are highly classified, some general concepts can be modeled in *War at Sea*.

Each Team can take Cyber actions to achieve specific *Offensive or Defensive* Cyber effects. Teams can only execute a limited number of Cyber actions in a game; the specific number will be listed in the Team **OAF Doc**. Two tables describing the menu of both the offensive and defensive cyber effects available to you are detailed in the tables on the following page.

Offensive Cyber Effects

Offensive cyber effects allow a Team to influence some portion of an opponent’s combined forces based upon one of the Operational Functions. These effects will either influence the entire opposing Order of Battle (marked “All” in the “Targeting…” column) or a specific force. Exactly *which* forces can be affected is also specified in the “Targeting…” column.

Defensive Cyber Effects

Defensive cyber effects do not harm your opponent, they are merely a shield against your enemy’s offensive cyber effects. As you will see at the bottom of the following page, there are only three types.



Important Note: It is not a given that your cyber-attack or defense will be successful. Within each Teams’ **OAF Doc**, you can find your “Cyber Effects Results Table,” which shows what rolls of the Facilitator dice makes your attack a success or a failure. You will not know it worked until the point in the TURN when it WOULD BE effective.

Offensive Cyber Effects Table:

Function to be Protected	Protecting...	Requires known location of target	Description The Team initiating the cyber-attack...
Command and Control	A single maritime force or stack	Y	...modifies the order to a specific <u>and known</u> maritime force on the opposing Team's Move Card. The attacking Team will indicate to the Facilitator the new movement they wish for the targeted force; the Facilitator will treat the movement as if it had originated from the opposing Team.
Communications	A single land or maritime force or stack	Y	...exploits a vulnerability in the opposing team's communications architecture. The affected enemy is not allowed to bring in reinforcements from any additional cups/CAP stations not touching it in any BR.
Intelligence	All	N	...degrades the opposing Team's ability to conduct ISR. The opposing Team suffers a -1 penalty to <u>all</u> of its ISR rolls during that TURN.
Movement and Maneuver	A single maritime force or stack	Y	...affects the opposing Team's ability to move its forces. The designated enemy maritime force is "Dead in the Water" for that TURN, but can still defend itself if attacked. <u>Submarine forces cannot be targeted by this effect.</u>
Fires	A single land or maritime force or stack	Y	...gains a +1 bonus on <u>all</u> rolls for one selected targeted force for that TURN
Sustainment	A single location or maritime force or stack	N	...affects the opposing Team's ability to sustain its forces. The opposing Team is unable to transfer Fuel Points for a designated maritime force or location (airfield or port) during that TURN. If targeting a location, this includes the stoppage of <u>all</u> FP transfers between all forces, ashore and afloat, being serviced by that location.
Protection	A single location or force or stack	Y	...affects the opposing Team's ability to conduct an effective defense. The affected enemy force or location will fight with a degraded -1 penalty on all rolls for the TURN, <i>i.e., if a  hit on a 5 & 6 before, it now only hits on a 6.</i>
Space	All	N	...has been able to exploit a weakness in the opposing Team's space surveillance system. The affected enemy will not receive ANY space-based ISR reports for that TURN.

Defensive Cyber Effects Table:

Function to be Protected	Protecting...	Description The Team initiating the cyber-defense will have the ...
Defend Force	A single air / land / maritime force or stack	...ability to protect a force (maritime, air, ground-based) from being targeted by an offensive cyber effect. Each TURN, the Team will have the ability to shield something from its opponent's Offensive Cyber Effects.
Defend Facility	A single port or airfield	...ability to protect a single point or airfield from being targeted by an Offensive Cyber Effect.
Defend Space	Connection with space based ISR	...ability to protect space based ISR from being targeted by an Offensive Cyber Effect.

Cyber Cards.

Each Team will be given some number of “Cyber Offense Cards” and “Cyber Defense Cards,” as noted in your **OAF Docs**. Each Cyber **Offense** Card is **expendable**. A Team CANNOT employ the **same** effect, on the **same** target, during the **same** TURN and receive compounding effects. A Team can, however, use **different** effects on the same target during the same TURN. There may be limits to how many each Team can use per TURN. Cyber **Defense** Cards are **reusable** throughout the game. Every card has the same go/no go criteria, a Facilitator roll of 1 or 2 is a failed card, 3-6 is a successful one.

During planning, the Team will assign its Cyber Defense Cards to two different forces (maritime or ground) or locations (airport or seaport) or to protect space-based ISR. The Team can select new forces to protect each TURN and can reallocate a Cyber Defense Card to a new force each TURN. Once the Planning Phase ends, the Card cannot be reallocated until the end of the TURN.

Rule 97: Offensive Cyber Cards are expendable while Defensive Cards are reusable.

Rule 98: Teams cannot employ the same Cyber effect more than once per TURN and receive compounding effects on a single target.

Rule 99: Teams are limited in the number of Cyber cards they can employ per TURN by their OAF Doc.

Rule 100: A force with a successful Cyber Defense Card assigned to it is impervious to all Cyber Offense Cards and will suffer no adverse effects from a cyber-attack.

You can deploy both cyber offense and defensive cards simultaneously. To do so, the Team will fill out a Cyber **Offense** Card and turn it in with the Team’s Move and ISR Cards. The Facilitator will adjudicate the Card and then action it during the appropriate step of that TURN.

Cyber Planning Example:

(Left Card) Team US expects its adversary to attempt cyber-attacks against its forces. It is concerned about the Ronald Reagan Carrier Strike Group (RR) because this is its main effort during TURN 1. During the Planning Phase, Team US completed a Cyber **Defense** Card and placed it on the Game Map under the RR Strike Group. If successful, this will ensure that the RR Strike Group is protected against all cyber-attacks throughout TURN 1.

(Right Card) During the Planning Phase for TURN 2, Team US decided to conduct a cyber-attack. It completed a Cyber **Offense** Card (**Right**) and submitted it to the Facilitator. Note that the ‘Information’ cyber effect does not target a specific force and the Team correctly entered ‘N/A’ in that spot. The Facilitator will adjudicate the Card at the appropriate point in the TURN.

US Cyber Defense Card #1 (Reusable Each TURN)	US Cyber Offense Card #1 (Expendable)
 TURN: <u>1</u>	 TURN: <u>2</u>
☒ Defend Force <u>RR</u>	☐ Command and Control ☐ Communications ☐ Intelligence ☐ Movement and Maneuver ☐ Fires ☐ Sustainment ☒ Protection ☐ Space
☐ Defend Facility _____ ☐ Defend Space _____	Targeted Force: <u>N/A</u> (if applicable)

Special Operations Planning

The missions of Special Operations Forces are another important aspect of modern warfare. While the specifics of the SOF capabilities of most nations are highly classified, some general concepts can be modeled in *War at Sea*. The following rules support player understanding of this concept. Your Team OAF Doc will provide more specifics with regard to your specific SOF capabilities. Each Team can conduct SOF missions to achieve specific ISR or Fires effects. Teams can only conduct a limited number of SOF missions per a game; this specific number is listed in the OAF Doc. Much like cyber-attacks, SOF missions require a roll of the dice to determine success (3-6) or failure (1-2).

Special Operations Missions.

SOF missions allow a Team to influence some portion of an opponent's combined forces during the Movement, ISR, and/or Fires step. These effects will influence a specific force or location (port or airfield) of the opposing force. The effects will last for a single TURN after which the effect goes away.

SOF Mission Menu

SOF Mission	Targeting...	Description. With this successful SOF Mission, the Team...
Strategic Reconnaissance	A Single Location	...receives a full report of the opposing Team's forces at a particular location during the ISR Step. These forces may be targeted in the subsequent FIRES portion of that TURN. This mission defeats any enemy Denial and Deception (D/D) force.
Sabotage Port	A Single Port	...shuts down a port, not allowing anything in or out for one TURN. Enemy forces intending to use this port will not be able to transfer any forces, refuel, or rearm for one TURN.
Sabotage Airfield & its Depot	A Single Airfield	...shuts down targeted airfield; enemy aircraft may not land or takeoff for one TURN. Enemy aircraft moving to this airfield must abort their landing and return to their initial airfield. Also removes a # of missiles determined by the SOF damage table in the OAF Doc.
Sabotage Ships	A Maritime Cup in Port	... removes a number of capability dice from the targeted maritime unit as indicated by the SOF card. The attacker does NOT have to follow the 1-1-1 Rule.
Sabotage Aircraft	Aircraft at an Airfield	...removes a number of aircraft dice from the targeted airfield as indicated by the SOF card. The attacker does NOT have to follow the 1-1-1 Rule.
Sabotage Missile Force	Land-based Missile Force Location	... removes a number of  Lg,  ,  , or  Lg from the targeted force as indicated by the SOF card. The attacker does NOT have to follow the 1-1-1 Rule when hits are apportioned.
Interdict Enemy Movement	A Ground Force	...the Facilitator modifies the move order of the targeted enemy ground force currently located in enemy territory; it cannot move for that TURN. (The SOF Team has done "something" to stop that force's onward movement for a TURN.)
Target Acquisition	A Ground Force or Location	...gains a +1 bonus on all rolls for the TURN during the Fires Step. Can either target a ground force anywhere, a port, or an airfield.

SOF Mission Cards.

Each Team will be given some number of “Special Operations Forces Missions Cards.” Each one is expendable. A Team CANNOT employ the same effect, in the same TURN, on the same target and receive compounding effects. However, a Team CAN employ different effects, in the same TURN, on the same target in order to receive compounding effects. This becomes more of a concern in Borneo. Teams are also limited to the number of cards that they can employ per TURN.

During planning for a TURN, the Team will allocate a Card(s) to a specific target(s). Once the Planning Phase ends, the Card cannot be reallocated.

The Facilitator will adjudicate the Card and then action it first. The owning Team will be informed of the results of the mission at that time. The opposing Team will be informed of any damage/degradation/destruction of Facilities/capabilities.

NOTE: Cyber and SOF cards are adjudicated first – *for example, a successful cyber or SOF effect that shuts down an airfield will prevent all sorties from launching from that airfield on that TURN.*

Rule 101: Teams cannot employ the same SOF effect more than once per TURN on a single target and receive compounding effects on a single target.

Rule 102: Teams are limited in the number of SOF cards they can employ per TURN by their OAF Doc.

Rule 103: SOF Card effects are not constrained by Rule 137: or Rule 218:. (Can only kill half the land force & the 1-1-1 rule, respectively)

Rule 104: SOF and Cyber cards can be combined on a single target to receive compounding effects.

The following provides examples of SOF Mission Cards for F/M. These get more complex and you have more options with the Borneo Scenario, but the concept is the same for both Games.

(Left Card) MC #1: Team UK needs to confirm that there are no forces currently at Pebble Island AFB due to a change it needs to make to its overall scheme. It will send a SOF team in to try and ascertain the ground truth of what, if any, forces are present.

(Right Card) MC #2: Team UK is also concerned about its ability to target enemy units on the enemy airfield at Port Stanley since it will be approaching soon for a landing nearby. It wants to increase its chances of a kill at that site.

(Both) During the Planning Phase, Team UK completed these Cards and turned them in with their Move and ISR Cards.

UK Special Operations Force Mission Card #1	UK Special Operations Force Mission Card #2
 UK Special Operations Force Mission Card #1 TURN: <u>1</u>	 UK Special Operations Force Mission Card #2 TURN: <u>1</u>
☒ Reconnaissance ☐ Sabotage Airfield ☐ Sabotage Aircraft at Airfield ☐ Target Acquisition	☐ Reconnaissance ☐ Sabotage Airfield ☒ Sabotage Aircraft at Airfield ☐ Target Acquisition
Location: <u>Pebble I.</u>	Location: <u>Stanley</u>



Helpful Hint: Combining the effects of Cyber, SOF, and Strike assets against a single target can greatly increase your chances of a successful attack.

Set Denial / Deception

This is the ability to “hide” forces from enemy detection and can only be used for forces on land. Each Team will have a specific number of Denial and Deception (D/D) Cups capable of holding a specific number of capabilities dice. More specifics are given to each Team within their **OAF Doc**, to include the Teams’ number and capacity of D/D Cups. Only a single D/D Cup can be employed at a location.

Rule 105: The location of D/D Cups are unknown to the opposing Team.

Rule 106: The contents of D/D Cups are invisible to Space-based and unmanned ISR.

Rule 107: The contents of D/D Cups can be detected by manned ISR in accordance with the normal ISR detection roll. The contents can then only be struck by manned strike missions.

Rule 108: The contents of D/D Cups cannot be targeted by Land Attack Cruise Missiles (including TLAM), unless detected first by a SOF Recon mission or that D/D Cup exposes itself by firing.

Rule 109: If contents of a D/D Cup are employed, they immediately lose the protection of the Cup and are targetable for the remainder of the TURN.

Rule 110: If a player elects to remove contents from a D/D Cup to employ them defensively, they enter combat on the second Battle Round, lose the protection of the Cup, and are targetable for the remainder of the TURN.

Designate Air Cargo Transfer

Many modern militaries often have an extensive air cargo delivery capacity. Transporting air cargo is a standard move that is annotated on a Team’s Move Card. Each Teams’ **OAF Doc** provides the number, capacity, and range of air transport units. These units are typically represented by Tiles on the Game Map.

Rule 111: Cargo flights can transport 1x ground die.

Rule 112: Cargo flights are assumed to be in EMCON but can experience Trip-Overs (be intercepted) in flight.

Rule 113: Unescorted Cargo flights intercepted by enemy fighter aircraft experience an Auto-Kill. Cargo flights intercepted by a maritime force are susceptible to engagement by , , or . Doing so will force the maritime unit into Emitting.

Rule 114: Ground forces transported via air cargo to an airfield are not available for use on that TURN. Those forces will be activated during TURN Closeout and are available for use the following TURN.

Rule 115: Airborne forces delivered to a drop zone via air cargo are available to fight in BR#2.

Escort Aircraft.

Any number of fighter aircraft can be used to provide escort as part of an Air Cargo, AAR, ISR, or AWACS mission as long as they are not carrying  or  Lg munitions. Escorts are specifically designed to bear the brunt of attacks by enemy aircraft and will suffer damage before their charge does.

Rule 116: If an escorted flight Trips Over an enemy force, the escorting fighter aircraft must be eliminated in normal combat BEFORE the escorted aircraft can be targeted by enemy fighter aircraft.

Rule 117: If an escorted flight contains more fighter aircraft dice than the enemy force, the escorted flight can split into two formations. The “escort formation” must contain at least as many fighter aircraft dice as the enemy force; it will remain and engage that enemy force. The “mission formation” will continue unimpeded to its intended target or destination.

Rule 118: Aircraft always operate Emitting when part of a CAP station, tasked to escort other aircraft, any AWACS aircraft or any aircraft conducting ISR missions.

Rule 119: Aircraft conducting Deliberate Fires missions are in EMCON unless otherwise specified.

Rule 120: Strike aircraft paired with a missile cannot also be escorts to another defenseless aircraft. Escort aircraft cannot be paired with another dice or have any other task that TURN.

Aggregate / Disaggregate Forces

Aggregating Forces

Combine already established forces, i.e., stacking maritime Cups to move and fight together. The largest element is the “Task Force” (TF), upon which you cannot stack any other force, except during an amphibious assault. Each Cup’s force size is annotated at the top left of its description in your **OAF Doc**.

Cup stacking guidelines:

Force (Cup) Size	Description	Stackable?
Task Element (TE)	Individual units, ~modern O5/O6 commands	Yes
Task Unit (TU)	Groups of 2-4 TEs	Yes – up to 4 high
Task Group (TG)	Groups of 2-4 TUs	Yes – up to 4 high
Task Force (TF)	Groups of 2-4 TGs	Only during an amphibious assault and only then up to 4 high (Rule 208:)
Ground Unit	A Division	No

Rule 121: Maritime forces are aggregated by stacking their Cups. Ground units cannot be aggregated.

Rule 122: An aggregated force can only move as fast as its slowest element.

Rule 123: ‘Task Force’ level units CANNOT be aggregated (stacked) together; they can only move adjacent to one another. Rule 208: is the exception during an Amphibious Assault.

Disaggregating Forces.

Splitting forces into smaller groupings occurs in one of two ways. It can be as simple as 1) ‘Unstacking’ previously aggregated Cups or as complicated as 2) pulling a new force combination from one of the larger Cups. These smaller forces can move independent of their parent Cup once split off as long as they retain their ☐. To select dice to remove from a single Cup, use the **Disaggregation Chart** below for the minimum requirements. These forces move at FAST (15”) speed distances.

Rule 124: Disaggregation is a special mission to support screening, scouting, or shifting capabilities between forces. It requires a ☐ for each disaggregated force.

*Rule 125: The disaggregated force must be composed of at least one complete capability grouping from the **Disaggregation Chart** below.*

Rule 126: If a disaggregated force loses its ☐ due to combat attrition, it will move to and re-aggregate with its parent force at 15” per TURN engaging only in actions of self-defense.

*			
CVE CVL CV Three possible carrier disaggregated groupings.	Battleship & Cruiser Capability All Battleships, All Cruisers One complete capability grouping.	Destroyer Capability DE, DD One complete capability grouping.	Amphib Capability DE, DD One complete capability grouping.

*This table only applies in the Leyte game since this kind of disaggregation is not a possibility in F/M or Borneo.

Set EMCON Status

Emissions Control (EMCON) is important in modern warfare. For the purpose of simplicity, the opposite of “in EMCON” for this game document and others is referred to as “Emitting.” Emitting with radars and other sensors is much like turning on a flashlight in a dark room; it has both positive and negative effects. Emitting vastly improves your ability to detect the enemy at greater range but it also makes you more visible to them. Your circumstances will help you dictate which forces should be in EMCON and which should be Emitting.

Clear blue chips (●) are placed on top of ground or maritime units to indicate that force is in EMCON. The Facilitator verifies for each Team that EMCON is set and correct on both the Game Map and the Move Card. EMCON is denoted on the Move Card by circling the Force’s symbol, as previous shown graphically on Page 17.

Organic ISR and EMCON.

For forces in EMCON, and because of the greater nm distance 1” models in this game, the first two rules below supersede Rule 60: and Rule 61: on Page 25, which provided different ranges for day and night. EMCON is now the higher concern vs. time of day.

*Rule 127: **The Organic ISR range for a force in EMCON is 1.5” Radius, day or night.*

*Rule 128: **The Organic ISR range for a force Emitting is 3” Radius, which accounts for radar and organic sensors.*

Rule 129: A force Emitting is detectable out to 5” along a line of bearing only. If the opposing Team comes to within 5” of your Emitting force, the Facilitator will give them a direction to your force, but not the distance.

Rule 130: Emitting ships will have their exact location and makeup reported to the opposing team.

ROLL FOR INITIATIVE

Initiative is decided at the beginning of each TURN and determines which Team acts first. A player on each Team rolls a die and announces the result to the Facilitator, who determines which Team won the Initiative. In the case of tie, the initiative goes to the Blue Team for Leyte **and F/M**, and to the Red Team for Borneo.

OPERATIONS PHASE

Now that planning is complete and initiative is established, Teams can expect a break in action as the Facilitators review the turned-in Move and ISR Cards. They will be judging what forces have come into contact with one another as well as what each Team's ISR assets discover. This will lead to "Trip-Over," which signify two or more opposing forces have "Tripped-Over" each other and now they will take a "Trip-Over" to the Battle Box to engage in combat.² It will also lead to intel reports of varying usefulness for the Fires Card.

OPERATIONS PHASE

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Move Forces (Conduct Organic ISR) - 41

ISR Report Step - 41

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Combat in Play & Battle Damage Assessment - 43

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Submarines and Trip-Over - 47

Fires Against Heliborne Transport - 48

Minefield Trip-Over - 48

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Account for Logistics (again) - 58

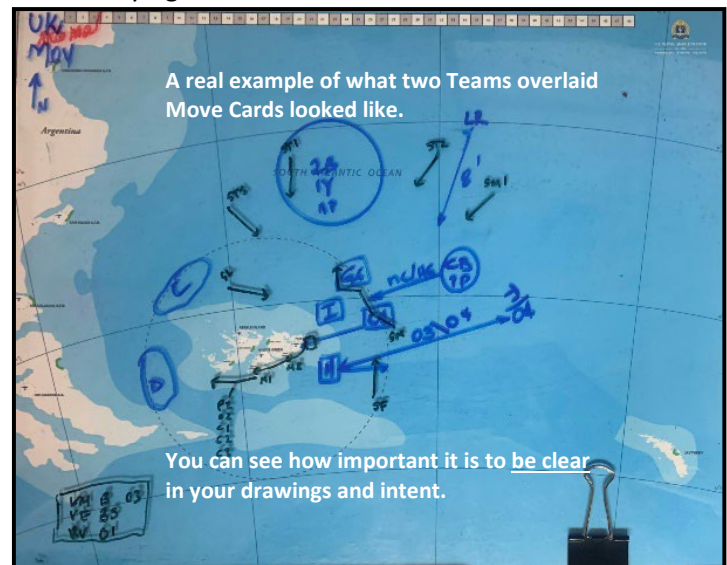
Activate Minefields

Cyber Effects Step

During the Cyber Effects Step, the Facilitators adjudicate the Cards but the results will not be provided to the owning Team until the appropriate step in the requisite phase (*i.e.*, *A Movement and Maneuver Offensive Cyber Effect would be provided to the attacked Team in their Movement Step*). Cyber effects are *in effect* for all portions of the subsequent steps in the TURN, (Movement, ISR, and Fires). For more information on Cyber Warfare, the majority of instructions were already described in detail starting back on Page 30.

SOF Mission Step

During this step, the Facilitator provides the owning Team the results of its SOF missions. The opposing Team is only informed of successful sabotage or interdiction missions. Reconnaissance and target acquisition mission results last the entire TURN (*i.e.*, *SOF target acquisition mission results are reported in this step to the*



If you are using this book in its PDF form, everywhere you see a Page Number, Rule Number, or Appendix Letter, you will find that the Number or Letter, itself, is hyperlinked to the Page it references. You can use Alt+← as your back button.

² Myths vary over the origin of "Trip-Over."

owning Team and would still apply to the surveilled force during the same TURN's Fires Step). For more information on SOF Missions, the majority of instructions were already described in detail starting back on Page 34.

Movement and Maneuver Step

Teams now move their forces, including newly arriving forces during specified TURNS, across the Game Map. The Facilitators will have reviewed the two Move Cards laid over one another and rolled to determine what contacts were made when opposing forces were in close proximity.³

ISR & Trip-Overs		Die Roll					
		1	2	3	4	5	6
Outcomes		No Report	No Report	No Report	Report	Report	Report

³ Close proximity denotes both an intersection, as in two forces paths cross one another (or one moving force comes across a static one), and those instances when opposing forces do not intersect but their visual range (Organic ISR) is such that one or both could have seen the other.

Move Forces.

The Initiative Team is first in conducting the Movement and Maneuver of its forces. It is directed by the Facilitator who informs the Team how far each force traveled along its intended route or if there were any Trip-Over that they detected. If there were, the Team will decide if it wants to engage in combat or not. For more on that, see “Conduct Trip-Over Fires” on page 42.

Subsequently, the Second Team executes Force Movement just as the Initiative Team did. If *either* team chooses to engage in combat based on detecting an enemy force, then the fight will occur. If *both* teams decline combat, then movement continues without fighting.

Rule 131: *The movement of forces will follow the planned moves in accordance with that Team’s Move Card. Deviations from the Move Card only occur as a result of interaction with enemy forces at which time both forces stop where the Trip-Over occurred.*

Rule 132: *In instances where a Game Map drawing does not match the Move, ISR, or Fires Card, the **Card** will be considered correct.*

ISR Report Step

The ISR Step is simply a brief from the Facilitators of what has been collected by a Team’s planned Reconnaissance assets. The real work by each Team was done in planning the ISR effort; the details for doing that start back on Page 24. The Facilitator informs each Team, starting with the Initiative Team, of the results of their ISR plan. All of the disclosed results will have been based on the **Detection Table** (also found on Page 24). If an ISR asset was lost at any point along its mission, it does not continue on to follow-on contacts. Subsequently, the Facilitator adjudicates for the Second Team.

Rule 133: *Only 1x ISR aircraft at a maximum, can be “Lost” to a Facilitator dice roll of “1” during a single TURN. However, multiple can be lost to an opponent’s CAP. Only CAP can shoot down ISR.*



Fires Step

During this step, Teams engage their opponents in combat. There are two stages within the Fires Step, the first deals with “Trip-Over Fires” and the second is “Deliberate Fires.” Trip-Over Fires are those that occur because two forces have come into some form of contact through their movements on the Game Map. Deliberate Fires are those that are planned with the information gathered from combat during the Movement and ISR Steps. Only one person per Team meets the Umpire at the Battle Box for each Trip-Over or Deliberate Fire.

Combat.

Combat occurs when directed by the Facilitator and overseen by the Umpire at the Battle Box. The Initiative Team completes all Trip-Over Fires first, followed by the Second Team. Deliberate Fires are conducted next, first by the Initiative Team and then the Second Team.

Combat is resolved at the Battle Box by rolling the forces’ appropriate dice and removing opponent’s dice from play based on hits generated in accordance with Appendix B. The number in each column indicates the number of Damage Points a given dice roll, of a given color, scores. Most of them are “1”.

The Team member that rolled dice and now has Damage Points to inflict on their opponent imposes them when the Umpire tells them to. They cannot look to their Teammates for advice once they are at the Battle Box. A single Damage Point can take any allowable die, or a Cup once that Cup is depleted of its capability dice. The chosen dice are physically removed from the opponent’s side and placed in the attacker’s own “kill Cup.”

Conduct Trip-Over Fires.

The Facilitators determine what is a Trip-Over and what is not. Prior to that, it is a *Potential* Trip-Over. Facilitators make this determination first by overlaying the drawings of the opposing Teams' Move Cards and annotating all places where opposing forces came into close proximity of one another. The Facilitator then has to judge each of these Potential Trip-Over.

The Facilitators identify Potential Trip-Over in one of three ways:

- Where two or more opposing movement lines touch / intersect.
- Where one moving force encounters a static force along its path.
- Where two opposing forces come close enough to potentially detect the opposition.

This last condition refers to a force's Organic ISR range.⁴ During Daytime TURNS, forces can "see" a 3" radius around themselves, which drops to a 1.5" radius during Night TURNS. Moving forces can Trip-Over stationary forces, maritime forces that get close enough to the land can Trip-Over nearby ground forces, and aircraft can Trip-Over both.

Rule 134: For each potential Trip-Over, the Facilitator rolls to determine whether each force detects the other.

Rule 135: Only the Tactical Commander of forces in combat reports to the Battle Box. The Tactical Commander is not allowed to interact with other members of their Team during combat.

Rule 136: Dice landing outside the Teams' allocated cardboard Battle Box are re-rolled until they land within it.

*Rule 137: "The 1-1-1 Rule" If multiple points of damage are scored, the Attacker must alternate the color of the killed dice up to sets of 3 prior to repeating a color until only a single color remains **See Page 44 for further explanation.***

*Rule 138: **When a naval force (Cup) is carrying embarked ground forces (ground force dice), all embarked dice must be eliminated prior to the destruction of the naval force Cup.*

*Rule 139: **After all dice in a Cup are eliminated, one additional point of damage is required to destroy the Cup. **This and the previous rule change with Rule 141:.***

Rule 140: A Cup Tripped-Over by armed reconnaissance is allowed to complete its movement for that TURN following the attack at the Battle Box.

*Rule 141: **Sinking a maritime Cup is done through eliminating all its Survivability Points (SP) regardless of its dice contents at the time of attack. (Explained thoroughly on pages 47 & 48)*

⁴ The range at which a force can detect another without having to run into it; i.e., a person on the bridge with a pair of binoculars sees the other force, or radar detected it. [Being in EMCON complicates this but that is explained in greater detail later.](#)

Facilitators Rolls for Determining Trip-Overs:

ISR & Trip-Overs		Die Roll					
		1	2	3	4	5	6
Outcomes		No Report	No Report	No Report	Report	Report	Report

Teams that receive a Report of the enemy presence can:

- Attempt to continue without engaging: If the other force is unaware or is aware but also elects to continue without engaging, then no combat will take place. Both forces will continue to the end of their planned movement. If either team elects to initiate combat, then the fight will take place.
- Stop Short: They can elect to stop shorter than planned after the Trip-Over location, but only along their planned route. To reiterate, they cannot change *direction*, they can only change *where they stop*.⁵
- Initiate Combat. The Facilitator will then assign the Trip-Over a number. Both forces mark the location on the Game map with a marker and places the Cup with its dice in the appropriate place on the Trip-Over placemat located adjacent to the game map. The Facilitator notes the location and circumstances of the engagement on the Move Card as a reminder for Fires Resolution and informs the Umpire of the situation.

Rule 142: If EITHER Team chooses combat during a Trip-Over, both forces stop at that location for the remainder of the TURN.

Rule 143: Once a force has engaged in a battle, it loses the ability to conduct a Deliberate Fire mission until the next TURN. It can still defend itself.

Combat in Play and Battle Damage Assessment (BDA).

Once combat begins, a Tactical Commander from each Team brings their engaged Forces' dice to the Battle Box. Only the engaged Tactical Commanders may be present at the Battle Box. Opposing Tactical Commanders have full visibility of each other's dice. BDA occurs by choosing opposing dice to destroy and observing any surviving dice. It is the Tactical Commanders' responsibility to report BDA to their respective Team. Following a roll, players do not touch the dice until the Umpire tells them they can. Any dice landing outside the box are re-rolled. The Umpire alone will judge what is a "hit" and what is a "miss" using the Damage Tables. Once complete at the Battle Box, players can then tell the rest of the Team the composition (to the best of their memory) of the force they faced.

Damage Point Effectiveness.

Damage Points are only effective against and between specific capabilities, (*i.e., Infantry dice cannot kill a submarine, but a submarine can kill infantry as long as they are embarked on a ship*). While many of these interactions are common sense, some have to be specified due to nuance.

⁵ A change in direction would create more than one additional step, and cause the Facilitators to reassess any enemy forces that cross the new line of bearing, which adds numerous complications; too many to list here. For the purpose of efficiency in game play, implementing this constraint outweighs its artificiality.

Reading the Fires Damage Tables.

The tables found in both Appendix B, and at the Battle Box are the Fire Damage Tables for each game, by Team. This is where Teams may notice that not all capabilities are equal between opponents, as was the case in 1944, 1982, and today. Below is an explanation of how to read the Damage Tables. Following this is the more thorough explanation of the 1-1-1 Rule and Battle Rounds.

How to read the Damage Tables in Appendix B:

		Die Roll					
Attack Type	Versus	1	2	3	4	5	6
 (G)	Maritime	-	-	1	1	1	1
 (G)	Land	-	-	-	-	1	2
 (G) (Submarine)	Maritime	-	-	-	-	1d6	1d6

For example: On the first roll, a  maritime dice roll of 2 (a “miss”) against another maritime force would impose zero Damage Points, whereas a roll of 3 (a “hit”) would have imposed 1x Damage Point.

Indicates the number of Damage Points you inflict on your opponent with the given dice roll.

“1d6” under a dice number indicates that you re-roll that dice and the number you get (1-6) is the number of Damage Points you inflict on your opponent.

The 1-1-1 Rule Explained. (Rule 137:)

The numbers in the example show the order in which they were chosen by the attacker:

Given 5 Damage Points to inflict...





12345





14253





1234Cup5

The “1-1-1 Rule” keeps a Team from being able to completely pinpoint and destroy a specific capability of their choosing without having inflicted a large amount of damage. **You must take nonrepetitive colors, in sets of three, unless there is only one color available.** In this example, you see that given 5 Damage Points to inflict, the rule has kept the attacker from taking all of the other’s aircraft without first spreading out the damage. As seen in the third example, the only time the damage does not have to be spread out is when there is no choice. An empty cup will be eliminated with 1x additional damage point in Leyte, as seen in the last example above.

Force-on-Force Trip-Overs.

Once a Trip-Over is determined to exist and one or both Teams decide that they want to engage in battle, the Facilitator will inform each Team which forces will be going to the Battle Box and when. Since there can usually be more than one Trip-Over happening in a single TURN, the Facilitator will tell the Teams that, *i.e.*, *Force A will be fighting in Trip-Over 1 and Force C will fight in Trip-Over 2.* These will coincide with the other Team's forces they will meet at the Battle Box. The Umpire will then call for "Trip-Over 1 to the Battle Box," at which time one Team member from each side will bring their Trip-Over 1 force for 2 (3) Battle Rounds of combat. This is explained in detail on the table below and the paragraphs that follow.

Battle Rounds Explained.

	Battle Round #0 (BR#0)	Battle Round #1 (BR#1)	Battle Round #2 (BR#2)
Stacked Cups	All forces in the stack are susceptible to ASCM-  attack damage. Must be Emitting at the time of the attack to use 1x  per  in defense, otherwise only  in defense and  in offensive response.	All forces within the stack are included in the engagement and are vulnerable to attack.	All forces remaining within the stack are included in the engagement. One cannot leave unless both red and blue sides agree to end combat.
Touching Cups	Only the Cup that the  would arrive to first is susceptible to the Damage Points earned. Same EMCON rules apply as above.	Only the force that was Tripped-Over is included in the engagement. The force that would be contacted first due to the direction of attack, is the only one that can fight in this BR.	The tactical commander can choose to bring their touching Cup(s) into the engagement as reinforcement. Submarines are the only forces that cannot reinforce. Once committed, these forces are susceptible to having capabilities taken from Damage Points. (*See C2  limitation during the Leyte game in the description below and Rule 153:)
Combat Air Patrols (CAP) Page 20	CAP have no involvement. See CAP involvement example for Borneo in table on Page 53.	CAP stations over the force or whose center point is within 10" of the engagement, can enter combat.	2x  from land- or carrier-based aircraft not airborne can be scrambled (Rule 155:) to enter the fight if within 10" of the Trip-Over. This does not automatically cause that carrier to be available for opponent Damage Point allocation.

BR#1: After the Umpire has announced "Trip-Over 1, to the Battle Box," and the individual earmarked as "Tactical Commander" for that Trip-Over moves to the concealed Battle Box location, the Umpire will inform the two opponents of the situation and what dice capabilities are available for use in BR#1. Stacked Cups will all go in BR#1. For Touching Cups, only the one that would be attacked first – given the direction of attack – will be the one that fights in BR#1. For this reason, players must know the position of the touching forces in relation to one another.

When  are being used, there will be a Battle Round Zero (BR#0) serving as a prequel to BR#1 in which only  can be fired offensively, while  and  can be used in defense against the incoming . If the force

being fired upon is in EMCON, however, they can only defend with , but can still fire back with  of their own if they have any. Being in EMCON only shuts down the use of that cup's . When there is a BR#0, no force is in EMCON following it.⁶ Defensive fires are typically rolled first to see how many  actually get through.

Rule 144: Opposing forces that Trip-Over one another enter the Battle Box for no more than 2x Battle Rounds.

Rule 145: Stacked Cups enter combat on BR#1 and are all susceptible to Damage Points.

Rule 146: In the event that a group of Touching Cups is Tripped-Over or attacked in a Deliberate Fire, the Cup that the attacking force would contact first - given the direction of attack - will be the one that fights in BR#1.

Rule 147: Stacked Cups in BR#0 are all susceptible to Damage Point allocation from the opposing force .

Rule 148: In a group of touching Cups, the  will target the Cup or stack it would contact first.

Rule 149: In BR#0, opponents not in EMCON choosing to roll  are only rolled in defensive mode and therefore limited to firing less than or equal to one per inbound .

Rule 150:  cannot be fired while in EMCON.

Rule 151: When a Team is rolling  in BR#1 or BR#2 when both air and ships are potential targets, they must inform the Umpire what they are targeting (air or maritime) before rolling.

Rule 152: During combat involving maritime forces employing , both forces are able to employ s only in BR#0 if they possess them. Any remaining  must wait until another trip to the Battle Box to be fired.

BR#2: Following BR#1, Tactical Commanders have the following five options heading into BR#2:

1. Continue combat. If EITHER Tactical Commander chooses to continue fighting, combat continues.
2. End combat. If BOTH Tactical Commanders decide to disengage, they do so without penalty. The forces do not move, combat ends, and they become undetected.
3. Reinforce. After BR#1, each Force with a  *can obtain reinforcements from Cups that were Touching it. Any Reinforcements that come in cannot then be utilized towards a Deliberate Fires event for the remainder of that TURN. They can, however, defend themselves if attacked.
 - a. Surface Reinforcement. Any number of surface capabilities can support the force in combat, if that Cup or Tile is touching the force in combat **and the supported force has a  ***, which provides the *Command and Control* function necessary to integrate the reinforcing maritime forces.
 - b. Air Reinforcement. The “Scramble Rule” (below) allows you to launch and bring in 2x  from nearby carriers or airfields. The supported force **does not** need a  *to receive Air reinforcements.

*The  is only utilized for the Command and Control function in the Leyte scenario.  with represent other capabilities in later games.

Rule 153: Touching Cups may enter combat on BR#2 if the Tactical Commander wants to bring them in and the force engaged in BR#1 has at least 1x  still alive at the end of BR#1. Once committed, the reinforcing forces are susceptible to Damage Points from the opposing force.

Rule 154: A force, which has already conducted fires against another target earlier in that TURN, CANNOT provide reinforcement.

*Rule 155: “Scramble rule” – During Day TURNS, Carrier- or land-based aircraft on deck within 10” of the current battle, can scramble 2x  to enter combat during BR#2. They will be unavailable for deliberate offensive fires the rest of the TURN.***

*Rule 156: **Scramble Rule applies during both Day and Night TURNS.*

⁶ BR#0 is necessary because of the dual-role nature of the  dice. Without this extra prequel step neither player nor Umpire would be able to determine in what role (offence or defense) each  dice was thrown, and if its intended target was the maritime force or the  during a regular Battle Round when everything else can be fired. The added step reduces complication.

Survivability Points (SP): THESE ARE A MAJOR CHANGE

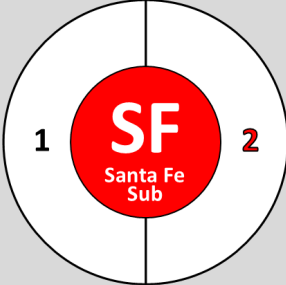
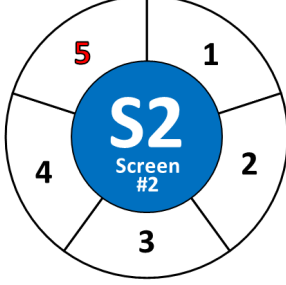
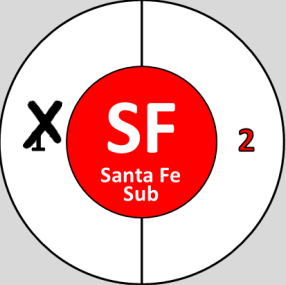
These determine how many hits a maritime force can take, its level of “survivability.” Each maritime Cup and Tile has an assigned number of SP and when they run out, it sinks. No longer does a maritime Cup’s survivability solely equal the number of dice it carries. Ships can now sink with fully functional capabilities, (written more plainly – Cups can be taken by an opponent with dice still in them) leaning more towards reality.

Old - Leyte: The W had 37x dice in it. = It took 38x DPs to sink it. (37 Dice + Cup)
New – F/M: The GS has 13x dice but a 7x SP lid. = It takes 7x DP to sink it.

The lid of every maritime force Cup has a ring of numbers on it; these are its SPs and one is crossed out for every Damage Point (DP) allotted against it. Additionally, that attacker takes the same number of dice out of that same Cup, just as in Leyte. (That is, unless the number of DP ≥ Cup SP. In that case just take the whole Cup and all its contents.)

The example below provides a look at a more complicated scenario to better explain SP. In it, the ARG SF Tripped Over the UK S2 and O3, and all three go to the Battle Box to fight it out WITH their Cup lids. They fight BR#1 and then (by the Umpire’s roll) the sub losses contact and is unable to go into BR#2.

**Question to Ponder:** After reviewing the example below, ask yourself, was the ARG dice roller thinking “operationally” when they made the decision for where to allocate their earned Damage Points?

Start of BR#1		Results of BR#1	
SF decides to fire 2x  and scores hits on both. On the re-roll they earn 6x DP.	S2 rolls its 2x  and earns 1x DP.	SF crosses out the “1” on its lid, and loses 1x  from the  hit. SF also discards its 2x expended  . It survives with 1x  and 1x SP remaining.	SF decided to allocate 5x of its 6x DPs to the S2, sinking it with all its 9x dice; and 1x DP to the O3. The S2 Cup goes to the ARG kill cup, the O3 survives but loses 1x SP and 3x  . ( on page 22)
	  	  Expended  Taken 	  

Submarines and Trip-Overs.

Submarine combat can result in forces being attacked with insufficient awareness to return fire in BR#1. The first indication a surface force may get that they have happened upon a submarine is when they observe “torpedoes in the water.” Similarly, a submarine may not be aware of an ASW threat until engaged with depth charges or ASW torpedoes (in modern combat).

Submerged subs cannot engage one another in Leyte or F/M. A surfaced submarine can be engaged by a submerged submarine for 1BR, just like any other surface combatant. The surfaced submarine is defenseless during this engagement. It is assumed that the target submarine will submerge for survival after BR1, which ends the Sub versus Sub combat.

Nuclear submarines (SSN) are difficult to detect, and they are lethal if they choose to strike. An SSN will likely enjoy a decision advantage, reflected in Rule 164:.

Rule 157: Surfaced submarines can be detected by any Force.

Rule 158: Submerged submarines can only be detected by forces with a  ASW die within 3”.

Rule 159: An UNDETECTED submarine enters combat submerged.

Rule 160: A DETECTED submarine force must enter combat in its present condition - Surfaced or Submerged for BR#1. It will submerge for BR#2, which means it is only vulnerable to .

Rule 161: A Surfaced submarine is vulnerable to the attacking force’s  AND 1x  for 1BR.

*Rule 162: ** A surfaced sub is vulnerable to 1BR as a Maritime Target by a submerged sub.*

*Rule 163: **A submarine in littoral waters (light blue on the map) is Surfaced. (Amended by Rule 167:)*

Rule 164: An UNDETECTED Nuclear Submarine (SSN) can engage an enemy surface force for BR#1 and then choose to disengage prior to BR#2. In that case, the enemy surface force will not detect the SSN, and does not have an opportunity to return fire.

Rule 165: If an UNDETECTED SSN chooses to continue to engage in BR#2, it is automatically detected by the enemy surface force which can return fire with all of its .

Rule 166: A DETECTED SSN is treated like any other force. Both the SSN and the surface force have an opportunity engage in BR#1 and can continue to engage in BR#2 if they so choose. If either force chooses to fight, the engagement continues.

*Rule 167: **A sub fighting in littoral waters is at periscope depth and can be detected by any force.*

Rule 168: Torpedoes are expendable.

Rule 169: Subs are restricted to firing 2x  maximum per Battle Round.

Rule 170: Subs remain in EMCON when firing missiles, but are an Auto-Detect for any platform within 3”.

*Rule 171: **Opposing subs can detect and engage one another while submerged.*

Fires against Heliborne Transport.

In addition to possible attacks from forces overflown by helicopters, those  that are detected by enemy forces at their landing zone can also be engaged as part of Trip-Over fires. It is assumed that any  shot down over land are able to execute a “hard landing” while carrying their Ground Forces. Over water, everyone dies.

Rule 172: During BR#1,  and any carried Ground Forces are subject to enemy fires.

Rule 173: During BR#2, any Ground Forces who survived are considered to have landed, offloaded the helicopters, and can participate in combat.

Rule 174: Surviving  are unable to leave the landing zone until the following TURN.

Rule 175: A helicopter kill over water will kill the pairing. If brought down over land, the ground dice survives at the point of the attack.

Minefield Trip-Overs.

A minefield is Tripped-Over when a maritime force contacts it without an MCM in the lead or stacked. Any minefields laid during Movement become active at the end of the Fires phase of that TURN. The range for minefield detection is a 1.5" radius circle around the MCM.

Rule 176: A maritime force that Trips-over an active minefield during its Movement automatically suffers 1d6 damage. The Facilitator will roll their own die and remove the appropriate number of dice. The force's movement for that TURN will stop at the edge of the minefield.

Rule 177: In the case of forces aggregated by stacking, each force in the stack will suffer 1d6 damage.

Rule 178: With an MCM leading or stacked with the force hitting the active minefield, no damage is sustained, but the entire stack still stops. If the MCM and force remains in place, the minefield will be clear after one more full TURN.

Rule 179: Only an MCM in the lead of, or stacked with, a force can carry out "mine hunting."

Visit, Board, Search, and Seizure (VBSS).

VBSS is a Fires action to determine if maritime traffic that appears to be neutral is in fact neutral, as opposed to being a disguised enemy intelligence asset or combatant. VBSS occurs as a Trip-Over fires action when a naval force Trips-Over suspect neutral traffic.

Rule 180: The force conducting VBSS must drive to within the 1.5" visual ISR range to the "neutral" force.

Rule 181: If the "neutral" force is a disguised enemy, the force conducting VBSS is informed of the true nature of the disguised force, and the disguised force is removed from play.

Rule 182: If the "neutral" force is actual neutral maritime traffic, the force conducting VBSS is informed of the true nature of the contact, and no further action is taken.

Rule 183: A force conducting VBSS can remain in EMCON.

Deliberate Fires

As stated previously, Deliberate Fires are planned targets, as opposed to the semi-unplanned nature of combat that comes from Trip-Over. When Teams amass their new-found knowledge gained from what they learned of the other Team through Trip-Over combat and their ISR report, they are frequently able to plan an attack against a specific target location. Teams are given a few minutes to draw their intentions on the Fires Card, usually in the form of an aerial Strike Package.



Helpful Hint: Attacking aircraft can usually go further if the plan is for them to land at an airfield closer to the target area instead of their point of origin.

Deliberate Aerial Attacks. (Strike Package).

With the exception of B-24s (page 51), deliberate aerial attacks occur only during Day TURNS when carrier- or land-based air departs and strikes at distant targets (within range of the aircraft). **In the two wargames that follow, these attacks can occur Day or Night.** Air dice (🎲, 🎲 or 🎲) can be allocated against specified targets. In doing so, however, they remain at risk to other opposing forces that are in range. Thus, aircraft committed to surface targets are susceptible to opposing 🎲 CAP and 🎲 on the way to the target, or in the target area. After 2x Battle Rounds, the surviving portions of the Strike Package returns to base (does not have to be their base of origin) and are unable to engage in further combat until the next TURN. The Facilitator informs the Defender of the direction from which Attacking aircraft arrived and departed.

Rule 184: Aircraft in a Strike Package can be engaged by enemy 🎲 CAP or 🎲 that it encounters en route to its intended target. The Strike Package can choose to fight at the intermediate target, or they can push through the enemy defenses to the intended target without fighting back. This rule is updated in later games with the additional mission of fighter "Escort" as described in Rule 116: and Rule 117:

Rule 185: Strike Packages that attempt to push through enemy 🎲 CAP or 🎲 encountered en route to their planned target are subject to 1x BR of enemy attacks without responding.

Rule 186: Aircraft do not have to return to their own base following their attack, but if the intended landing site is a carrier, there must be room to land. The OOB chart for each CV Cup has its aircraft capacity listed.

Rule 187: Aircraft conducting Deliberate Fires missions are in EMCON.

Rule 188: Strike aircraft paired with a missile can defend themselves for 1x BR while transiting an opponent's CAP station if they so choose.



Explanation of Rule 184:

Strike Packages may unknowingly overfly a defended target area on the way to attack their assigned target. The commander of the Strike Package must choose whether attacking the intermediate target is a better use if the available force, or if they should stick to the original plan. Either way, the enemy has a vote and forces defending the intermediate base may choose to engage the Strike Package if it transits within range of defending CAP or anti-aircraft defenses.

Kamikaze (aka Divine Wind - aka 🎲): Only Japan has 🎲, which models a squadron of this proto-guided-munition capability. They are usually used only during a planned Deliberate Fire strike as part of a Strike Package. The rules are slightly different for them as opposed to other aircraft. 🎲 **become Anti-ship Cruise Missiles in later games.**

Rule 189: 🎲 can only attack maritime targets.

Rule 190: 🎲 are expendable, you lose them once you throw them whether they roll a Hit or a Miss.

Rule 191: 🎲 can be flown to and parked at other bases day or night.

Rule 192: 🎲 can only be shot down by 🎲 CAP or 🎲 if it crosses thru those defenses while en route to its intended target. (Defenses in the target area are already included in the Damage Table probabilities).

B-24 Strikes. Team US has B-24 Bombers modeled through  **Lg**. These are used exclusively during the Fires Phase of any TURN, Day or Night. Since they fly at a greater altitude than the vertical range of Japanese ground air defenses,  cannot touch them – only CAP over the target area or along the Bombers flight path may engage them. Though  **Lg** can conduct bombing during Day or Night TURNS it can only conduct ISR during the Day. This models the B-24's ability to strike at a known direction and distance to a given target though it could not see it at night. During nighttime B-24 strikes, the U.S. team does not get to see what enemy forces are present in the target area, nor select the dice that are destroyed if they score a hit. The Umpire will choose at random what will be taken from the targeted area using the 1-1-1 Rule.

*Rule 193:  **Lg** can bomb Day or Night, but only conduct ISR during the Day.*

*Rule 194: For Night attacks, the Umpire will choose the dice, out of view of Team US, from the number of damage points inflicted by a  **Lg** strike. They will be chosen at random using the 1-1-1 Rule.*

Attack in the Blind. This is a subset of the Deliberate Aerial Attack in which an attack air capability (, , ) flies directly to a point to conduct fires without knowing if enemy forces are present at that location. Organic ISR along the route and at the end point apply to Attacks in the Blind, but enemy forces must be detected in order to be engaged. These are the only attacks that  can come back from, and only if there were no enemy present at the targeted location.

Rule 195: When adjudicating ISR Results for Attacks in the Blind, the Facilitator will only roll one die against the Detection Table, regardless of the total number of dice on the attack.

Deliberate Aerial Attacks Including Missiles (F/M).

The table on the next page outlines the rules governing an Argentinian Exocet Aerial Strike Package against a UK Surface Force. This will be the first time many players will have the opportunity to plan a missile strike on a Naval force. This procedure is more complex in Borneo, so it is important to have an understanding of it and experience it in F/M.

*Rule 196: Only the Super Étendard version of the  **Lg** can fire the , and only against a maritime surface force.*

*Rule 197: A  **Lg** returns to base (RTB) after firing its .*

*Rule 198: CAP aircraft dice are unable to stop or chase a  **Lg** firing a .*

Rules Governing a Super Étendard-Launched Exocet Attack Against UK Surface Forces

		BR#0	BR#1	BR#2
Argentinian Forces (Attacking)	Super Étendard Lg	-Launches Exocets and Returns to Base (RTB). -Cannot follow-up with a direct attack behind Exocets. (Rule 197:)	-Gone.	←Same
	Exocets	- rolls against first Cup or stack along its flight path within the targeted area. (Rule 148:)	-Expend.	←Same
	Follow-on ARG Aircraft Lg, ,	-Not yet in the fight.	-Conducts Airstrike against Targeted Force at the Battle Box.	-Decide to re-attack or leave.
UK Forces (Defending)	UK Force (Cup or Stack)	<u>In EMCON</u> -Defend only with . (Rule 150:) <u>Emitting</u> -Defend with and/or . Can fire a # of ≤ # of incoming . (Rule 149:)	-Now Emitting automatically. -Can employ and/or against any incoming enemy aircraft. ≤ ARG .	-Can bring in touching forces as desired. -Can scramble 2x from CV within 10" of combat. (Rule 156:)
	UK CAP within 10" of Targeted Force	-Cannot engage or Lg. (Rule 198:)	Follow-on enemy aircraft can be engaged.	←Same

Deliberate Aerial Attacks Including Missiles (Borneo).

The proliferation of long-range anti-ship cruise and land attack missiles creates more challenging interactions between enemy units. Engagements are still limited to 2x BR per TURN (BR#1 and BR#2), however, like engagements involving Exocets in the **F/M scenario**, those engagements that involve the use of Anti-ship cruise missiles (, , Lg, Lg) or Land Attack Cruise Missiles () will include the additional BR#0. Subsequent BRs include all attack capabilities and, often, engagements will be complete with just BR#0.

Missile strikes from multiple platforms and locations may be synchronized so they are programmed to impact the target simultaneously or in a sequence desired by the initiator. The table on the next page gives a detailed explanation of how to conduct a Strike Package in Borneo with aircraft and missiles.

To read the table on the next page, notice that the columns are setup to model the range to target (Space) from 10" out, and the rows model the steps (Time). Within each box are what each capability (Force) can do within that given Space/Time. Missiles and their direction of flight are denoted by a flame and an arrow (*i.e.*, Lg→ is an LRASM traveling left to right, towards the Target). The blue boxes within the table span a specific set of columns, referring to what Space is the main concern of that row/step. Notes on the right of that blue box are for the defender (or target), while notes to the left are for the attacker.

Employment of Missiles in Borneo Against a Maritime Target.

This table provides the “how to” for sending, or defending against, a Strike Package that includes missiles. The inside lettering of this table has been kept black as a courtesy to your eyes. Skip this table if you are just learning Leyte.

Range to Target:		10"	9"	8"	7"	6"	5"	4"	3"	2"	1"	Target			
BR#0 - <u>Step 1</u> Long-range ASCM shot		Attacker fires  Lg from 10" and aircraft RTB.  Lg can accompany this strike also but it comes from 50" out. Neither rolls until Step 3, but # of each inbound missile is provided to the Umpire and opposing Team.				Defender choses # of CAP aircraft to place against the inbound  Lg, and  paired with  , they cannot stop the  Lg. Hits in the next step will reduce the # of inbound threats. CAP committed to this will be out for the rest of the engagement following their 1x BR outside the 5" range ring of the targeted vessel.									
						 x? #  Lg x? #  Lg→  Lg→									
BR#0 - <u>Step 2</u> Short-range ASCM shot	Since aircraft firing  cannot do so until 5" out from the target, they are susceptible to 1x BR of a two-sided engagement (in the 1" overlap) from a # of defending CAP. Survivors of the single BR, RTB while their fired  are added to the  Lg that survived in Step 1. Only  Lg cannot defend itself.					 Lg→  Lg→  Lg→		The Defender's CAP chosen in Step 1 are committed to attacking incoming aircraft and  Lg from 6" out for 1x BR of a two-sided attack. A single Damage Point kills the pairing before it fires. CAP committed to this are now out for the rest of the engagement.			Target still waiting to engage with protective fires and remaining CAP.				
BR#0 - <u>Step 3</u> Missile on missile		ON THIS STEP - ATTACKER ROLLS 2nd					DEFENDER ROLLS 1st								
		Read the other side first, since the defender rolls first this time. In the final step of BR#0, the attacking Team rolls its surviving  Lg,  , &  Lg after the last layer of defense is rolled with its final protective fires listed on the right. → Following this final attrition, the surviving missiles roll and apply their earned Damage Points against the targeted force(s).					This last layer of defense is rolled first to determine what ordnance gets though. The targeted force (and any Cups touching or stacked) not in EMCON can engage the incoming missiles with both  &  in defensive mode, any CAP remaining, and  .								
BR#1	From this point on, all offensive missiles that will be used for this engagement have been expended. What flows in now will be the aircraft-only portion of the Strike Package. Any aircraft used previously in launching missiles have all RTB'd. The engagement from here is like any other Deliberate Fire Strike Package from the previous games.					 →  Lg  →  →		Remaining CAP can now engage the incoming enemy fighters with remaining BR(s).* <u>Any</u> used in BR#0 cannot.							
BR#2	Attacking Team has the option to stay and fight or attempt to disengage.					 →  Lg  →  →		Defending Team can now apply the scramble rule to any carriers within 10" of the target.							
Range to Target:		10"	9"	8"	7"	6"	5"	4"	3"	2"	1"	Target			

*CAP which stayed within 5" of targeted vessel has 2x BRs to give per TURN, those outside have only 1.

Initiating and Conducting Ground Combat

This section discusses how to get your ground forces from ship to shore and then how to use them once you gain a foothold. It is also valuable for the defending Team since the procedures will advise them how to deny these actions. There is also more team specific information within the **OAF Docs**.

Rule 199: On land, Ground force (🟩 & 🟩) can target opposing ground forces (🟩, 🟩, 🟩) dice on land as well as aircraft (🟩, 🟩, 🟩, or 🟩), when those dice represent parked aircraft.

Rule 200: Ground force (🟩) can only target aircraft in flight and only once disembarked from a vessel.

Rule 201: Ground forces may provide support to touching forces, for purposes of reinforcement.

Maritime Transport of Ground Forces (Friendly or Hostile).

In order for ground forces to be able to secure a beach, they must be embarked on maritime vessels and transported from one port or beach to another. In cases where the landing occurs on a friendly beach, this is done during the **Movement and Maneuver Step**. If against an enemy-held beach, the landing is classified as an Amphibious Assault and occurs during the **Fires Step**.

Rule 202: Ground Forces only embark/ disembark at "Green Beach Landing Zones," and only 12 dice of ground forces per TURN may embark/ disembark at a single beach.

Rule 203: Embark/ disembark takes six hours (Half a TURN), therefore a Maritime Force can embark or disembark ground forces and move one-half its range during a TURN.

Rule 204: Airfields are available for use the TURN following the presence of a friendly Ground Force. Once aircraft are established at the field, the Ground Force's presence is no longer required starting the following TURN.

Rule 205: Movement between East and West Falklands can be done by ground forces, but that ground force can only move half the amount it has remaining after crossing over.

Conducting an Amphibious Assault.

An amphibious assault occurs when a Team attempts to land ground forces ashore on territory that it does not possess. The landing must occur at a Green Beach Landing Zone. Though preparation for an Amphibious Assault takes place during the Movement and Maneuver Step, the Assault itself is a Deliberate Fires action and takes place during the Fires Step. In order to establish conditions for the Amphibious Assault, the Attacking Team will move its amphibious forces into position as part of the Movement and Maneuver Step.

Stopping outside of ground force detection range will allow the assault to occur during the Fires Step. During the Fires Step, the Team will inform the Facilitator that it is conducting the Amphibious Assault. The amphibious force can move up to 3" to get to a Green landing beach.

On the next page is a chart showing the steps for both defender and attacker leading up to, and during, an Amphibious Assault followed by the rules governing the procedure.

Steps for Conducting an Amphibious Assault.

	Movement Step	Deliberate Fires Step	
		BR#1	BR#2
Defending Force	Ground Units	- 🟩 & 🟩 can engage any landing ground forces. - 🟥 can engage any supporting aircraft. - 🟩 hits can also count against Maritime forces within 2" of them.	- Continue to engage landing force that can now fight back.
	Air Units	- CAP w/in range of Assault Force can cause a Trip-Over.	- 🟦 can still attack ground forces on beach.
Assaulting Force	Ground Units	- Moving from ship-to-shore "on amphib," cannot fight back yet, but can die. - Only 12x Dice /TURN.	- Surviving 🟩 & 🟩 are now "on the beach" and can engage defending ground forces. - Surviving 🟥 can now engage enemy aircraft.
	Maritime Cups	- Must end the Movement Step within 3" of the desired landing beach. - Can be stacked 4 high.	- Provides Naval Gun 🟩 not used during separate prep fires and (if Daytime) Air Support 🟦. Declare # of each dedicated to this task. ← Same.

If forces from both Teams remain, combat will re-commence during the following TURN. Surviving amphibious vessels remain in contact with the beach.

Rule 206: Infantry Forces-(🟩, 🟥) within 1", or Field Artillery-🟩 within 2" of a Green beach have the ability to oppose an Amphibious landing.

Rule 207: The defending force must declare which ground forces within the ranges listed above are opposing an Amphibious Assault. Those forces will defend the beach and are also targetable by the landing force.

Rule 208: During amphibious assaults, standard aggregation (stacking) rules do not apply in waters immediately adjacent to "Green Beach Landing Zones." Stacking of 4 'Task Force' level units is allowed.

Rule 209: If assaulting force is further than 3" at the start of the Fires Step, the assault is not possible.

Rule 210: Landing Force must declare the number of 🟩 and 🟦 dice dedicated to supporting the landing; those dice are committed to the landing and unavailable for other uses.

Rule 211: Any maritime 🟩 in range of the beach can be used during a Trip-Over with the defending ground force, or during the landing itself, but not both.

Rule 212: Captured airfields are available for use on the TURN following the presence of a Ground Force. Once aircraft are established there, the Ground Force is no longer required.

Rule 213: 12x ground force dice may move ashore per TURN during an Amphibious Assault but can only do so at "Green Beach Landing Zones." Those dice will take position at the beach if the beach is actively defended by opposing ground forces. If the beach is not actively defended, attacking ground forces may move 1" inland.



Explanation of Rule 211: This rule exists to keep the assaulting force from being allowed to attack the same target with the same dice in the same TURN. To allow such a thing to occur would lend an unfair advantage to the assaulting force, as they would be able to reduce the defending ground force to an unrealistic size through indirect fire alone.

Naval Attacks Against Ground Forces.

Prior to any good landing, Teams will most likely want to execute some level of preparatory fires of the objective area. Maritime forces can conduct shore bombardment, but they can

only destroy a maximum of half the enemy force present at the landing site per TURN. Naval gunnery- and air ( or ) supporting fires can target ANY ground forces or 'parked' aircraft within range.

Rule 214: A naval force must be within 20 nm (1") of land to conduct naval gunfire support.

Rule 215: Only ground forces within 20 nm (1") of the sea (touching water) can be attacked by naval gunnery.

Rule 216: Coastal defense artillery  can attack naval surface forces within their 2" range.

Rule 217: Coastal defense artillery  can attack naval forces transiting a chokepoint 20nm (1") wide or less, or conducting a landing or naval gunfire.

Rule 218: A shore bombardment by any platform, surface or air, can attrite no more than half the enemy ground force present at the targeted area per TURN.



Explanation of Rule 218: Let's start with an example; if the maximum allowable 5 of 10 ground dice were destroyed through prep fires in TURN 3, only 2 may be destroyed in the next TURN by more prep fires, not the other 5. The reason you can only destroy half of the defenders of a defended beach has to do with a lean towards historical accuracy. Dug in troops and their equipment cannot be removed by artillery and air prep fires alone – a lesson learned on countless occasions. While it can be very effective and highly desirable of those who will have to land at the beach, the only thing that can feasibly remove all resistance is infantry.

Heliborne Transport

Helicopters- can be used to move any single ground die per movement; however, they can do it only on land. The only ground force  can move from ship-to-shore are SOF.⁷ Helicopters can move themselves from ship-to-ship and ship-to-shore, but not while carrying any conventional troops/equipment. Helicopters that pass over an enemy airfield are subject to one BR of attacks from any CAP flying over that airfield and any Anti-Air capability protecting that airfield *on the way to their LZ*. These attacks do not count against the two maximum BRs of possible fires for the onboard forces.

Rule 219: A single  can transfer 1 dice of ground force capability per TURN up to a radius of 6" or 12" one-way.

Rule 220:  can only transfer a SOF from ship-to-shore. All other ground dice can be transported via  from one ground LZ to another.

Rule 221:  can transfer ground dice from ship-to-shore.

Rule 222: Similar to other air dice,  have a 1.5" visual ISR radius.

Rule 223: Ships launching Helicopters can move the full range of their movement and then launch helicopters for additional range.

Rule 224: Only one pick-up/drop-off cycle of forces can be completed per TURN.

Rule 225:  transferring ground forces can be detected via Trip-Over or Organic ISR by  as described in ISR. (See Page 25)

Conducting Airborne Assaults

An airborne assault occurs when a Team attempts to infiltrate ground forces by parachute into enemy territory using transport aircraft. The assault can occur on any ground area (it is assumed that given the scale of the Game Map there is an appropriate Drop Zone (DZ) within the area selected for the airborne assault). The Airborne Assault itself is a Deliberate Fires action and takes place during the Fires Step much like an Amphibious Assault. Only "Airborne units," as specified in the **OAF Docs**, are capable of conducting one.

Airborne Assault flights are planned and conducted similar to a Deliberate Aerial Attack and are done in the Fires step of the TURN and plotted on the Fires Card. Ground Forces may detect and attack the flight en route to the drop zone as a normal Aerial Attack. During the Fires Step, the Team will inform the Facilitator of its intent to conduct an Airborne Assault and draw a 1" diameter circle modeling the intended DZ. Opposing Ground

⁷ The reason for this has to do with deck space and numbers of ground forces and helicopters these dice model. It is not feasible that such quantities could have been airlifted from ships such as the amphibious or cruise liners the UK used. This changes with Borneo as the ships are purpose built and the  also represent Osprey to move Marines.

Forces within 1.5” of the Airborne Assault’s proposed 1” DZ circle creates an Opposed Landing. The Landing Force must declare AT THIS TIME the number of  and  dice in support of the drop and the number in support of aircraft escorting the Air Transport; the latter are unavailable for any other use.

If the designated DZ is defended, (forces within 1.5” of it) the table on the next page applies:

Executing an Airborne Assault

Airborne Force Element	On Approach Flight	BR#1	BR#2	End of Fires Step
Air Transport	If drawn flight path from origin to DZ on Fires Card crosses any  or CAP stations, 1x BR will be adjudicated for every instance that the defender elects to attack.	Drop Paratroopers.	Return to Base.	Arrive back at Base.
Aircraft Escorting Air Transport	Escort rules on Page 36 apply.	Remain with Air Transport for return flight.	Remain with Air Transport for return flight.	Returned to origin.
Aircraft Supporting Ground Forces	Fight enemy CAP or carry on to DZ to support infil and ground units.	If enemy CAP is on station within 3” of DZ, determine # of  to split to attack enemy ground forces and to fight the CAP. Lg can also be in support of infil with bombing. Defending , , and  can target these aircraft.	Same as BR#1	Lg Returns to origin. Fighters from both sides can remain on station as CAP until TURN Closeout.
Ground Forces	-	Can be attacked by defending ground forces, but cannot yet defend themselves.	Surviving ground force elements can now engage defending air and ground forces.	If both sides have forces remaining, combat will recommence as a Trip-Over in the following TURN.

Account for Logistics following Combat.

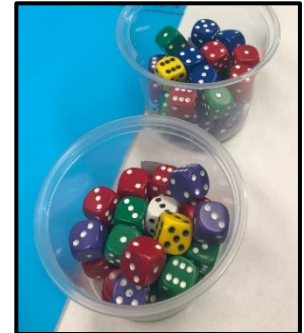
Participation in combat depletes logistics for all forces. Teams must track logistics expenditures upon completion of all Fires' actions.

CLOSE-OUT PHASE

Once Operational Actions are completed by both Teams, administrative TURN close-out actions are conducted simultaneously, directed by the Facilitator.

Recover ISR Assets. Each Team recovers any airborne ISR missions deployed during the TURN.

Facility Repair. Each Facility has a “repair capacity number” associated with it. If you have a Facility that has some of its Survivability Points marked out from taking damage, this is the time that repair number comes in. *If that number is 3, erase the markings over 3 of your marked out Survivability Points.* These repairs are done on damage points that were sustained in previous TURNS, **not the current TURN.**



of

Newly Arriving Forces. Newly arriving forces are placed at the edge of the Game Map Edge for each Team, as directed by the Facilitator. The position at the edge of the Map is the starting point for subsequent movement by that force.

Advance TURN Marker. Both Teams advance their TURN marker on the Game Maps. The Facilitator announces the beginning of a new TURN, and whether it is Day or Night. Play then proceeds.

APPENDIX A

This is a quick reference for the capabilities and limitations of particular die. Note that some dice are repeated if they serve more than one platform; (i.e., ) act differently depending on if they are part of a ship or an infantry division.)

Key:

-Any box in **this format** signifies a change in that category from the previous game.

-A/C = Aircraft.

Blue Team Dice

Blue Team

Leyte / United States			Falklands / United Kingdom			Borneo / United States			
MARITIME SURFACE									
Dice - "GREEN"	 (G)	Small	-Damage point hits cannot go against aircraft in flight.	 (G)	Small	-No change.	 (G)	Small	-No change.
Capability Represented	Naval Gun Fire			Naval Gun Fire			Naval Gun Fire		
Expendable / Reusable	Reusable			Reusable			Reusable		
Day / Night / Both	Both			Both			Both		
Targets	Surface forces*		*One dice can be rolled against a surfaced sub in BR#1.	Surface forces*			Surface forces*		
Re-Roll for Damage?	No			No			No		

Dice - "RED"	 (R)	Small	-Damage Point Hits can only go towards aircraft in flight.	 (R)	Small	-Addition 1: Can now engage Blk ASCM. -Addition 2: Cannot shoot down Gr missiles.	 (R)	Small	-Addition 1: Can now engage LgBlk and O missiles in addition to Blk. -Addition 2: Cannot shoot down Gr, LgT, or LgPk missiles.
Capability Represented	Anti-Air			Anti-Air/SeaRAM			Anti-Air/SeaRAM		
Expendable / Reusable	Reusable			Reusable			Reusable		
Day / Night / Both	Both			Both			Both		
Targets	Aircraft in flight			A/C & Blk in flight			A/C & MsIs in flight		
Re-Roll for Damage?	No			No			No		

Dice - "PURPLE"	 (P)	Small	-Gives a cup the ability to Trip-Over a submerged submarine. -Each models a spread of depth charges when rolled. -Can be used against submerged and surfaced subs.	 (P)	Small	-No change.	 (P)	Small	
Capability Represented	Anti-Sub			Anti-Sub			Anti-Sub		
Expendable / Reusable	Reusable			Reusable			Reusable		
Day / Night / Both	Both			Both			Both		
Targets	Submarines			Submarines			Submarines		
Re-Roll for Damage?	No			No			Yes		

SUBMARINES									
Dice Color - "GREEN"	 (G)	Small	*Submarines can only attack each other when one or both opponents are surfaced.	 (G)	Small	-Change 1: Torpedoes are now expendable.	 (G)	Small	-Change 1: Two opposing submerged sub forces can now fight, but only with G.
Capability Represented	Torpedo spread			Torpedo			Torpedo		
Expendable / Reusable	Reusable			Expendable			Expendable		
Day / Night / Both	Both			Both			Both		
Targets	Maritime Forces*			Maritime Forces			Maritime Forces		
Re-Roll for Damage?	Yes			Yes			Yes		

Blue Team

Leyte / United States			Falklands / United Kingdom		Borneo / United States		
GROUND FORCES							
Dice - "RED"	 (R) Small	-If embarked on a ship, cannot be used to defend the ship. Ships have their own R dice for that.	 (R) Small	-Change 1: Distance of ground movement is cut in half due to a lack of vehicles.	 (R) Small	-Change 1: 2" per TURN again along the TB Hwy. 1" per TURN when off of it.	
Capability Represented	Anti-air Battalion		Anti-air Battalion		Anti-air Battalion		
Expendable / Reusable	Reusable		Reusable		Reusable		
Day / Night / Both	Both	-Can only target B, Blk, and Y aircraft while in flight. No "ground mode" usage.	Both		Both		
Ground Movement	2" per TURN		1" per TURN		2" per TURN*		
Attack Range	1.5" Radius		1.5" Radius		1.5" Radius		
Re-Roll for Damage?	No		No		No		
Dice Color - "BROWN"	 (Br) Small	-Can attack other ground dice within 1".	 (Br) Small	-Change 1: Distance of ground movement is cut in half due to a lack of vehicles.	 (Br) Small	-Change 1: 2" per TURN along the TB Hwy. 1" per TURN when off of it.	
Capability Represented	Infantry Regiment		Infantry Regiment		Infantry Regiment		
Expendable / Reusable	Reusable	-Can target parked aircraft.	Reusable		Reusable		
Day / Night / Both	Both	-Must be within 1" of a Green beach to oppose an enemy landing occurring there.	Both		Both		
Ground Movement	2" per TURN		1" per TURN		2" per TURN		
Attack Range	1" Radius		1" Radius		1" Radius		
Re-Roll for Damage?	No		No		No		
Dice - "GREEN"	 (G) Small	-Can attack other dice on the ground within 2".	 (G) Small	-No change. Artillery was not lacking vehicles since they cannot move without them.	 (G) Small	-Change 1: 2" per TURN along the TB Hwy. 1" per TURN when off of it.	
Capability Represented	Artillery Brigade		Artillery Battalion		Artillery Battalion		
Expendable / Reusable	Reusable	-If within 1" of shoreline, can attack surface ships within 1" of shoreline.	Reusable		Reusable		
Day / Night / Both	Both	-Must be within 2" of a Green beach to oppose a landing.	Both		Both		
Ground Movement	2" per TURN		2" per TURN		2" per TURN		
Attack Range	2" Radius		2" Radius		2" Radius		
Re-Roll for Damage?	No		No		No		
AIRCRAFT							
Dice - "YELLOW"	 (Y) Small	-Unarmed.	 (Y)	Does not exist.	 (Y) Small	-Change 1: UAV.	
Capability Represented	Recon Squadron	-The more dice placed in a recon ring and/route, the higher the probability of detecting enemy forces present in those areas.			MQ-4C Triton	-Change 2: Flight range extended.	
Expendable / Reusable	Reusable				Reusable	-Change 3: Can now operate Day and Night.	
Day / Night / Both	Day Only				Both		
Re-Roll for Damage?	No				No		
Air Re-fuelable (AC:Tank)	No				No		
Carrier / Land / Both	Both				Land Only		
Range	40"				120"		
Can be Paired with	Nothing			Nothing			

Leyte / United States			Falklands / United Kingdom		Borneo / United States		
AIRCRAFT							
Dice - "LARGE YELLOW"	 (LgY)	Does not exist.	 (LgY) Large	-When paired with a P, becomes a sub hunter/killer. Can attack a Sub surfaced or submerged.	 (LgY) Large	-Change 1: Can be paired with either 1xP or 1xBlk. The P are still reusable.	
Capability Represented			MPRA - Nimrod		MPRA - P-8		
Expendable / Reusable			Reusable		Reusable	-Change 2: Maritime mine-laying capable. Requires 2x LgY converging on the targeted area in the same	
Day / Night / Both			Both		Both	TURN to emplace it.	
Re-Roll for Damage?			No		No	-See range restriction notes in OAF.	
Air Re-fuelable (AC:Tank)			Yes (1:1)		Yes (3:1)		
Carrier / Land / Both			Land Only		Land Only		
Range			70"		70"		
Can be Paired with				1xP	-Loss of the Tanker = loss of the LgY and P.	1xP or 1xBlk	
Dice - "BLUE"	 (B) Small	-Can attack anything except SUBMERGED subs.	 (B) Small	-Change 1: Can now operate at night.	 (B) Small	-Change 2: Buddy AAR: (2:1) Single B provides 2xB with 20" extra total flight range each. Cannot refuel LgB.	
Capability Represented	Fighter Squadron		Harrier Sqdn	-Change 2: CAP can be established over any geographic position but still within 10" of its origin.	4th Gen (4-6)		
Expendable / Reusable	Reusable	-Can be used for CAP, Strike, or ISR during the DAY.	Reusable		Reusable	Fuel giver cannot be paired with a missile.	
Day / Night / Both	Day Only		Both		Both	-Change 3: 3 fighters/tanker for normal AAR. Resets full fighter range.	
Re-Roll for Damage?	No	-Can fly CAP (3"diameter) over any force or airfield within 10" of its origin.	No		Target Dependent*	-Change 4: Escort tasking.	
Air Re-fuelable (AC:Tank)	No		No		Yes B(2:1) - T(3:1)	-Change 5: Can now engage Blk, LgBlk, and O missiles in flight.	
Carrier / Land / Both	Both		Both		Both		
Range	40"	-2 dice can be "scrambled" to reinforce a maritime tripped-over force within 10" on BR#2.	40"		40"		
Can be Paired with	Nothing		Nothing		1xBlk or 1xLgBlk		
Dice - "LARGE BLUE"	 (LgB)	Does not exist.	 (LgB)	Does not exist.	 (LgB) Large	-Enemy ISR suffers -2 penalty when trying to detect them.	
Capability Represented						5th Gen (4-6)	-Can only AAR from Tanker (3:1), no Buddy Tanking.
Expendable / Reusable						Reusable	-Same CAP rules as B.
Day / Night / Both						Both	-Can engage Blk, LgBlk, and O missiles.
Re-Roll for Damage?						Target Dependent*	
Air Re-fuelable (AC:Tank)						Yes (3:1)	
Carrier / Land / Both						Both	
Range						40"	
Can be Paired with						1xBlk or 1xLgBlk	

Leyte / United States			Falklands / United Kingdom		Borneo / United States		
AIRCRAFT							
		Basics	Notes			Basics	Notes
Dice - "LARGE ORANGE"	 (LgO) Large	Large	-Long-range strike bomber. (B-24)	 (LgO) Large	Large	 (LgO) Large	Large
Capability Represented	LR Bomber Sqdn		-Cannot fight air-to-air.	LR Bombers - Vulcan		LR Bombers - B-52	
Expendable / Reusable	Reusable		-Can attack land only. No maritime targets.	Reusable		Reusable	
Day / Night / Both	Both			Both		Both	
Re-Roll for Damage?	No		-Cannot be engaged by R, but can be	No		No	
Air Re-fuelable (AC:Tank)	No		by B en route if the flight path goes	Yes (1:1)		Yes (2:1)	
Carrier / Land / Both	Land Only		thru enemy CAP or there is CAP over	Land Only		Land Only	
Range	70"		the intended target.	70"		70"	
Can be Paired with	Nothing			Nothing		2xBlk, 2xLgBlk, 3xO	
		Basics	Notes			Basics	Notes
Dice - "WHITE"	 (W)			 (W) Small	Small	 (W) Small	Small
Capability Represented	Reference "SPECIAL" banner below			Helicopter Sqdn		Helo / Osprey	
Expendable / Reusable		Reusable		Reusable		Reusable	
Day / Night / Both		Both		Both		Both	
Re-Roll for Damage?		No		No		No	
Air Re-fuelable (AC:Tank)		No		No		No	
Carrier / Land / Both		Both		Both		Both	
Range		12"		12"		12"	
Can be Paired with				1x Ground die		1x Ground dice	
		Basics	Notes			Basics	Notes
Dice - "LARGE WHITE"	 (LgW)			 (LgW)		 (LgW) Large	Large
Capability Represented	Does not exist.					AWACS	-Auto-detect on all non-5th Gen enemy aircraft and missiles.
Expendable / Reusable						Reusable	-Never in EMCOM.
Day / Night / Both						Both	-Is auto-detected by SAT ISR assets.
Re-Roll for Damage?						No	-When escorted by B or LgB, can
Air Re-fuelable (AC:Tank)						Yes	immediately strike found targets.
Carrier / Land / Both						Land Only	-Its escorts die before it dies.
Range						120"	-An AAR tanker traveling with it adds 70" to its range.
Can be Paired with						Nothing	

Leyte / United States			Falklands / United Kingdom			Borneo / United States						
SPECIAL												
Dice Color - "WHITE"	Basics		Notes		Basics		Notes		Basics		Notes	
	(W) Small		-C2 element required to call in		(W)				(W)			
	Leadership		Maritime reinforcements for BR#2. If		Becomes Helicopters. Reference "AIRCRAFT" banner above				Becomes Helicopters. Reference "AIRCRAFT" banner above			
	Expendable / Reusable		no C2 is present in the engaged force,									
	Both		they can only get air support for									
Day / Night / Both		BR#2, not support from touching										
		cups.										
		-Required to disaggregate a force.										

MISSILES												
Dice - "BLACK"	Basics		Notes		Basics		Notes		Basics		Notes	
	(Blk)				(Blk) Small		-Fired during BR#0 only.		(Blk) Small		-Change 1: Greater range.	
	Capability Represented				Anti-Ship Cruise Msl		-Only targets maritime surface		Anti-Ship Cruise Msl		-Change 2: Can now also be fired	
	Expendable / Reusable				Expendable		vessels.		Expendable		from specific aircraft.	
	Day / Night / Both				Both		-Can be shot down by R or		Both		-Change 3: Can now be intercepted	
	Re-Roll for Damage?				Yes		intercepted by Gr.		Yes		by fighter aircraft.	
	Max Range				1"		-Cannot be intercepted by aircraft.		5"			
	Firing Platform Domain				Sea				Sea, Air			
	Can be Paired with				Nothing				LgY, LgO, LgB, B			
	Maneuverable				No				No			
Dice - "LARGE BLACK"	Basics		Notes		Basics		Notes		Basics		Notes	
	(LgBlk)				(LgBlk)				(LgBlk) Large		-Much the same characteristics as the	
	Capability Represented								LRASM		Blk; exceptions: twice the range and	
	Expendable / Reusable								Expendable		cannot be paired with the LgY.	
	Day / Night / Both								Both			
	Re-Roll for Damage?								Yes			
	Max Range								10"			
	Firing Platform Domain								Sea, Air			
	Can be Paired with								B, LgB, LgO			
	Maneuverable								No			
Dice - "GRAY"	Basics		Notes		Basics		Notes		Basics		Notes	
	(Gr)				(Gr) Small		-Cannot be fired in EMCON.		(Gr) Small		-Change 1: Range increase.	
	Capability Represented				Dual-Role Missile		-Used in defensive mode in BR#0 to		Dual-Role Missile		-Change 2: Can engage O, Blk, and	
	Expendable / Reusable				Expendable		intercept Blk (1 for 1).		Expendable		LgBlk in BR#0 in defensive mode but	
	Day / Night / Both				Both		-Used in offensive mode in BR#1&2 to		Both		cannot engage LgPk. Still limited to no	
	Re-Roll for Damage?				Off-Y; Def- N		engage aircraft or surface vessels. No		Off-Y; Def- N		more than 1-for-1 in incoming	
	Max Range				1"		more than 1-for-1 against aircraft		5"		missiles and aircraft.	
	Firing Platform Domain				Sea		(rolled separately from those going		Sea		-Still cannot be intercepted once	
	Can be Paired with				Nothing		against surface vessels)		Nothing		fired.	
	Maneuverable				No		-Cannot be intercepted.		No			

Leyte / United States		Falklands / United Kingdom		Borneo / United States		
MISSILES						
Dice - "ORANGE"	 (O)		 (O)		 (O) Small	
Capability Represented	Does not exist.		Does not exist.		TLAM "Tomahawk"	*Exception to the re-roll for damage is that it only happens on a 6 when attacking a Land Facility.
Expendable / Reusable					Expendable	-As the "LA" in "TLAM" suggests, only good against targets on land.
Day / Night / Both					Both	-Maneuverable means you can choose waypoints for it to follow along its 50" of travel to the target.
Re-Roll for Damage?					No*	
Max Range					50"	
Firing Platform Domain					Sea, Sub, Air	
Can be Paired with					LgO	
Maneuverable					Yes	
Dice - "PINK"	 (Pk)		 (Pk)		 (Pk) Small	
Capability Represented	Does not exist.		Does not exist.		Dual-Role Missile	-Dual-Role Missile similar to the Gr. Only difference is that the Pk is the ONLY missile that can intercept LgPk in flight--can fire up to one per incoming LgPk.
Expendable / Reusable					Expendable	-It cannot intercept a Gr or LgGr.
Day / Night / Both					Both	-Cannot be fired in EMCON.
Re-Roll for Damage?					Off-Y; Def- N*	
Max Range					5"	
Firing Platform Domain					Sea	
Can be Paired with					Nothing	
Maneuverable					No	
Dice - "CLEAR ORANGE"	 (CO)		 (CO)		 (CO) Small	
Capability Represented	Does not exist.		Does not exist.		Maritime Strk Tmwk	-This is a land-based, anti-ship cruise missile targeting only surfaced vessels.
Expendable / Reusable					Expendable	-Maneuverable means you can pick waypoints for it to follow along its 50" of travel to the target.
Day / Night / Both					Both	-Can be shot down by R, fighter aircraft, and interceptor missiles.
Re-Roll for Damage?					Yes	
Max Range					50"	
Firing Platform Domain					Land	
Can be Paired with					Nothing	
Maneuverable					Yes	
Dice - "TEAL"	 (T)		 (T)		 (T) Small	
Capability Represented	Does not exist.		Does not exist.		Patriot Missile	-This is a land-based anti-aircraft and missile interceptor missile. There is no defense against it and it cannot be intercepted.
Expendable / Reusable					Expendable	
Day / Night / Both					Both	
Re-Roll for Damage?					No	
Max Range					5"	
Firing Platform Domain					Land	
Can be Paired with					Nothing	
Maneuverable					No	

Red Team Dice

Red Team

Red Team	Leyte / Japan	Falklands / Argentina	Borneo / Kalamantan			
MARITIME SURFACE						
Dice - "GREEN"	 (G) Small	Cannot engage aircraft in flight.	 (G) Small	No change.	 (G) Small	No change.
Capability Represented	Naval Gun Fire	-Considered to have a 2" range when engaging targets on land.	Naval Gun Fire		Naval Gun Fire	
Expendable / Reusable	Reusable		Reusable		Reusable	
Day / Night / Both	Both	*One dice can be rolled against a surfaced sub in BR#1.	Both		Both	
Targets	Surface forces*		Surface forces		Surface forces	
Re-Roll for Damage?	No		No		No	

Dice - "RED"	 (R) Small	-Damage Point Hits can only go towards aircraft in flight.	 (R) Small	-Addition 1: Can now engage Blk ASCM. -Addition 2: Cannot shoot down Gr missiles.	 (R) Small	-Addition 1: Can now engage LgBlk and O missiles in addition to Blk. -Addition 2: Cannot shoot down Gr, T, or Pk missiles.
Capability Represented	Anti-Air		Anti-Air		Anti-Air	
Expendable / Reusable	Reusable		Reusable		Reusable	
Day / Night / Both	Both		Both		Both	
Targets	Aircraft in flight		A/C & ASCM in flight		A/C & MsIs in flight	
Re-Roll for Damage?	No		No		No	

Dice - "PURPLE"	 (P) Small	-Gives a cup the ability to Trip-Over a submerged submarine. -Each models a spread of depth charges. -Can be used against submerged and surfaced subs.	 (P) Small	No change.	 (P) Small	No change.
Capability Represented	Anti-Sub		Anti-Sub		Anti-Sub	
Expendable / Reusable	Reusable		Reusable		Reusable	
Day / Night / Both	Both		Both		Both	
Targets	Submarines		Submarines		Submarines	
Re-Roll for Damage?	No		No		Yes	

SUBMARINES						
Dice - "GREEN"	 (G) Small	*Submarines can attack each other when one or both opponents are surfaced.	 (G) Small	-Change 1: Torpedoes are now expendable. -Addition 1: Can rearm subs by going to any port on the mainland for a TURN.	 (G) Small	-Change 1: Two opposing submerged forces can now fight, but only with G.
Capability Represented	Torpedo spread		Torpedo spread		Torpedo	
Expendable / Reusable	No		Yes		Yes	
Day / Night / Both	Both		Both		Both	
Targets	Maritime Forces*		Maritime Forces		Maritime Forces	
Re-Roll for Damage?	Yes		Yes		Yes	

Red Team

Red Team

Leyte / Japan

Falklands / Argentina

Borneo / Kalamantan

GROUND FORCES

	Basics	Notes	Basics	Notes	Basics	Notes
Dice - "RED"	🔴 (R) Small	-If embarked on a ship, cannot be used to defend the ship. The ship has its own R for that.	🔴 (R) Small	-Change 1: Distance of ground movement is cut in half due to a lack of vehicles.	🔴 (R) Small	-Change 1: Can move 2" per TURN again as long as they stay on the TB
Capability Represented	Anti-air Battalion		Anti-air Battalion		Anti-air Battalion	Hwy. 1" per TURN when off of it.
Expendable / Reusable	Reusable	-Can only target aircraft in flight. No "ground mode" usage.	Reusable		Reusable	-Change 2: Can target any aircraft in flight.
Day / Night / Both	Both		Both		Both	
Ground Movement	2" per TURN		1" per TURN		2" per TURN*	
Attack Range	1.5" Radius	-Cannot range B-24 bombers due to their high altitude.	1.5" Radius		1.5" Radius	
Re-Roll for Damage?	No		No		No	

	Basics	Notes	Basics	Notes	Basics	Notes
Dice - "BROWN"	🟤 (Br) Small	-Can attack other ground dice within 1".	🟤 (Br) Small	-Change 1: Distance of ground movement is cut in half due to a lack of vehicles.	🟤 (Br) Small	-Change 1: Can move 2" per TURN again as long as they stay on the TB
Capability Represented	Infantry Regiment		Infantry Regiment		Infantry Regiment	Hwy. 1" per TURN when off of it.
Expendable / Reusable	Reusable	-Can target parked aircraft.	Reusable		Reusable	
Day / Night / Both	Both	-Must be within 1" of a Green beach to oppose an enemy landing occurring there.	Both		Both	
Ground Movement	2" per TURN		1" per TURN		2" per TURN	
Attack Range	1" Radius		1"		1"	
Re-Roll for Damage?	No		No		No	

	Basics	Notes	Basics	Notes	Basics	Notes
Dice - "GREEN"	🟢 (G) Small	-Can attack other ground dice within 2".	🟢 (G) Small	No change. Artillery was not lacking vehicles since they cannot move without them.	🟢 (G) Small	-Change 1: 2" per TURN along the TB
Capability Represented	Artillery Brigade		Artillery Battalion		Artillery Battalion	Hwy. 1" per TURN when off of it.
Expendable / Reusable	Reusable	-If within 1" of shoreline, can attack surface ships within 1" of shoreline.	Reusable		Reusable	
Day / Night / Both	Both		Both		Both	
Ground Movement	2" per TURN	-Must be within 2" of a Green beach to oppose a landing.	2" per TURN		2" per TURN	
Attack Range	2" Radius		2" Radius		2" Radius	
Re-Roll for Damage?	No		No		No	

Red Team	Leyte / Japan	Falklands / Argentina	Borneo / Kalamantan																																																																																	
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Red Team

Leyte / Japan

Falklands / Argentina

Borneo / Kalamantan

AIRCRAFT

Dice - "LARGE BLUE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	Basics (LgB) Does not exist.	Notes	Basics (LgB) Large Fighter Sqdn Reusable Both No Yes *See notes -> 40" 1xBlk (Super E only)	Notes -Models Super Étendard & Skyhawk. -Only Étendards can be paired with Exocet missiles. *-Only Skyhawks can be launched from a CV. -All other capabilities and restrictions of Small BLUE apply.	Basics (LgB) Large 5th Gen (4-6) Reusable Both Target Dependent* Yes (3:1) Both 40" 1xBlk or 1xLgBlk	Notes -Enemy ISR suffers -2 penalty when trying to detect them. -Can only AAR from Tanker (3:1), no Buddy Tanking. -Same CAP rules as B. -Can engage Blk, LgBlk, and O missiles.
Dice - "ORANGE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	Basics (O) Small LR Bomber Sqdn Reusable Bomb-Day/ISR-Both No No Land Only 70" Nothing	Notes -Twin-engine strike bomber. -Cannot win an air-to-air fight. -Can attack anything on the surface; land or maritime.	Basics (O) Small CAS Fighter Sqdn Reusable Day Only No No Land Only 20" Nothing	Notes -Change 1: No longer a bomber, now a small Close Air Support aircraft. -Change 2: Reduced Range. -Change 3: Cannot operate at night. -Still no air-air combat capability against fixed wing aircraft. -Addition OAF Doc info available.	Basics (O) Becomes Land Attack Cruise Missile during Borneo. Reference "MISSILES" banner below.	Notes
Dice - "LARGE ORANGE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	Basics (LgO) Does not exist.	Notes	Basics (LgO) Large Long-Range Bomber Reusable Both No No Land Only 120" Nothing	Notes -Can bomb both land and maritime targets. -Can conduct armed ISR. -The only Large dice that cannot conduct air-to-air refuel.	Basics (LgO) Large Bomber-H6K Badger Reusable Both No No Land Only 70" 1x Blk, 1xLgBlk, 1xO	Notes -Change 1: Reduced Range. -Change 2: Can either drop bombs on land, be paired with a single LgBlk (for maritime targets), or a single O for standoff against land targets. No mixed munition loads.

Red Team

Leyte / Japan

Falklands / Argentina

Borneo / Kalamantan

AIRCRAFT

Dice - "BLACK" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	(Blk) Small Kamikaze Sqdn Expendable Land Only No No Both 40" Nothing	-Can attack only surface ships. Not air-to-air combat capable. -Target area defenses are calculated into hit probability. -CAP station en route to its target can intercept. -Can only be flown to other friendly airfields or to attack. -Can only fly back from an "Attack in the Blind" if no enemy was present.	(Blk) Becomes ASCM during Falklands. Reference "MISSILES" banner below	(Blk) Reference "MISSILES" banner below
Dice - "WHITE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	(W) Reference "SPECIAL" banner below.	-Can carry 1x ground die across land or SOF ship to shore. -Getting shot down over land does not automatically kill ground dice it carries, but it does leave them at the point of the downed helo. Getting shot down over water does kill the forces onboard. -If landing at its full range, must have fuel there to continue from there on the next TURN.	(W) Does not exist.	(W) Small Helicopters Reusable Both No No Land Only 12" 1x Ground die
Dice - "LARGE WHITE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Air Re-fuelable Carrier / Land / Both Range Can be Paired with	(LgW) Does not exist.	-Provides auto-detect on all missiles and aircraft except 5th Gen Fighters. -Never in EMCOM. -Auto-detected by SAT ISR assets. -When escorted by a fighter, can immediately strike detected targets. -Its escorts die before it does.	(LgW) Does not exist.	(LgW) Large AWACS - SAC Y-9 Reusable Both No No Land Only 120" Nothing

Red Team

Leyte / Japan

Falklands / Argentina

Borneo / Kalamantan

SPECIAL

	Basics	Notes	Basics	Notes	Basics	Notes
Dice - "WHITE"	(W)	Small	(W)		(W)	
Capability Represented	Leadership	-C2 element required to call in Maritime reinforcements for BR#2. If no C2 is present in the engaged force, they can only get air support for BR#2, not support from touching cups.	Becomes Helicopters. Reference "AIRCRAFT" banner above.		Becomes Helicopters. Reference "AIRCRAFT" banner above.	
Expendable / Reusable	Reusable					
Day / Night / Both	Both					
		-Required to disaggregate a force.				

MISSILES

	Basics	Notes	Basics	Notes	Basics	Notes
Dice - "BLACK"	(Blk)	Kamikaze Aircraft. Reference "AIRCRAFT" banner above	(Blk)	Small	(Blk)	Small
Capability Represented	Anti-Ship Cruise M.					
Expendable / Reusable	Expendable		-Fired during BR#0 only.	-When delivered by LgB, aircraft RTB after ASCM launch.	Anti-Ship Cruise M.	-Change 1: Greater range.
Day / Night / Both	Both		-Can only attack maritime surface vessels.	-Can be shot down by R or intercepted by Gr or aircraft.	Expendable	-Change 2: Can now also be fired from three specific aircraft and some subs.
Re-Roll for Damage?	Yes		-Maritime version can be replenished at any mainland port.		Both	-Change 3: Can be intercepted by some enemy aircraft.
Max Range	1"				Yes	-Change 4: Can be intercepted by Pk in addition to Gr.
Firing Platform Domain	Sea, Land, Air				5"	
Can be Paired with	LgB (Super E)				Sea, Sub, Land, Air	
Maneuverable	No				B, LgY, LgB	
Target Domain	Sea				No	
				Sea		

	Basics	Notes	Basics	Notes	Basics	Notes	
Dice - "LARGE BLACK"	(LgBlk)	Does not exist.	(LgBlk)	Does not exist.	(LgBlk)	Large	
Capability Represented						LR ASCM / CDCM	-Long-range cruise missile that can be fired from land, some subs, surface forces, and paired with specific air assets.
Expendable / Reusable						Expendable	
Day / Night / Both						Both	-Can be intercepted by some enemy aircraft.
Re-Roll for Damage?						Yes	-Can be intercepted by Gr and Pk.
Max Range						10"	
Firing Platform Domain						Sea, Sub, Land, Air	
Can be Paired with						B, LgO, LgB	
Maneuverable				No			

Red Team	Leyte / Japan		Falklands / Argentina		Borneo / Kalamantan	
MISSILES						
Dice - "GRAY" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Max Range Firing Platform Domain Can be Paired with Maneuverable	 (Gr)	Does not exist.	 (Gr) Small Dual-Role Missile Expendable Both Off-Yes; Def- No* 1" Sea Nothing No	-Only in BR#0 is it fired in defense against incoming ASCMs: 1-for-1. -Only in BR#1&2 is it fired in offense role against surfaced maritime forces or aircraft. -There is no defense against it. -It cannot be fired while in EMCON. -Expend missiles can be replenished at any mainland port.	 (Gr) Small Dual-Role Missile Expendable Both Off-Yes; Def- No* 5" Sea, Land Nothing No	-Change 1: Greater range. -Change 2: In BR#0 can now target Blk, LgBlk, and O. Still cannot target other Gr dice or the new Pk. -Change 3: Also now models a missile and aircraft interceptor on land.
Dice - "ORANGE" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Max Range Firing Platform Domain Can be Paired with Maneuverable	 (O)	Small Bomber during Leyte. Reference "AIRCRAFT" banner above	 (O)	Small Fighter during Falklands. Reference "AIRCRAFT" banner above	 (O) Small Land Atk Crz Msl Expendable Both No* 50" Land, Sea, Sub, Air LgO Yes	-As the name suggests, only good against targets on land.
Dice - "LARGE PINK" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Max Range Firing Platform Domain Can be Paired with Maneuverable	 (LgPk)	Does not exist.	 (LgPk)	Does not exist.	 (LgPk) Large ASBM - DF-21D Expendable Both Yes 50" Land Nothing No	-Can only be intercepted by T or Pk. -Follows a ballistic trajectory, not steerable.
Dice - "LARGE TEAL" Capability Represented Expendable / Reusable Day / Night / Both Re-Roll for Damage? Max Range Firing Platform Domain Can be Paired with Maneuverable	 (LgT)	Does not exist.	 (LgT)	Does not exist.	 (LgT) Large Land Atk Msl DF-21C Expendable Both No* 50" Land Nothing No	-Can only be intercepted by T or Pk. -Follows a ballistic trajectory, not steerable. -Only good against facilities.

APPENDIX B

Leyte Gulf Damage Tables

American Damage Table vs. Japanese Capabilities:

		Dice Roll					
US Attack Type	Versus Japanese	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine)	Maritime*	-	-	-	-	1d6	1d6
 (R)	 (B)  (O)  (Blk)**	-	-	-	1	1	1
 (P)	Submarine	-	-	-	-	1	1
 (B) CAP	 (Blk) en route**	-	-	-	-	1	1
 (B)	Maritime  (B)  (O)	-	-	-	-	1	1
 (B)  (LgO)  (Br)	Land	-	-	-	-	-	1

Japanese Damage Table vs. US Capabilities:

		Dice Roll					
Japan Attack Type	Versus US	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine)	Maritime*	-	-	-	-	1d6	1d6
 (R)	 (B)	-	-	-	-	1	1
 (P)	Submarine	-	-	-	-	1	1
 (Blk)	Maritime	-	-	-	1	1d6	1d6
 (B)	 (B)  (O)	-	-	-	-	1	1
 (B)  (O)	Maritime	-	-	-	-	1	1
 (B)  (O)  (Br)	Land	-	-	-	-	-	1

*A surfaced sub is vulnerable to 1BR as a Maritime Target by a submerged sub.

**Rule 192:  can only be shot down by  CAP or  if it crosses thru those defenses while en route to its intended target.

Falklands / Malvinas Damage Tables

Note:  (Submarine) /  /  are **EXPENDABLE** and removed from play upon employment.

Subs may launch only 2x  per Battle Round.

UK SSNs have a decision advantage due to stealth & speed. (Ref Rule 164:, Rule 165:, Rule 166:)

British Damage Table vs. Argentinian Capabilities:

		Dice Roll					
UK Attack Type	Versus Argentina	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land*	-	-	-	-	1	2
 (G) (Submarine) (2/BR)	Maritime**	-	-	-	1d6	1d6	1d6
 (R)  (Gr)	 (Blk)	-	-	-	-	-	1
 (R)  (Gr)  (B)	 (B)  (LgB)  (LgO)  (Y)	-	-	-	1	1	1
 (R)  (Gr)  (B)	 (O)	-	-	1	1	1	1
 (Blk)  (Gr)	Maritime	-	-	-	1d6	1d6	1d6
 (P)	Submarine	-	-	-	-	1	1
 (B)	Maritime	-	-	-	1	1	1d6
 (B)	Land*	-	-	-	-	1	1
 (LgO)	Land*	-	-	-	-	1	1
 (Br)	Land*	-	-	-	1	1	1

Argentinian Damage Table vs. British Capabilities:

		Dice Roll					
Argentina Attack Type	Versus UK	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine) (2/BR)	Maritime**	-	-	-	-	1d6	1d6
 (R)  (Gr)	 (Blk)	-	-	-	-	-	1
 (R)  (Gr)  (B)  (LgB)	 (B)  (LgO)  (Y)  (W)	-	-	-	-	1	1
 (Blk)  (Gr)	Maritime	-	-	-	1d6	1d6	1d6
 (P)	Submarine	-	-	-	-	-	1
 (B)  (LgB)  (LgO)	Maritime	-	-	-	-	1	1d6
 (O)	Maritime	-	-	-	-	-	1
 (B)  (LgB)  (LgO)  (O) 	Land	-	-	-	-	-	1

*Hits against forces at Port Stanley can be applied against Logistics Points (-3 LP/damage point)

**A surfaced sub is vulnerable to 1BR as a Maritime Target by a submerged sub.

Borneo Damage Tables

JTF Damage Table vs. Kalimantan Capabilities:

 US Attack Type		Dice Roll					
		1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	-	-	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine) (2/BR)	Maritime	-	-	-	1d6	1d6	1d6
 (G) (Submarine) (2/BR)	Submarine	-	-	-	-	1d6	1d6
 (R)  (Gr)  (Pk)  (B)  (LgB)  (T)	 (O)	-	-	-	-	-	1
 (Gr)  (Pk)  (LgB)  (T)	 (Blk)  (LgBlk)  (B)  (LgO)  (Y)  (LgY)  (W)	-	-	-	1	1	1
 (R)	 (Blk)  (LgBlk)  (B)  (LgO)  (Y)  (LgY)  (W)	-	-	-	-	1	1
 (B)	 (Blk)  (LgBlk)  (B)	-	-	-	-	1	1
 (B)	 (LgO)  (Y)  (LgY)  (W)	-	-	-	1	1	1
 (B)  (LgB)	Maritime	-	-	-	1d6	1d6	1d6
 (B)  (LgB)  (O)	Land	-	-	-	-	1	1
 (Blk)  (LgBlk)  (Gr)  (Pk)  (CO)	Maritime	-	-	-	1d6	1d6	1d6
 (B)  (LgB)  (O)	Facilities (No 1-1-1 Rule)	-	1	1	1	1	1d6
 (LgO)	Land	-	-	2	2	3	3
 (P)	Submarine	-	-	-	-	-	1d6
 (Br)	Land	-	-	-	-	-	1
 (Pk)  (T)	 (LgPk)  (LgT)	-	-	-	1	1	1

Kalimantan Damage Table vs. JTF Capabilities:

		Dice Roll					
KAL Attack Type	Versus American	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	-	-	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine) (2/BR)	Maritime	-	-	-	1d6	1d6	1d6
 (G) (Submarine) (2/BR)	Submarine	-	-	-	-	1d6	1d6
 (Gr)  (B)	 (Blk)  (LgBlk)  (CO)  (B)	-	-	-	-	1	1
 (R)	 (Blk)  (LgBlk)  (CO)  (B)  (Y)  (LgY)  (W)	-	-	-	-	1	1
 (Gr)  (B)	 (LgO)  (Y)  (LgY)  (W)	-	-	-	1	1	1
 (R)  (Gr)  (B)	 (LgB)  (O)	-	-	-	-	-	1
 (Blk)  (LgBlk)  (Gr)  (LgPk)	Maritime	-	-	-	1d6	1d6	1d6
 (P)	Submarine	-	-	-	-	-	1d6
 (B)	Maritime	-	-	-	-	1d6	1d6
 (O)  (B)	Land	-	-	-	-	1	1
 (O)  (LgT)  (B)	Facilities (No 1-1-1 Rule)	-	1	1	1	1	1d6
 (Br)  (LgO)	Land	-	-	-	-	-	1

APPENDIX C

Quick Start Guide – Leyte

This Appendix is a consolidated summary of the basic ruleset with references to the expanded explanations available in the full instruction book.

Primary Roles. (Page 9) Moderators divide their seminar into two Teams for the War Game, (Japan vs. US) using the Naval War College’s case study of the Battle of Leyte Gulf as its foundation. Moderators will assign each Team a “Team Captain,” also referred to here as the “Operational Commander.” Other players act as Tactical Commanders or staff elements within the Team.

- **Operational Commander (Student):** Leader of the Team, responsible for developing the Operational Idea (Op Idea) and ensuring that their teammates operate the Team’s forces in accordance with that Op Idea. Make decisions, set priorities.
- **Tactical Commanders (Student):** Teams can allocate these as needed. In the past, some have designated Tactical Commanders for land forces, air forces, surface forces and sub forces, while others have had just a ground and a maritime commander. These individuals can be responsible for moving their forces, deciding to engage enemy forces, and/or fight at the Battle Box. (More on that, later)
- **Moderator (Faculty):** The Moderator observes Team actions and decision making while interacting with players, ensuring the achievement of learning outcomes. Following the game, Moderators lead the all-important debrief.
- **Umpire (Faculty):** The Umpire adjudicates combat at the Battle Box, ensuring the proper conduct of the game’s combat mechanics. The role may be fulfilled by a Moderator. Dice rollers take all commands from the Umpire.
- **Facilitator (Faculty):** The Facilitator “runs” the game as an impartial arbiter like the umpire. It is through the Facilitator that Teams interact with the game and with their opponent. Players will inform the Facilitator of their intentions and the Facilitator will report the outcomes of those actions back to the Teams. It is important to have a dialogue with your Facilitator if you have an especially complex Move or plan at the start of a TURN.

Game Components: *War at Sea* is not automated and uses physical components to support game play, including:

- Cups: Model a force by containing a specific group of capabilities (dice).
- Planning Cards: Plexiglass rectangles upon which Teams draw to convey their Movement, ISR, and Fires Plans to the Facilitator.
- Tiles: Model a force or used to maintain situational awareness.
- Black and Blue **thin** markers: Used to draw on the **planning cards**.
- Black and Blue **thick** markers: Used to draw on the **acetate covering the Game Map**.
- 6” and 9” Brass rings: Convey an ISR Search area on the Game Map.
- Red Ruler with two holes: For converting drawings and rings on the Game Map onto the planning cards.
- Different colors and sizes of dice (Appendix A) model different operational **capabilities**:



Dice	Capability Modeled	Explanation
 (G)	Naval Guns / Torpedoes / Artillery	In Surface Force Cups these are Naval Guns, in Submarines - Torpedoes, on land - Artillery.
 (R)	Anti-Air Defense	Air defense for both land and maritime forces.
 (Br)	Ground Units	Models the capabilities of infantry forces on land.
 (P)	Anti-Submarine Warfare (ASW)	Gives a Cup the ability to detect (sonar) and strike (depth charges) a submarine – submerged or not.
 (Y)	Airborne ISR	Recon aircraft that have no offensive or defensive capability.
 (B)	Combat Air	Fighter aircraft that can conduct air combat, strike, CAS, ISR, and CAP.
 (O) and  Lg (LgO)	Long-range Bombers	Bombers with no air-to-air combat capability.
 (Blk)	Kamikaze	Highly lethal Proto-guided aircraft.
 (W)	Command & Control	Required to bring in maritime reinforcements during BR#2. Needed in order to split off a smaller formation from a maritime Cup.

Days leading up to Game Day – Student Preparation:

1. Read this Appendix, be familiar with the rest of the instruction book.
2. Once moderators break you into Team US and Team Japan, Team Captains will hand out the Objectives and Forces (OAF) Document to read and understand. Do not let your opponent see them.
3. Watch instructional videos.
4. Teams will complete a Commander's Estimate of the Situation (CES). This will assist in developing a C2 structure for the Team as well as an Operational Idea. Some ideas for staff/commanders are:
 - a. Intel Officer (Fills out ISR Card)
 - b. Movements Coordinator (Fills out Move Card)
 - c. Targeting Officer (Fills out Fires Card)
5. Develop an Operational Idea consisting of:
 - a. A narrative that reflects COG analysis, operational functions, and the tenets of operational art.
 - b. A supporting sketch with major force movements.
6. Conduct preliminary planning for the first TURN of Game Day.

Game Day:

The remainder of this Appendix steps the reader through a "TURN." A full game typically lasts anywhere from 4-7 TURNS (alternating between Day and Night) and the Facilitators control the pace of each one. The sequence within each TURN is:

1. **Plan**
 2. **Roll for Initiative**
 3. **Move (Conduct Movement and Maneuver)**
 4. **Look (ISR Report from Facilitator)**
 5. **Shoot (Conduct Fires)**
 6. **Close**
1. **Plan.** *This is a 12-minute period at the start of every TURN, when each Team draws out where*



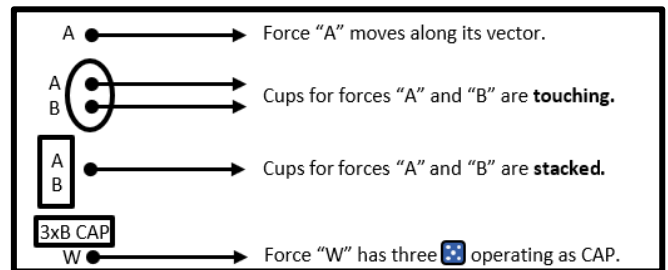
You may have noticed that the sequence to the left has you doing the Move Step before the ISR Step, which is clearly against the ideal situation of looking before you move. Fear not. The ISR step is intended to prepare you for your Move in the next TURN, and the Fires Step of the current one. All of your planning set conditions for your Turn 1 move.

they want their forces to move and where they want their aerial recon assets to observe.

During planning, Teams plan the Movement/Maneuver and ISR at their Game Map. They then transpose that plan onto their Move and ISR Cards. Accuracy is important, but so is time. You will have 12 minutes to plan and draw all your forces (moving and stationary) onto the first two Cards.

A. Move Planning and the Move Card: (Page 17)

Each Team transposes its Movement and Maneuver plan from the Game Map to the Move Card. For now, players leave the actual forces at the start of their path on the Map. They will later be able to move them once directed by the Facilitator at the Movement brief. Players draw on the Card with a common set of symbols. →



- Surface Force Movements: (Page 17) Different maritime forces (Cups) move at different speeds, modeled through the number of inches they can travel in a single TURN. FAST= up to 15"; SLOW= up to 8"; CREEP= up to 3". Among a host of other pertinent information, each Team's Objectives and Forces Document (OAF Doc) provides the speed of each force.
- Submarine Movements: (Page 18) Subs move up to 8" per TURN when surfaced and up to 3" per TURN when submerged. When Subs run on the surface, anything can detect them, but it takes a Cup possessing a [blue die with 4 dots] to Trip-Over and detect a submerged sub, since [blue die with 4 dots] models an ASW capability.
- Aircraft Movements: (Page 19) Aircraft dice movements are governed by the total range their gas will allow, (i.e., [blue die with 4 dots] has 40" of total range, meaning you can send it on a mission 20" out and back, or fly it to another place to land 40" away.) Aircraft movements (not strikes) from one airfield or carrier to another are planned on the Move Card. Aircraft strikes are planned on the Fires Card.
- Ground force Movements: (Page 21) All ([green die with 6 dots], [red die with 6 dots], [yellow die with 6 dots]) move across land at 2" per TURN. Greater detail on the movements, limitations, and constraints of each dice color is listed in Appendix A.
- Maritime transport of Ground forces: (Page 54) From the beginning, Team Japan has the option to move ground forces from one **friendly green beach** to another with maritime troop transports. Since loading or unloading troops and equipment (up to 12x dice at a time) takes "half a TURN," the transports' movement distance is halved to make up for either the loading or unloading activity. If the **beach is unfriendly** (i.e. the US is trying to land on a Japanese held beach) then this is an Amphibious Assault discussed below during the Fires Step vice the Move Step.

Aircraft note: Your dice that model aircraft can only perform one of four tasks per TURN. *For example, if you choose to put a [blue die with 4 dots] on CAP it cannot then be used in a strike mission during that TURN's Fires Phase.*

An individual aircraft die can only do one task per TURN. Choose wisely.

ISR	CAP	Strike (Only one Strike per die per TURN)	Move
-----	-----	--	------

Combat Air Patrols (CAP): (Page 20) Deciding which of your [blue die with 4 dots] are flying CAP is part of your Move plan, (Function Protection). The chart above shows how to convey a CAP station on the Move Card. To do so on the Game Map, Teams place their CAP [blue die with 4 dots] on top of the Cup they are meant to protect – it must be within 10" of their point of origin carrier. To show CAP on land, place [blue die with 4 dots] on a Yellow Tile on the airfield you have tasked them to over-watch. If CAP is a [blue die with 4 dots]'s mission for a TURN, that same [blue die with 4 dots] cannot be used for a Deliberate Strike until the next one, but they can engage any enemy coming to within 10" of their charge. Remember, CAP can only engage in 2x BRs per turn, so you must prioritize when they are committed to fight.

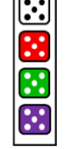
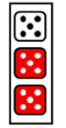
Aggregating and Disaggregating (Page 37) This is the final piece of Move planning discussed here.

Aggregating forces means you group Cups so that they are considered moving together. Teams can do this by stacking Cups or having them touch.

When you **Disaggregate a force**, you remove a specific set of dice, creating a new force, free to move independently around the Map at up to 15" per TURN.

Remember, Cups do not model a single ship, but a large grouping of them. This modeling is what allows you to

break off a smaller force, but you must have a  to do so and you must fulfill the specific minimums shown in the table on the right. A disaggregated force that losses its  must begin moving back to its parent Cup at top speed, without engaging in offensive combat en route.

Carrier Capability	Battleship & Cruiser Capability	Destroyer Capability	Amphib Capability
CVE CVL CV 	All Battleships, All Cruisers 	DE, DD 	DE, DD 
Three possible carrier disaggregated groupings.	One complete capability grouping.	One complete capability grouping.	One complete capability grouping.

B. ISR Planning and the ISR Card:

There is an overlap where ISR and Movement planning converge, **Organic ISR**. (Page 25) Each force on a Move Card can “see” (*i.e., binoculars on the bridge*) what is around it along its planned path.

Organic ISR Ranges:

TURN	Surface Forces	All others
Night	1.5" Radius	1.5" Radius
Day	3" Radius	1.5" Radius

ISR Card: (Page 24) The heart of ISR planning and the ISR Card lies in **Airborne ISR**. This can be conducted by any aircraft capability dice except . Aerial ISR is the ’s primary role. Sending out a  or  on an ISR sortie creates an Armed Reconnaissance mission, on Page 25. The table below shows the ranges of airborne reconnaissance assets. It gives the distances of their total range, a straight out-and-back round trip, a round trip to the entry point of a 6" Search Area, and a round trip to the entry point of a 9" Search Area. Intel will be collected along the route and within the Search Areas (brass rings). Page 25 provides an example of what this can look like.

Aircraft Dice	Total Range	Night Limitations
	70"	Carrier-based cannot fly at Night but land-based can.
	40"	Carrier-based cannot fly at Night. Land-based can only move from one airfield to another but no ISR can be gathered nor fighting occur.
	70"	ISR can be done Day or Night, but it can only Bomb (land and maritime targets) during the Day. (Japan only)
 Lg	70"	Bombing Day and Night against only land targets. ISR Day only. (US Only)

C. Fires Planning and the Fires Card:

Combat occurs in two general situations – when opposing forces move/maneuver into range of one another (a Trip-Over) and choose to engage in combat, or when aircraft are purposefully sent out to strike targets (Deliberate Fires). Fires planning describes the latter as well as amphibious assaults, which typically come in later TURNS of a game

Fires Card: (Page 27-28) Not every TURN warrants one. Teams only draw/plan a Fires Card if they know (or at least have a good idea of) a **worthy** target's location. That location data typically comes from the information gathered on Trip-Overs and ISR reports. This is why most Teams want to fill out a Fires Card after completing both the Move and ISR Steps. Teams that want to strike a target with less than full-targeting quality information can conduct an **Attack in the Blind**. (Page 51) These are historically less successful.

- 2. Roll for Initiative.** *This decides which Team a Facilitator briefs first and (more importantly) when two forces see each other in a Trip-Over, which Team first gets to decide to engage or not.*

At the close of planning (Move and ISR Cards turned in) the Facilitator will announce "Roll for initiative." Each Team rolls a single die of any color on their own Game Map and announces its number. The Team with the highest number gets initiative for that TURN. A tie goes to the US since forces on the offensive typically have the lion's share of initiative.

3. Move (Conduct Movement and Maneuver).

Movement Report from Facilitator: (Page 40-41) After reviewing each Teams' Move Cards, (the pieces of plexiglass layered over each other) Facilitators show where US and Japanese forces paths' have either crossed or come into Organic ISR range of one another. They will report to each Team (Initiative Team first) how far each of their forces moved along their planned path. They will do this for maritime Cups, ground forces going over land or via troop transport from one friendly beach to another, and aircraft dice that are only flying from one friendly airfield or CV to another.

Trip-Over Decision: (Page 42) Friendly forces that detect an enemy force will have to choose whether to engage the enemy force, ignore it and keep moving, or stop at some other point along the planned path. In the case of a mutual detection, either force may choose to engage and force combat.

Submarines and Trip-Overs: (Page 48) An UNDETECTED submarine (whether on the surface or submerged) that chooses to attack will report to the battle box *submerged* and gets to attack during BR#1 with no return fire. This is a major advantage of submarine warfare. During BR#2 the target force can defend itself with .

A DETECTED submarine reports to the Battle Box in its current state, surfaced or submerged. A submerged Sub is only susceptible to  dice, a Surfaced sub is **also** vulnerable to 1x  from its attacker (simulating naval gunfire). A surfaced submarine that survives BR#1 automatically submerges for BR#2 and is only vulnerable to .

A surfaced submarine can be engaged by a submerged submarine for 1x Battle Round, just like any other surface combatant. The surfaced submarine is defenseless during this engagement. It is assumed that the target submarine will submerge for survival after BR#1, which ends the Sub versus Sub combat.

- 4. Look (ISR Report from Facilitator).** *This is a report from the Facilitator informing each Team of the results of their planned ISR missions. Use this intel to inform that TURN's Fires Card and the next TURN's Move Card.*

For each die along each leg and each Search area ring, the Facilitator has already rolled a chance die to determine if the Recon die (again, typically ) was shot down (17%), it saw nothing due to bad weather (33%), or it saw what enemy passed under it (50%). *Note: Even if the Facilitator rolls nothing but shoot downs for each ISR dice, a Team can only lose 1x die to this circumstance per TURN (Rule 133:).* If there is more than 1 die assigned to a mission, the Facilitator will roll for each and go with the most advantageous for ISR, increasing the chances for targetable data. *Probability of at least one of 2x  on the same mission seeing something vice only one by itself goes from 50% to 75%. 3x bumps that up to 87.5%, and so on.*

Facilitators tell each Team the results of their ISR Plan as it if were a quick intel brief: "This mission saw nothing, this one saw 50-60 combat vessels traveling from here to here before it lost contact, this one was shot down, and this one saw 20 combat ships including aircraft carriers stop right here." Teams are encouraged to

have their assigned N2 draw/write out the results on the acetate covering the Game Map in red marker as the Facilitator briefs it. Players can ask follow-up questions for clarification, which may or may not be answered.

5. Shoot (Conduct Fires). *This is where rolling capabilities dice at the Battle Box decides who lives and dies.*

Most forces can engage each other but there are some exceptions. For example, maritime forces can all engage each other, but embarked ground force dice cannot assist, they can only die. Naval and air forces can bombard a beach as preparatory fires for a landing but they can only kill half the ground force dice present per TURN. There are two different kinds of engagements, 1) those derived from Trip-Overs and decisions made during the Move Step, and 2) those derived from Deliberate Fires and intel gathered during the Move and ISR Steps.

Battle Damage Assessment (BDA): (Page 43) Once combat begins, one player (possibly the Tactical Commander for that force) from each Team brings their attacking or attacked force(s) to the Battle Box. They will have full visibility of each other's forces and it will be up to them to report back to their respective Teams on what they saw of the engaged forces' remaining composition, **(no photos allowed)**. They roll dice within their respective box lid, those landing outside of it will not count and will be re-rolled into the correct box lid.

Combat Resolution: (Page 42-44) Combat is resolved by rolling certain capabilities applicable to the situation against the other, (i.e., *A Japanese Submarine Cup would roll its  Torpedo against a US Surface Force Cup, which in turn would roll its  ASM (depth charge) dice against the Sub.*) These rolls will generate Damage Points when they land on a "hit." Each Damage Point earned allows the roller to move a capability dice from their opponent's Force Cup to their own "kill Cup," where it remains for the rest of the game. This is how you attrit each other's forces. The Fires Damage Tables for each game are in Appendix B and show you what constitutes a hit and how many Damage Points they earn; most of which are one-for-one. Screenshots of the Leyte ones are below for this explanation:

Japan Attack Type		Dice Roll					
	Versus US	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine)	Maritime*	-	-	-	-	1d6	1d6
 (R)	 (B)	-	-	-	-	1	1
 (P)	Submarine	-	-	-	-	1	1
 (Blk)	Maritime	-	-	-	1	1d6	1d6
 (B)	 (B)  (O)	-	-	-	-	1	1
 (B)  (O)	Maritime	-	-	-	-	1	1
 (B)  (O)  (Br)	Land	-	-	-	-	-	1

US Attack Type		Dice Roll					
	Versus Japanese	1	2	3	4	5	6
 (G) (Maritime)	Maritime	-	-	1	1	1	1
 (G) (Land)	Maritime	-	-	-	-	1	1
 (G) (Maritime & Land)	Land	-	-	-	-	1	2
 (G) (Submarine)	Maritime*	-	-	-	-	1d6	1d6
 (R)	 (B)  (O)  (Blk)**	-	-	-	1	1	1
 (P)	Submarine	-	-	-	-	1	1
 (B) CAP	 (Blk) en route**	-	-	-	1	1	1
 (B)	Maritime  (B)  (O)	-	-	-	-	1	1
 (B)  (LgO)  (Br)	Land	-	-	-	-	-	1

As you can see – continuing the Japanese Sub vs US Surface Force example – the US Surface Force would have to roll a 5 or a 6 with its  to get a "hit" and earn 1x Damage Point for each of those hits. The Japanese  also needs a 5 or a 6 to get a "hit," but it earns "1d6" Damage Points per "hit." 1d6 indicates that you re-roll any dice that are a "hit" and the number that comes up is how many Damage Points that "hit" earned. i.e., *If the Japanese Sub rolled 3x , and they came up 2, 5, and 6 – the 2 would be a "miss," but the 5 and 6 would get re-rolled. If those two came up 2 and 6, then the Sub would have earned 8x Damage Points, thus allowing the Sub Commander to remove 8 dice from their opponent's Surface Force Cup to their own kill cup. If there were only 7 dice in the Cup, the*

Given 5 Damage Points to inflict...



1

2

3

4

5

6

7

8

9

10

11

12



1

4

2

3

5

6

7

8

9

10

11

12



1

2

3

4

Cup

5

The "1-1-1 Rule" keeps a Team from being able to completely pinpoint and destroy a specific capability of their choosing without having inflicted a large amount of damage. **You must take nonrepetitive colors, in sets of three, unless there is only one color available.** In this example, you see that given 5 Damage Points to inflict, the rule has kept the attacker from taking all of the other's aircraft without first spreading out the damage. As seen in the third example, the only time the damage does not have to be spread out is when there is no choice. Only an empty Cup can only be taken with a Damage Point in Leyte.

remaining Damage Point would be used to take the actual Cup now that it is empty.

Allocating Damage Points: (Page 44) **There is one constraint on which dice you can choose.** The 1-1-1 Rule (Rule 137:) exists to keep the attacker from pinpointing the destruction of a single capability. You must take away an opponent's dice in sets of three non-repetitive colors unless there is only one color available. Damage Points are not "wasted" unless there is nothing left of the attacked force when Damage Points remain. The picture above is an illustrative example that more thoroughly explains how this rule works.

Battle Rounds: (Page 45-47) The two rounds of dice throwing that occur in each trip to the Battle Box are called "Battle Rounds," abbreviated throughout as BR#1 and BR#2.

BR#1: Forces that enter BR#1 are any Cups or Tiles that were Stacked on the Tripped-Over or attacked force, and any CAP within 10" of the engagement.

BR#2: Following BR#1, the two opposing players decide with the Umpire if there is to be a BR#2. They will tell the Umpire if they want to either end combat or re-attack. If re-attack is happening, they must decide if they are going to reinforce (and how much) prior to BR#2:

- **End or Continue Combat:** Both must decide to end, otherwise one wanting combat will bring the other into BR#2.
- **Reinforce:** (Table on Page 45) Any Cups that were Touching the engaged Cup/Stack can now come in if the force engaged in BR#1 has at least 1x  still alive. In addition, 2x  from any carrier or airfield within 10" of the engagement can be "scrambled" to fight in BR#2. Air support for BR#2 does not require a .

Kamikaze Attacks: (Page 50)  can only attack maritime targets. Japan rolls them in BR#1 only and then they go into the kill cup, hit or miss. **Neither** a  nor a  can stop a Hit in the target area, those defenses are accounted for with their low hit probability. The only way  is subject to air defense dice rolls is if the kamikaze transits a  CAP or overflies  en route to its intended target.

Aerial Combat:  can engage other aircraft but not vice-versa. All Aircraft in flight, except US  Lg, are susceptible to  from land and sea.

Ground Combat: (Page 54) Ground dice consist of  - Infantry,  - Artillery, and  - Air Defense and they are all useless when embarked on a ship.  and  on land can target any other ground dice and parked aircraft.  can only target aircraft in flight.  can additionally target any maritime force within 2" of it.

Deliberate Aerial Attack: (Page 50) aka Deliberate Fires, aka Strike Package. This is a grouping of aircraft dice, probably from multiple locations, converging on a single or multiple targets. Aircraft cannot go on a planned strike if they already completed another task that TURN, (move, CAP, ISR, strike).

En route to its intended target, the strike force can "Trip-Over" a different force and decide to attack it instead. Flying over enemy  or through enemy  CAP en route exposes the Strike Package to being attacked. Dice in the Strike Package can either fight the intermediate target of opportunity for 2x BR or they can continue to the original target, in which case they are subject to 1xBR of attacks without responding as they pass the intermediate defenses. The surviving aircraft of the Strike Package will not be subject to the same attacks en route back to their base. It is assumed they would find a way to go around.

Attack in the Blind: (Page 51) This is like a Deliberate Aerial Attack, only the attacker is unsure if an enemy force is at the targeted location. All the same en route rules apply.

Shore Bombardment: (Page 55-56) Naval Gun Fire can conduct an attack on land forces but must be within 1" of the shore to do so, impacting targeted land forces within 1" inland. Air support can also assist in these preparatory fires. Only half the ground forces are susceptible to being eliminated per TURN; this models the inability of indirect fires to completely destroy a dug-in force. Any  land dice within 1" of the water, and thus 2" from the origin of the shore bombardment, are capable of returning fire. Any  used during the prep fires cannot then also be used for the actual landing, (see page 55 for explanation).

	Movement Step	Deliberate Fires Step	
		BR#1	BR#2
Defending Force Ground Units	- 🟩 & 🟥 within 1" or 🟩 within 2" makes this an opposed landing.	- 🟩 & 🟩 can engage any landing ground forces. - 🟥 can engage any supporting aircraft. - 🟩 hits can also count against Maritime forces within 2" of them.	- Continue to engage landing force that can now fight back.
Defending Force Air Units	- CAP w/in range of Assault Force can cause a Trip-Over.	- Strike Package can attack Daytime landing.	- 🟩 can still attack ground forces on beach.
Assaulting Force Ground Units	- Embarked on ships at least 3" from beach. Cannot fight.	- Moving from ship-to-shore "on amphib," cannot fight back yet, but can die. - Only 12x Dice /TURN.	- Surviving 🟩 & 🟩 are now "on the beach" and can engage defending ground forces. - Surviving 🟥 can now engage enemy aircraft.
Assaulting Force Maritime Cups	-Must end the Movement Step within 3" of the desired landing beach. -Can be stacked 4 high.	- Provides Naval Gun 🟩 not used during prep fires and (if Daytime) Air Support 🟩. Declare # of each dedicated to this task.	← Same.

Amphibious Assault: (Pages 54-56) This is how to land, and defend against, Ground Forces attempting to secure an enemy-held green beach. If both forces remain, combat will re-commence during the following TURN. Surviving amphibious forces remain in contact with the beach.

6. **Close:** *Clean the slate and get ready for the next TURN.*

This step starts following the last action at the Battle Box and is largely administrative. Teams will erase all movement and ISR paths off the Game Map and Cards, recover all their ISR assets (brass rings and 🟩), place newly arriving forces for the upcoming TURN at their appointed place at an edge of the Map, and advance the TURN marker to the next numbered TURN. (Try not to erase what you **learned** from ISR)

After a few minutes, the facilitator will start a new 12-minute timer and the process starts again.

APPENDIX D

Quick-Start Guide – Falklands /Malvinas

The purpose of this Appendix is to prepare Leyte-experienced players for the increased complexity and changes of the Falklands / Malvinas (F/M) Game. All the basics of the Leyte Game remain unless otherwise noted here or in the main text's rules. What follows are the 12 most substantial modifications that players need to be aware of to help make themselves a more competent member of Team UK or Argentina. The Page numbers listed throughout point to the location of greater detail on the subject at hand and are hyperlinked *unless the number is in italics*. Remember, all things dealing with the F/M game in the main text are in blue. In order to save your eyes from reading so much blue lettering, all font in this appendix is back to black lettering on white background.

Change #1 – Emissions Control (EMCON): (Page 38) Putting a force “in EMCON” is the act of hiding the signals it emits (radar & comms) from the enemy by shutting them off. Maritime forces can now be detected in this game through your electronic emissions, causing a Trip-Over from further out if you are “emitting,” (the opposite of being “in EMCON”). Though it is not a pinpoint location, it does give your opponent a direction to your force from theirs, just not the distance. Some key points:

- Place a clear, blue, plastic chip on top of any Cup you want to be in EMCON and put a circle around it on the Move Card.
- While emitting, you can detect an enemy force further out than in EMCON.
- While emitting, you can be detected by an enemy force further out than in EMCON.
- Being attacked while in EMCON keeps you from using  – a dual-role missile that requires the ship's fire control radar.
- If attacked while in EMCON, that Force comes out of EMCON after the first Battle Round, (either BR#0 or BR#1, whichever comes first).
- If an attacker is in EMCON when it finds a force, it can start emitting prior to the first Battle Round.
- Aircraft Carriers can have CAP up and be in EMCON.
- CAP is always emitting.

Change #2 – Missiles: (Pages 28, 51-52) Missiles were more prolific in 1982 and feature in this game as expendable dice. Like the Kamikaze  in Leyte, you only get to throw them once. There are just two missiles,  (Blk) and  (Gr). This will increase in the Borneo game.

-  – Exocet Anti-ship Cruise Missiles (ASCM): (See also Appendix A)
 - French-made subsonic cruise missile that possesses its own radar-seeking head. Since it is self-guiding, it **can be** fired when the Cup is in EMCON.
 - Can only target Maritime Surface Forces.
 - Targeted Cup can attempt to intercept it with  and .
 - Team ARG also has limited numbers of the Air- and Truck-launched versions. The air-launched Exocet attack can be complex and is described via table on Page 52.
 - A Team ARG Surface Force Cup can resupply its Exocets at any mainland port.
 - Only fired during BR#0. (More on that later)
-  – Dual-role Missiles: (See also Appendix A) The “dual roles” are 1) offensive and 2) defensive.
 - Supersonic missiles that require a target lock from the ship before firing, meaning it cannot be fired from a Cup in EMCON.
 - In defensive mode (BR#0 only) – targets incoming .
 - In offensive mode (BR#1 & BR#2 only) – targets other ships or aircraft in flight.

- In both modes, limited to firing a number of  less than or equal to the number of targets available.
- Unlike ,  is traveling too fast to be intercepted in flight.
- Like , ARG Ships can replenish their  back to their initial loadout at a mainland port.

Change #3 – Battle Round Zero (BR#0): (Pages 45-47) This is a new prequel Battle Round made necessary to reduce complication for three reasons: 1) the addition of the dual-role nature of the . BR#0 keeps the roller from forgetting (or changing their mind about) which  are serving which roles, against which targets if they were to be rolled together in a normal Battle Round. 2)  can be intercepted in flight by  or .  cannot, so this allows it to be fired offensively in BR#1 & BR#2, but not BR#0. 3) It serves as a “negative” factor when balancing whether to keep certain forces in EMCON or not, since  cannot be used to help defend against an incoming  while in EMCON. Only , , and  can be rolled during BR#0 and in the following sequence:

- 1) Attacker announces their number of outgoing  to their opponent and the Umpire.
- 2) If emitting, defender rolls all available  within the targeted Cup as well as (up to) an equal number of  to incoming . Each  or  roll that comes up 6, intercepts an incoming  before the attacker gets to throw it – instead, placing the “intercepted ” in the kill cup.
- 3) Attacker then rolls any  that “got thru” the defensive wall of munitions.
- 4) All forces are emitting following a BR#0 and thus able to fire  in offensive mode. Any remaining  cannot be fired until the next trip to the Battle Box.

Change #4 – Survivability Points: (Pages 47-48) Maritime Cup and Tile survivability no longer rests on the number of dice it brings to the Battle Box. It now depends on its number of “Survivability Points” annotated by the red number on its lid and the upper left corner of its explanation in the Team OAF Docs. Teams will find a ring of numbers on each Cup lid. These are crossed off by the Umpire with every Damage Point an opponent allocates to it at the Battle Box. This is in addition to the usual taking of dice from those Cups.

Start of BR#1		Results of BR#1	
SF decides to fire 2x  and scores hits on both. On the re-roll they earn 6x DP.	S2 rolls its 2x  and earns 1x DP.	SF crosses out the “1” on its lid, and loses 1x  from the  hit. SF also discards its 2x expended . It survives with 1x  and 1x SP remaining.	SF decided to allocate 5x of its 6x DPs to the S2, sinking it with all its 9x dice; and 1x DP to the O3. The S2 Cup goes to the ARG kill cup, the O3 survives but loses 1x SP and 3x . ( on page 22)
			

Change #5 – BR#2 Maritime Reinforcements:  no longer represent C2, therefore any Cup touching an engaged force can enter combat in BR#2. There is no longer a separate C2 capability required to coordinate support from adjacent maritime forces.

Change #6 – Space-based ISR (UK Only): (Page 26) The US provided this to the UK and it comes in two forms in gameplay.

- Electro-optical/Infrared (EO/IR) Satellites: Day TURNS only and provides targetable data on ARG land-based force locations.
- ELINT/SIGINT Satellites: Day or Night, will pick up the general area of any ARG maritime force not in EMCON. Location data is not precise enough to be targetable.

Change #7 – Logistics Planning & Tracking: (Pages 22-24 & *Back of OAF Docs*) Previously omitted during the Leyte game due to the stair-step nature of learning *War at Sea*, Logistics has long been one of the most difficult Functions of warfare to do well. It is now modeled in this, the second game. Team Captains, *please* assign a Team member to be your J4. Logistics are tracked in three categories: Fuel, Supplies (ARG Only), and expendable munitions. Each Team is responsible for its own logistics tracking sheets that come with the OAF Docs along with a thorough explanation and examples of how to use them.

- Tracking Fuel: Fuel Points () apply to most surface forces. Both Teams have Oilers that can transfer  to Surface Forces or other Oilers. Team UK has Shuttle Tankers that can refuel the Oilers, but not Surface Forces. ARG forces can refuel at a mainland port. Forces burn  by being stationary, more by moving, and more again by getting engaged in combat and taking damage. A ship that reaches Zero , can only defend itself and provide Organic ISR until refueled. Oilers refuel a force by moving stacked with them (up to three at once) for a TURN. 8" Top speed.

	Moving > 8"	Moving 8" – 0"	Taking Damage	Engaging in Combat	Zero  Cannot...	Zero  Can still...
Force Cup*	2x 	1x 	1x 	1x 	Move, Offensive actions	Organic ISR, Defend self
Oiler	-	1x 	Up to 3x  per DP	1x 	Move	Organic ISR
Shuttle Tanker	-	1x 	Up to 3x  per DP	1x 	Move	Organic ISR
Cargo Ship	-	-	1x  per DP	-	-	-

- Tracking Supplies (ARG Only): Port Stanley cannot run out of Logistics Points () without penalty. The Facilitator will take a ground dice away every TURN that starts with Stanley down to Zero . They are replenished by shuttling  via Cargo ships from the mainland to Stanley.
- Replenishing Expended Munitions (ARG Only): An ARG Surface force that ends its movement at a mainland port will leave that port after staying there for one full TURN, not only with a full tank of gas, but also see its  and  replenished to their starting numbers if they were lost as a result of being fired. Submarines can do the same (though their fuel is not tracked) to have their  Torpedoes brought back up to their starting number as well. None of the air-launched Exocets, truck-launched Exocets, nor *any* of the UK's expendable munitions can be replenished. Cups cannot pass munitions between one another, nor can an air- or truck- launched  become a sea-launched one.

Change #8 – Submarines Updates: (Page 17.) Fuel levels are not tracked for the UK nor ARG teams. UK Subs are SSNs and can now remain Submerged while moving at up to 15" per TURN. ARG Sub movement limitations do not change from Leyte. As noted above, torpedoes are now expendable for all, only ARG can replenish them, and both Teams are restricted to firing no more than 2x  per Battle Round. Two submarines still cannot fight each other while submerged.

UK SSNs now have a decision advantage due to their speed and stealth. Refer to Rule 164:, Rule 165:, Rule 166:

Change #9 – Background Traffic, Spy Ships and VBSS: (Page 17, 49) Team ARG has Civilian Vessel Tiles called Spy Ships that provide Organic ISR. The Facilitators have neutral Background Traffic Tiles, which they spread randomly on the ARG Game Map every TURN; these model non-combatants in the area. UK Maritime Forces have the option to conduct a Visit, Board, Search, and Seizure (VBSS) if they come within 3” of one of these two kinds of Tiles, or ignore them. If they choose to stop and inspect, and the Tile is a Spy Ship, it is removed from play and Team ARG typically does not get an ISR report from it, only that it has gone silent. A Background Traffic Tile if inspected, however, will remain on the board.

Change #10 – Minefields: (Page 18, 48, *OAF Docs*) Team ARG can emplace minefields in littoral waters with its MCM Tiles, while the UK has MCM Tiles that can find and clear them but cannot emplace their own. All ARG Maritime Forces know the Q route of its minefields. For the UK, only its MCMs can conduct mine hunting and clearing operations. If an MCM Tile is stacked with another force, it will stop the force before it hits it and suffer no damage. If a force traveling without an MCM (either in the lead or Stacked) Trips-Over a minefield, that force, and any Stacked on it, each suffer a 1d6. A UK MCM clears a lane thru after remaining at the site for a full TURN following its arrival at the minefield, (i.e., *A UK MCM that finds a minefield in TURN 3, spends TURN 4 stationary and clearing a lane, so that its forces can pass thru in TURN 5*).

Change #11 – Aircraft Updates: (Primarily found in *OAF Docs*) 1982 is well into the jet age and both combatants used a variety of modern combat aircraft. Listed below are many of the changes that will affect game play:

1. Flying at Night: All aircraft can fly Day and Night except the new ARG Pucaras (O), which are limited to Daytime operations only.
2. (Y) Lg (LgY) – Maritime Patrol and Reconnaissance Aircraft (MPRA, “Nimrod” UK Only): (Page 20 & 60) The Nimrod operates from Ascension Island, off the Map. It is both an ISR platform and Sub-hunter that arrives paired with a (P) it can roll against any ARG Sub it finds.
3. (O) Lg (LgO) – Long-range Bomber (Vulcan – UK Only): (Appendix A) The Vulcan also arrives from off the Map and is only good for bombing pre-planned land targets – it cannot be used for ISR.
4. (B) Lg (LgB) – Long-range Fighters (Super Étendards & Skyhawks – ARG Only): (Appendix A) Both aircraft are modeled by the (B) Lg, but they have different limitations. The (B) Lg Super E’s can be paired with the (R) air-launched Exocet; the (B) Lg Skyhawks cannot, but they can operate from the carrier, while the Super-E cannot.
5. CAP: (Page 21) Operate emitting, can be established over any force or land within 10” of its origin, but cannot intercept incoming (R) or the (B) Lg that fired it.
6. Air-to-air Refueling (AAR): (Page 19) Both Teams have this capability modeled through a Tile, but only ARG has Fighters (B) Lg not (B) that can use it. The UK’s (O) Lg and (Y) Lg use the Tankers arriving from off the Game Map, but **due to the distance they have to travel, Team UK must decide to fly either 1x (O) Lg or 1x (Y) Lg per TURN, not one of each.** See the OAF Doc for distances. When a Tanker is shot down while loitering, if the aircraft it was supporting cannot reach a friendly landing location, those aircraft are lost as well.
7. Island Airfield Limitations (ARG Only): (B) Lg can operate from mainland airfields only. (B) Lg Skyhawks can operate from any of the mainland airfields and the carrier, while the (B) Lg Super E’s are mainland only. The (O) operate exclusively from the islands, and are the only ones that can.
8. Helicopters (UK Only): (Pages 20, 48, 56, Appendix A) (H) is no longer C2. (H) now represent a Helicopter capability and can lift 1x ground die from one land LZ to another. They cannot move forces (except SOF) from ship-to-shore.

Aircraft	Munition	Details
Lg Super Étendards	1x Exocet Missile	This pairing is only for Argentina and the rules for it are spelled out in chart form in more detail than room will allow here on Page 52.
Lg	1x ASW	If it detects a submarine, it can attack with a roll of the at the Battle Box for 2x BRs. If the Lg is intercepted by enemy aircraft, only 1x Damage Point is required to kill the pairing.
	1x or or or	Helicopters can move only 1x ground die or 1x SOF Chip at a time. A Damage Point over water will kill the pairing. If brought down over land, the ground die or SOF team survives at the point of the kill.

Change #12 – Amphibious Assault Update: The UK must transfer its ground dice from its cruise liners to military amphibs prior to landing troops. Details on how to do this are a matter of OPSEC and found within the Team UK OAF Doc.

Change #13 – SOF Mission Cards: (Pages 34-35): Both Teams have a specific number of expendable SOF Cards they can employ against ground targets. Each is marked by the Team to execute a specific mission from a menu of options on the Card at a specific location. The options are written on the front of the Cards, but their explanations are on the back. These are turned in with your Move and ISR cards

Finding Other Notable Dice Changes: (Appendix A) When reading (the now center column of) Appendix A, any cells under “Basics” that are in filled in **black** instead of either gray or white denotes an attribute that changed from the previous game. The notes section of each provides further details to changes or additions to the capability.



	Basics	Notes		Basics	Notes
Dice - "BLUE"	(B) Small	-Can attack anything except	(B) Small	-Change 1: Can now operate at	
Capability Represented	Fighter Squadron	SUBMERGED subs.	Harrier Sqdn	-Change 2: CAP can be established	
Expendable / Reusable	Reusable	-Ca	Reusable	over any geographic position but	
Day / Night / Both	Day Only	SR during the DAY.	Both	still within 10" of its origin.	
Re-Roll for Damage?	No	-Can fly CAP (3" radius) K110over	No		
Air Re-fuelable (AC:Tank)	No	any force or airfield within 10" of	No		
Carrier / Land / Both	Both	its origin.	Both		
One-way Trip Range	40"	-2 dice can be "scrambled" to	40"		
Return Trip Range	20"	reinforce a maritime tripped-over	20"		
Recon Range to 6" Ring	15"	force within 10" on BR#2.	15"		
Recon Range to 9" Ring	5"		5"		
Target Domain	Sea, Air, Land		Sea, Air, Land		
Can be Paired with	Nothing		Nothing		

Appendix E

Quick-Start Guide – Borneo

The purpose of this Appendix is to prepare Leyte and F/M-experienced players for the increased complexity and changes of the Borneo Game. What follows are the 12 most substantial modifications that players need to be aware of to help make themselves a more competent member of Team JTF Pacific Teak or Kalimantan.

Airfield			Port		
100% - 51%	50% - 1%	0%	100% - 51%	50% - 1%	0%
Fully operational	Can only support a number of aircraft equal to ½ of its MOG.	Cannot support any air operations until repaired.	Fully operational	Can no longer rearm forces at port.	Can no longer rearm or refuel forces. Can no longer load or unload ground forces.

Change #1 – Facilities Planning and Repair: (Pages 29-30, 58) Facilities in this game refers to Ports and Airfields, and they can now be attacked during the Fires Phase. Like Cups and Tiles have Survivability Points, so to do Facilities, though they can be resurrected over time. Each facility has a number of repair points associated with it, which bring back some of its Survivability Points at the end of each TURN. Those incurred during a TURN cannot be repaired until the next TURN. A facility loses its full capacity to operate with a certain number of Survivability Points lost. While all the **ports** your Team will deal with are on one card, each individual **airfield** has its own.

Airfields have an additional number associated with each of them called their “Maximum on Ground” (MOG). This is the maximum number of aircraft dice or tiles a Team can place at a given airfield. Once it hits 100% of its MOG no other aircraft cannot land there. The MOG number is affected when an airfield losses Survivability Points. When an airfield is attacked that has both dice present and Survivability Points available for the taking, a new 1-1 Rule (Rule 95:) applies in which half of earned Damage Points must go towards the dice and half must go towards the airfield’s Survival Points. Like the 1-1-1 Rule, this occurs until one is depleted and the remaining points all go towards the other.

ISKANDAR Airfield
MOG: 12 Aircraft Dice / Tiles

Current Airfield Damage Status

2	2	2	2	3	3	7	8	9	10	11	12
100% MOG Capacity						50% MOG Capacity					
Repair Capacity = 3/TURN											

Airfield at START

42nd Fighter Sqdn
51st Attack Sqdn
5th Coastal Patrol Sqdn

Aircraft Currently Airborne

Ground Dice

1-1 AD
2-2 CD

Callouts:

- Airfield Name and MOG Number.
- Location and MOG Number again.
- Airfield Survivability Points. With 6 depleted here, Iskandar can only now operate 6 aircraft dice/tiles, no matter how many are present.
- Number of SP “repaired” at the end of each TURN. In this case “3”.
- Ground dice present on the airfield are stored here. The halo over a group signifies that they are in a Cup and so can be moved around the Map. Without that, they are fixed to the site.
- The number of aircraft dice/tiles cannot add up to be more than the MOG.

Change #2 – Cyber Cards: (Pages 31-33, 39) There are two types of Cyber cards at your disposal, offensive and defensive. Neither of them is guaranteed to work - the offensive ones are expendable, and the defensive ones are reusable. The offensive cards allow you to attack your opponent from a menu of eight different effects. A short description of what each does is on the back of the cards, and a more thorough one is found on page 32. Teams cannot employ the *same* effect more than once against the *same* target in the *same* TURN. Teams **can** employ multiple different offensive effects against the same target in the same TURN and have those effects compound. Also, using a specific kind of cyber-attack does not “burn it” for future use. It is assumed the same effect could be achieved through another type of hack. Turn in Cyber Cards with Move and ISR Cards. Keep in mind that these model reality in that they may or may not work.

Change #3 – SOF Card Updates: (Pages 34-35, 39) There are two main changes to the SOF Cards for this game. 1) There are no dice representing the SOF teams; 2) There are many more options available to each Team. For many of the available options, the 1-1-1 Rule limitation is lifted due to the surgical nature SOF.

NOTE: Cyber and SOF cards are adjudicated first – *for example, a successful cyber or SOF effect that shuts down an airfield will prevent all sorties from launching from that airfield on that TURN.*

Change #4 – Deception and Denial Cups: (Page 36) This is how a Team can better hide a specific force on land. Teams get a certain number of Cups that each have a 13-dice capacity. Teams can only put **one** D/D Cup on any given site, making its contents invisible to space-based ISR (Rule 70:). It can still be seen by any **manned** ISR and then struck by manned strike missions. It remains invisible to unmanned ISR.

Change #5 – Aircraft Update:

1. **New Pairings:** There are several new pairings with Borneo. Though most of them can now carry more than one type of munitions, aircraft cannot mix their loads in a single sortie.

Aircraft	PACTEAK	Kalimantan
 Lg	1x  or 1x  A maritime minefield. (Requires 4x  Lg loitering over the targeted area for a TURN)	1x  or 1x 
	1x  or 1x  Lg	1x  or 1x  Lg
 Lg	1x  or 1x  Lg	1x  or 1x  Lg
 Lg	2x  or 2x  Lg or 3x  or Organic Bombing or a maritime minefield (Requires 2x  Lg over the target area for a TURN)	1x  or 1x  Lg or 1x  or Organic Bombing
	1x ground dice - Can now fly them ship to shore.	1x ground dice

2. **Escort Aircraft:** (Page 36) Escort duty is now the fifth mission an aircraft can execute in a single TURN. Any number of fighter aircraft can be used to provide escort as part of an Air Cargo, Air-to-air refuel, ISR, or AWACS mission. Escorts are specifically designed to bear the brunt of attacks by enemy aircraft and suffer damage before the aircraft they are protecting does (Rule 116:).
3. **Air Transports:** (Page 36) These are Tiles marked with AT#. They are capable of moving 1x ground die from one area to another. They are automatically dead if caught unescorted (Rule 113:).
4. **CAP:** (Page 21) CAP stations can be set at any location (land or sea) and must be annotated on the Team's Move Card. They still must be within 10" of their station of origin, but if it is a carrier, that CAP station can now move in parallel with the carrier's movement, offset up to 10", (*i.e., having a CAP*

station remain 10" ahead of a carrier's movement). This is **annotated on the Move Card with a dotted line** running along with the solid line of the carrier. The station will have a 3" radius ring of organic ISR (6" circle) around it, which can be Tripped-Over. An incoming enemy Strike Package can commit the friendly CAP to one direction, doubling its range to 6" from its center point. Any CAP beyond 5" from a targeted force can only have 1x BR per TURN, whereas CAP within 5" will have 2x BRs.

5. **AWACS:** (Pages 27, Appendix A) These are modeled through  **Lg** and have a 120" of total range to roam around the map and they provide ISR in a 5" radius around their center point as they move. If they are found by enemy aircraft, unescorted, they are automatically shot down (Rule 80:).
6. **Helicopters:** (Page 56)  can still only carry 1x ground die per, and they can move those dice from ship-to-ship, ship-to-shore, as well as from one land LZ to another.  model Ospreys as well.

Change #6 – MWAS Satellites: (Page 26) All satellites are now available to both sides but there is also the addition of the Maritime Wide Area Search (MWAS) Satellite. This satellite only looks in the maritime domain. It provides targetable data and has the added benefit of being able to follow a specified target across TURNS, once located.

Change #7 – Missiles Updates: (Pages 28)

 **Lg (LgBlk)** – Long-range Anti-ship Missile (LRASM): This is much the same as the , with the exception of its greater range - from 5" previously, to 10" now. When fired from land-based systems, it is assumed that they can "see" as far as they can shoot.

 **(T)** – Ground-based Air Defense Missile (Patriot): Think of it as an expendable (but more effective) missile version of the . The  can shoot down aircraft and intercept other missiles but **cannot** target surface vessels or ground equipment. Like the , however, there is no intercepting it once fired and both have a 5" range. It, and the  are the only dice that can intercept the  **Lg** and  **Lg** in flight.

 **(O)** – Land-attack Cruise Missile (LACM / Tomahawk): Only good against targets on land. This is a maneuverable missile, meaning you can plot waypoints on the Fires Card and Map for it to follow along its 50" of range to the target. These can be launched from specific surface forces, submarines, and aircraft.

 **C (CO)** – Maritime Strike Tomahawk (JTF Only): These orange dice are different in that they are clear and have white pips. It is a land-based maneuverable cruise missile with a 50" range that targets surface vessels.

 **(Pk)** – Ballistic Missile Defense (JTF Only): This new missile has all the same properties as the  always has had with the exception that the  cannot intercept the  **Lg** - the  can. Only the JTF has them because only Team Kalimantan has the  **Lg**. Just like , they can also be used against aircraft and surface forces in offensive mode.

 **Lg (LgPk)** – Anti-ship Ballistic Missile (DF-21, KAL Only): This is a long-range ship-killer missile that can only be stopped by  or . No aircraft can engage it in flight, nor can any roll of  intercept it along its 50" of range. Unlike the , which can be steered along waypoints, the  **Lg** follows a straight ballistic trajectory from its point of origin to its target.

 **Lg (LgT)** – Land Attack Missile (DF-21C, KAL Only): This is a land-based missile that follows a ballistic trajectory up to 50". It is only good against enemy facilities and can only be intercepted by  or .

Change #8 – Fortifying (Digging in) Ground Units: (Page 21) Dense jungle areas and rough terrain provide the opportunity for ground units to dig in and resist destruction from enemy attacks. "Fortified" ground units (denoted on the Move Card as shown on the right) may not move, but they may conduct ISR and Fires. If attacked, fortified ground forces May attempt a saving dice roll to prevent destruction.

- For each hit on a ground dice, the owning player rolls the die that would be destroyed.
- If it comes up with a 5-6, the die survives.
- If it comes up 1-4, the die is destroyed as normal.
- This only works for 2x BRs per position.



Force "3D" is fortified.

Change #9 – Replenishment of Expendable Munitions: (OAF Docs) Maritime expendable munitions may be replenished. The rules for doing so are Team dependent and a matter of OPSEC for them. Reference your Team OAF Doc for how and where this can be done.

Change #10 – Submarines: (OAF Docs) Opposing submarine forces can now attack each other when both are submerged. They can also fire various missiles that can be reloaded at specific ports and emplace minefields. Each Team has slight differences that are, by necessity, mentioned in their respective OAF Docs.

Change #11 – Deliberate Aerial**Attacks with Missiles:** (Page 52-53)

This is possibly the most complex addition and is best described within the main text with the corresponding table to the right. This table models the attacking enemy's aircraft and ordnance from left to right, and the defending force from right to left. The width covers 10" out from the target, denoted on the top and bottom rows. The light blue shaded areas are where the action happens along that 10" scale to the target.

Range to Target:	10"	9"	8"	7"	6"	5"	4"	3"	2"	1"	Target
BR#0 - Step 1 Long-range ASCM shot	Attacker fires Lg from 10" and aircraft RTB. Lg can accompany this strike also but it comes from 50" out. Neither rolls until Step 3, but # of each inbound missile is provided to the Umpire and opposing Team.					Defender chooses # of CAP aircraft to place against the inbound Lg, and paired with they cannot stop the Lg. Hits in the next step will reduce the # of inbound threats. CAP committed to this will be out for the rest of the engagement following their 1x BR outside the 5" range ring of the targeted vessel.					
BR#0 - Step 2 Short-range ASCM shot	Since aircraft firing cannot do so until 5" out from the target, they are susceptible to 1x BR of a two-sided engagement (in the 1" overlap) from a # of defending CAP. Survivors of the single BR, RTB while their fired are added to the Lg that survived in Step 1. Only Lg cannot defend itself.					The Defender's CAP chosen in Step 1 are committed to attacking incoming aircraft and Lg from 6" out for 1x BR of a two-sided attack. A single Damage Point kills the pairing before it fires. CAP committed to this are now out for the rest of the engagement.					Target still waiting to engage with protective fires and remaining CAP.
BR#0 - Step 3 Missile on missile	ON THIS STEP - ATTACKER ROLLS 2nd Read the other side first, since the defender rolls first this time. In the final step of BR#0, the attacking Team rolls its surviving Lg, , & Lg after the last layer of defense is rolled with its final protective fires listed on the right. → Following this final attrition, the surviving missiles roll and apply their earned Damage Points against the targeted force(s).					DEFENDER ROLLS 1st This last layer of defense is rolled first to determine what ordnance gets through. The targeted force (and any Cups touching or stacked) not in EMCON can engage the incoming missiles with both & in defensive mode, any CAP remaining, and . Lg → (≤ + Lg) Lg → x All CAP Lg →					
BR#1	From this point on, all offensive missiles that will be used for this engagement have been expended. What flows in now will be the aircraft-only portion of the Strike Package. Any aircraft used previously in launching missiles have all RTB'd. The engagement from here is like any other Deliberate Fire Strike Package from the previous games.					Lg → Lg → Lg → Lg → Lg Remaining CAP can now engage the incoming enemy fighters with remaining BR(s). * Any used in BR#0 cannot.					
BR#2	Attacking Team has the option to stay and fight or attempt to disengage.					Lg → Lg → Lg → Lg → Lg Defending Team can now apply the scramble rule to any carriers within 10" of the target.					
Range to Target:	10"	9"	8"	7"	6"	5"	4"	3"	2"	1"	Target

Airborne Force Element	On Approach Flight	BR#1	BR#2	End of Fires Step
Air Transport	If drawn flight path from origin to DZ on Fires Card crosses any or CAP stations, 1x BR will be adjudicated for every instance that the defender elects to attack.	Drop Paratroopers.	Return to Base.	Arrive back at Base.
Aircraft Escorting Air Transport	Escort rules on Page 35 apply.	Remain with Air Transport for return flight.	Remain with Air Transport for return flight.	Returned to origin.
Aircraft Supporting Ground Forces	Fight enemy CAP or carry on to DZ to support infil and ground units.	If enemy CAP is on station within 3" of DZ, determine # of to split to attack enemy ground forces and to fight the CAP. Lg can also be in support of infil with bombing. Defending , , and can target these aircraft.	Same as BR#1	Lg Returns to origin. Fighters from both sides can remain on station as CAP until TURN Closeout.
Ground Forces	-	Can be attacked by defending ground forces, but cannot yet defend themselves.	Surviving ground force elements can now engage defending air and ground forces.	If both sides have forces remaining, combat will recommence as a Trip-Over in the following TURN.

Change #12 – Airborne Assaults:

(Pages 56-57)

These occur during the Fires Step and are drawn on the Fires Card, much like an amphibious assault. The assaulting Team starts by drawing a 1" diameter circle to indicate the intended drop zone (DZ). If that DZ is defended by enemy on the DZ or enemy or within 1.5" then the table on the left applies. Each air transport Tile can carry 1x ground die, each.

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